When Silver Isn't Gold: Quantifying and Auditing Group-Based Disagreement in Human and Persona-Driven LLM Annotation

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LLM annotation is increasingly used as a scalable substitute for human annotations, including in polarized, subjective tasks where humans systematically disagree along group lines. Whether these systems reproduce, ignore, or distort human disagreement patterns is largely untested, in part because most existing methods for attributing Systematic Disagreement (SD) have not yet been tested in realistic, subjective annotation settings. This work represents the first empirical study which applies proposed, theoretical methods on large scale, realistic annotation settings. Observing the weaknesses of these methods, we develop a method that not only detects, but also quantifies attribution of Attributable SD (ASD) to annotator SocioDemographic Dimensions (SDDs) and use it to audit nine open-source LLMs against five real-life datasets on subjective Natural Language Processing (NLP) tasks. Our results show that while LLMs using persona-prompting can produce polarized annotations, the exhibited disagreement patterns are not aligned with those of human annotator SocioDemographic Dimensions (SDDs), even when tuning instruction prompts for this purpose. This limitation indicates that applications, studies and resources using persona-prompting should include appropriate scoping and evaluation in order to avoid further model miscalibration. Our study also suggests that the structure of current subjective NLP datasets is insufficient, requiring more annotators per item, as we find that not all SD can be attributed to any combination of known or latent annotator SDDs.

CCS Concepts: • **Computing methodologies** → **Natural language processing**; **Unsupervised learning**; Language resources; • **Human-centered computing** → **Empirical studies in collaborative and social computing**; • **Social and professional topics** → *Race and ethnicity*; *Gender*; Sexual orientation; Age.

Additional Key Words and Phrases: Polarization, Clustering, Annotation, Sociodemographic Characteristics, Persona-Prompting, LLM Bias

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1 Introduction

The rise of internet spaces has necessitated research in subjective tasks such as the detection of harmful content. Annotations are needed to support social science research, the training of automated classifiers and the proactive identification of vulnerable groups [71]. These annotations are frequently released as datasets, which due to their subjective nature typically contain items with very low Inter-Annotator Agreement (IAA) [71]. If ignored, this disagreement may lead to loss of valuable information [52] and provide us with the wrong signal, leading to significant misconfiguration in trained AI systems [69]. For example, toxicity detection models have disproportionately marked African American (AA) speech as "toxic" [60], or systematically ignored minority opinions [55] in the past.

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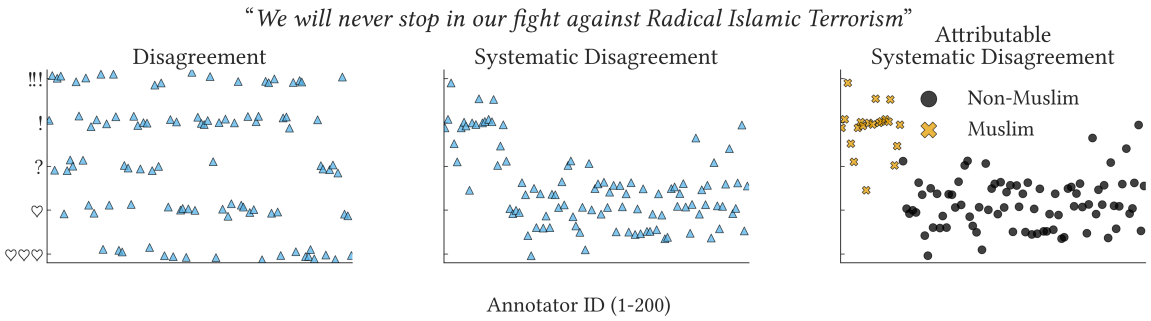


Fig. 1. Three kinds of disagreement in a hypothetical hate-speech annotations of a controversial comment. Each point is one annotator's rating, from ♡♡♡ (not hateful) to !!! (very hateful). *Left:* ratings spread uniformly across the scale, an extreme case of GD. *Middle:* SD, where a subset of annotators consistently rate the comment differently from the rest, although the cause is not visible from the ratings alone. *Right:* the same ratings as the middle panel, grouped by religion. Religion, one of the observable SDDs, acts as a "polarizer" that explains the observed differences in opinion, marking the disagreement as ASD.

Current literature lacks precise terminology to describe opinion differences between groups. We thus use the following definitions in this study: (1) **General Disagreement (GD)** represents any variance among annotators' ratings for the same item. (2) **Systematic Disagreement (SD)** is non-random and recurring disagreement. (3) **Attributable SD (ASD)** is SD that can be attributed to differences in a specific SocioDemographic Dimension (SDD) (e.g., gender, age, education level). (4) **Unnattributtable SD (USD)** is SD that cannot be attributed to *any* SDD (both observed and latent). We posit that **polarization** is the mechanism that distinguishes SD from ASD. The intuition comes from physics, where a *polarizer* is a filter that selects light along one chosen direction and blocks the rest, turning a mixed beam into a "pure" one. In our setting, SD is the beam, and the polarizer is each SDD used to split annotators into groups (e.g., left-wing vs. right-wing for political orientation). If SD is reduced once annotators are grouped by an SDD, the disagreement is attributable to that dimension, and SD becomes ASD. Fig. 1 demonstrates this distinction.

Researchers traditionally attempted to distinguish GD from SD, but few studies moved towards discovering ASD [15, 52, 74]. In both cases, IAA metrics are typically used both to distinguish between GD and SD and to attribute it to specific annotator groups [15, 74]. Traditional metrics however hold significant limitations, especially when few annotations per item are available and when the studied SDDs contain imbalanced groups. Both of these issues are typically present in subjective tasks featuring minorities, which is the exact case where our measures should be reliable.

Furthermore, as human annotation is generally expensive, LLM-as-judge systems are increasingly deployed as scalable substitutes for human annotators [33, 58] to acquire a cheaper, "silver" supervision signal in place of human-annotated "gold" labels.[1] It is therefore imperative that ASD by human annotators be preserved by LLMs whose annotations by default gravitate toward the statistical average [8]. Persona-prompting is sometimes used to bypass this limitation, but suffers from its own challenges such as stereotyping and inconsistent annotations [41, 43]. We therefore ask:

RQ 1: *How Do Current Theoretical Polarization Attribution Approaches Behave in Realistic Annotation Scenarios*? Our work is the first empirical study applying measures of polarization detection and attribution in real-life subjective datasets, finding that they tend to severely underperform. One of the reasons for this collapse is likely the assumption that annotations can be aggregated across items [2, 10, 52] (Finding 1).

[1]What constitutes "gold" or "silver" labels is not standardized and the terms frequently change definitions between publications. Some advocate against using the term "gold labels" for subjective, human annotations [5, 21].

RQ 2: *How Can SD be Quantitatively Attributed to Annotator SDDs*? Since we can not reliably use established approaches, we introduce a new measure which *quantitatively* attributes SD to specific SDDs, while being conscious of the limitations present in realistic annotation scenarios (Fig. 2).

RQ 3: *Do LLM Annotators Exhibit ASD, and if so, Which Patterns are Shared Between Them and Humans*? We find that human ASD is explained by gender and ethnicity across five real-life subjective datasets (Finding 2). On the other hand, while LLMs using persona-prompting *can* produce polarized annotations, they fail to reproduce any ASD patterns regardless of instruction prompt, confirming literature suggesting that persona-prompting does not consistently influence LLM outputs in the examined tasks (Finding 3).

In prior work, the question of whether human and LLM opinions with persona-prompting are aligned was answered by measuring task performance [29], or by directly comparing human and LLM annotations [16, 24, 44]. The first case produced non-generalizable results, while the second was made arduous by the insufficient distinction of GD and SD by traditional metrics, especially in the case that SD between human groups was not accounted for [9, 10, 19, 70]. Our methodology instead first filters out GD from SD, then analyzes inter-group ASD for human and LLM annotations separately and then compares patterns of disagreement between the two.

While our results are consistent with prior work [33, 41, 43, 58, 78], our methodology provides an answer that is more robust (as it accounts for the aforementioned confounding issues of IAA metrics and group disagreement) and explainable (as we provide quantifiable values for each step of the process). It also answers the question on a deeper level: previous work had only confirmed that persona-prompting meaningfully increases the variance (here, GD) of LLM annotations [23], while we show that persona-prompting can not only produce SD but also ASD, although the direction of ASD is inconsistent, both between LLMs and humans and across different LLMs. These inconsistencies call for careful evaluation and calibration of LLM outputs in sensitive applications such as content moderation [12] and facilitation [34], which currently assume LLMs can act autonomously. They also suggest that persona-prompting to increase annotation diversity [22, 23] may produce miscalibrated data outside narrow, well-validated scenarios, amplifying existing biases across model generations, should LLMs be trained on biased LLM annotations [57, 63].

Our methodology also provides a quantifiable argument for a paradigm shift in the creation of subjective Natural Language Processing (NLP) datasets. Given the discovery of USD (Finding 4), our work argues that even with recent methodological improvements (e.g., phasing out majority voting), the field of subjective NLP has not adequately addressed the challenge of analyzing multiple perspectives. Specifically, annotator sparsity on the item level leads to the emergence of USD, indicating the absence of enough information for disagreement attribution. We describe how this finding impacts dataset design, LLM training and social science analysis in §6. We also release a PyPI library implementing our measure, our LLM annotated datasets totaling more than 3,000,000 annotations, as well as the code and reproducibility details to aid future research in this topic.[2]

Content Warning: Controversial and potentially harmful speech is used for demonstration purposes.

2 Background and Related Work

2.1 Systematic Disagreement Detection

Prior work on analyzing annotations in subjective tasks has shown that SD may be caused by different points-of-view between groups [25, 47, 69]. While current research mostly rests on using IAA metrics between groups of a

[2]Library https://anonymous.4open.science/r/apunim-5F70, HTML documentation https://anonymous.4open.science/r/apunim-page, replication code https://anonymous.4open.science/r/aposteriori-unimodality-E8C1. See Ethical Considerations for export measures relating to gated-release datasets.

single annotator SDD [31, 35, 39, 47, 68, 70], these metrics suffer from severe limitations, which have motivated the development of new approaches.

Limitations of Inter-Annotator Agreement. IAA metrics such as Cohen's kappa, Krippendorff's alpha, and Fleiss's kappa aim to distinguish SD from GD through statistical chance-correction. However, these metrics are often difficult to apply to intergroup IAA without extensions or restrictive assumptions (e.g., no missing labels or equal numbers of annotations). They have also been criticized for failing to account for differences arising from social, cultural, and demographic backgrounds, producing results that are difficult to compare across studies [9, 10, 19, 70], being sensitive to class imbalance [19], and relying on questionable assumptions about chance agreement [10]. While recent work has addressed some of these limitations with methods that improve applicability and robustness [31, 70, 77], they are much more complex and less intuitive, indicating that IAA may not be the correct tool for analysis of minority viewpoints.

Alternatives. A different approach [2] utilizes χ^2 to estimate in-group vs out-group variance. Others frame SD detection as a cluster identification problem, applying either parametric (Gaussian Mixture Model (GMM)) [10], or non-parametric [46, 52] approaches to the annotation histograms. For our work, we use a non-parametric disagreement measure, normalized Distance From Unimodality (nDFU), which has been shown to correlate better with human judgment of disagreement than established measures [52], as we also verify in Fig. 2. nDFU (Eq. 1) takes values in the range $[0, 1]$, where values close to 0 indicate no polarization (i.e., opinions are concentrated around a single mode). Intuitively, nDFU measures how prominent the maximum secondary peak is compared to the main peak.

$$\mathrm{nDFU}(X) = \frac{\max_{i \neq m} \left(f_i - \begin{cases} f_{i-1}, & \text{if } i > m \\ f_{i+1}, & \text{if } i < m \end{cases} \right)}{f_m}, m = \arg\max_i f_i \tag{1}$$

where, X denotes the set of annotations, f_i the relative frequency of the i-th annotation (e.g., the frequency of 2 in a scale of 1–5), and m the histogram peak.

What About Attribution? IAA analysis may be useful for exploration and identification of some controversial items, but fails to explain the reason *why* they are controversial [1, 72]. A purpose-designed method for the identification of ASD is the A Posteriori Unimodality (AU) method [52], which builds upon nDFU. It states that if a single item's annotations are multimodal (contains SD), but unimodal for all annotator subgroups in a specific SDD (contains disagreement but not SD), then polarization is explained by those groups (ASD). Though promising, this mechanism operates on the assumption that *all* polarization can be explained by a single SDD, which we prove to be impossible in many realistic cases (§6).

2.2 Using LLMs for Subjective Annotation

Persona Prompting. To address subjectivity in annotation, several NLP approaches incorporate annotator metadata into LLM annotation [14, 20, 28, 59]. A significant limitation is the requirement for annotator metadata, which can be side-stepped through embedding clustering [73], at greater computational cost and harder evaluation. Our approach instead directly clusters the annotations, which we believe provides a comparatively simpler and more interpretable mechanism.

Persona Engineering. At the time of writing, there is no established taxonomy of persona-prompting techniques. Generally, personas may be authored by hand [45, 81], generated using observed annotator attributes [3], composed of

Table 1. Datasets used in this study. *#Ann* refers to the number of annotators per item, while the **#SDDs** refers to the number of SDDs (e.g., age, gender, education). *Guid.* describers whether annotation guidelines are provided in the publication introducing the dataset (including documentation and supporting material) and the last two columns denote whether the datasets were usable for analysis of the original (human) annotations and LLM annotation respectively.

Dataset	#Items	#Ann	#SDDs	Text Source	Task	Guid.	Human	LLM
Kumar [36]	1000	5	10	Twitter, Reddit, 4chan	Toxicity	Yes	No	Yes
Sap [61]	585	5–6	3	Twitter	Hate Speech	No	Yes	Yes
DICES-350 [4]	350	123	4	LLM responses	Hate Speech	No	Yes	No
DICES-990 [4]	990	70–76	4	LLM responses	Hate Speech	No	Yes	No
POPQUORN [53]	1500	7–9	5	Reddit	Hate Speech	Yes	Yes	Yes

answers to extensive questionnaires [51], or recently, generated entirely using LLMs [64]. The prompt itself may be structured [50, 75, 80] or described through natural language [40, 64, 81]. The effects of these methods on LLMs have not yet been studied and are largely outside the scope of this work. Since our goal is to explore disagreement patterns across LLMs and humans (RQ 3), we select attributes from the human annotators of each dataset and format them using JSON instead of natural language, in order to facilitate comparisons across experimental conditions.

Persona Reliability. Prior work suggests that LLMs are not necessarily reliable annotators when prompted with personas, as they tend to reproduce stereotypes [41, 43] and perform unreliably for minority groups [33, 43]. Additionally, some evidence suggests that task-specific improvements may be not be replicable outside of the datasets on which models are trained [38]. Our work ultimately supports the view that persona-prompting is not reliable by itself in all tasks and must be separately evaluated when used.

3 Datasets

Finding appropriate subjective datasets to test LLMs on is difficult for many reasons: (1) very few include annotator metadata [73], (2) most include only three annotators per item (e.g., [13, 37, 48, 66, 67]), (3) many datasets are sampled from the same source, negatively impacting the breadth of possible generalizable findings [71], and (4) annotation guidelines are frequently not included [71], complicating replication. Additionally, meta-studies on subjective datasets [54, 71] do not give special importance to the sparsity of annotations per item, which is an important consideration.[3]

We use five datasets from current NLP research that contain SDD information at the level of individual annotators:

- The *Kumar* dataset [36] which contains annotations for Toxicity detection.
- The *Sap* "Breadth of Posts" dataset with the "*toyou*" labels [61] for Hate Speech (HS) detection.
- The *DICES-350* (*Q3_bias_targeting_inherited_attributes* label) and *DICES-990* (*Q3_bias_incites_hatred* label) datasets [4], which pertain to LLM response safety. Our choice of labels allows us to use them as HS datasets, capturing conceptually similar SDDs despite the lack of shared labels between them. There is little-to-not information around these labels, since neither a description nor annotation guidelines are provided.
- The POPQUORN [53] dataset using the *"offensiveness"* label for HS detection.[4]

[3] §6 shows that the real bottleneck in many NLP datasets is not the amount of recorded SDDs nor the amount of items, but rather the amount of information (number of annotators) per item.
[4] While "politeness" labels are included, they may not directly translate to toxicity; comments can simultaneously be toxic and phrased politely. Indeed, toxicity is not mentioned in the methodology of the original POPQUORN dataset and is considered a separate NLP task in other work [42, 82].

Table 2. Statistical significance of each SDD feature per method. ✓: significant at $p < 0.05$ after Bonferroni correction across the features of the dataset; ×: not significant.

	DICES-350				DICES-990				Sap			Kumar												POPQUORN				
SDD	Age	Education	Gender	Race	Age	Education	Gender	Race	Age	Ethnicity	Gender	Age	Education	Ethnicity	Gender	Has Been Targeted	Is Parent	Is Transgender	Political Affiliation	Religion Important	Seen Toxicity	Sexual Orientation	Technology Impact	Toxicity Problem	Age	Education	Gender	Occupation	Race
apunim (ours)	×	✓	×	✓	✓	✓	✓	✓	✓	×	✓	×	×	×	×	×	×	×	×	×	×	×	×	×	×	✓	×	×	✓
Krippendorff $\Delta\alpha$	×	×	✓	✓	×	✓	✓	×	×	×	✓	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×
AU	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×
χ^2	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	×	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

A brief overview of the dataset characteristics is provided in Table 1. The Kumar dataset is notable for its large scale, featuring over 100,000 annotated items and 10 distinct SDDs.[5] In contrast, the DICES datasets have fewer annotated items but involve more than an order of magnitude more annotators per item than the others. An exploratory analysis of these datasets is available in App. B. Note that not all datasets can be used for all stages of our study; the DICES datasets do not include enough reproducibility information to re-annotate using LLMs, while the Kumar dataset does not include enough human annotations for any methodology to find ASD (Table 2).

4 Empirically Testing Proposed Methods

4.1 On Real Data

We first test prior methods of attributing ASD to our real-life, human-annotated datasets. Through this comparison we aim to both explore the datasets themselves by finding SDDs on which multiple methods attribute ASD and as a sanity-check on the methods themselves.

Methods Considered. We compare the effectiveness of our method (introduced in §5), the AU framework proposed by Pavlopoulos and Likas [52], the χ^2 method presented in Akhtar et al. [2], GMM clustering used in Checco et al. [10], and a baseline using IAA ($\Delta\alpha$ Krippendorff) on simulated distributions. In this case, the GMM method is not run, as it requires the exact same annotators to annotate the items in each dataset, which is not the case for any of our datasets.[6]

Results. Table 2 shows whether each of the methods attributes ASD and on which SDDs. We observe that: (1) Two of the proposed methodologies do not yield any meaningful results. The AU method fails to find any ASD attribution pattern in *any* of the SDDs in any of the datasets, while the χ^2 method reports that ASD can be attributed *all-but-one* SDD. (2) In contrast, both IAA and our own method report some shared findings across the DICES-350, DICES-990 and Sap datasets, while neither reports any attributable SDDs in Kumar. In fact, we observe that attributable SDDs (namely *Education, Race, Gender*) are commonly shared between datasets. (3) Unlike IAA, own method finds that the

[5]In order to allow for a full comparison between human and LLM annotations, we use a sample of 1,000 items from the Kumar dataset (§5.3).
[6]Regardless of empirical results, this limitation makes the GMM method extremely restrictive in realistic annotation settings.

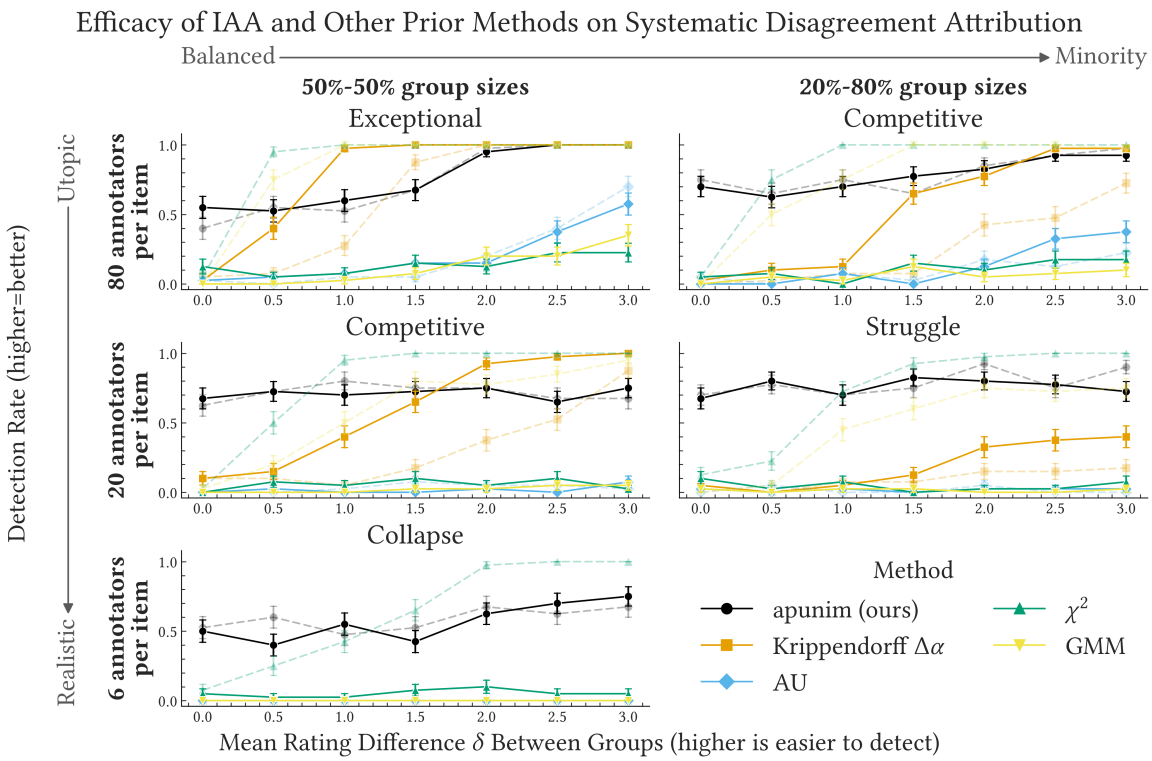


Fig. 2. ASD attribution quality on synthetic distributions. The x-axis represents the gap in opinion between groups (δ parameter used in p_j). Higher is easier to detect. The y-axis shows the accuracy of each method when attempting to recover the (known) SDD causing ASD. We find that both IAA and other proposed methods struggle when presented with realistic annotation scenarios, while our method remains consistent. Dashed lines show the results of a simplified experimental setup, where each group always prefers higher ratings than the other in all items and bars indicate standard error.

POPQUORN dataset does have SDDs on which ASD can be attributed. The fact that *three of the four currently proposed ASD attribution metrics are not usable* in any of our datasets is troubling and requires investigation.

4.2 Diagnosing Proposed Methods With Synthetic Data

Generating ASD. In both real-life human and LLM annotation we lack a ground truth for ASD. This limitation can be circumvented by constructing annotation distributions by hand, since we can know which SDD explains ASD and can test whether it is correctly attributed by each method. Additionally, by constructing these distributions we can test these methods in specific annotation scenarios reflected by our choice of parameters. We define an annotation y_{ij} by annotator i for item j as $y_{ij} = b_j + g_i p_j + \epsilon_{ij}$, where:

- b_j represents a baseline rating around which all annotations for an item j are clustered (e.g., $b_j = 5$ for an obviously toxic comment on a toxicity scale of 1–5).
- $g_i \sim \text{Uniform}(0, \frac{1}{2})$ denotes group identity. We use sampling to represent each annotator internalizing group identity differently.

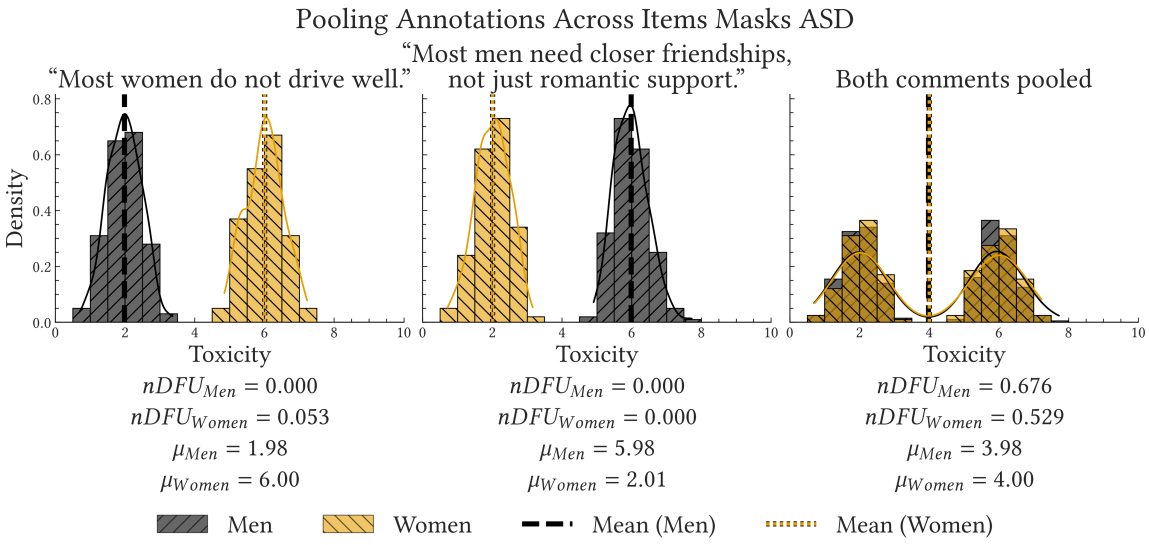


Fig. 3. Synthetic experiment showing two comments with annotators disagreeing about which one is toxic. Instead of comparing two unimodal distributions (comments viewed separately) with two bimodal distributions (their aggregated annotations), we end up comparing two bimodal distributions during aggregation, obscuring the source of the polarization (explained by gender, but not shown). This is true both for means-effects models, and nDFU comparisons between groups.

- p_j represents real-life group polarization. We use two different formulations: (1) **Unidirectional**: $p_j \sim \text{Uniform}(0, \delta)$. The group always favors higher ratings across all items, though its magnitude varies per item. (2) **Bidirectional**: $p_j \sim \text{Uniform}(-\delta, +\delta)$. Annotators of the same group use systematically higher ratings in some items and lower in others. See Fig. 3 for an example.
- $\epsilon_{ij} \sim N(0, 1.8)$ represents GD.

We use bootstrapping to determine whether a score is statistically significant and counts as a detection. For each experiment, we create 200 items across 40 independent runs, used to calculate standard error. Our hypothetical SDD features two groups, although ablations for $k = \{3, 5, 10\}$ groups per SDD are shown in Fig 10 in App. E.

Sensitivity to per-Item Annotation Counts. Fig. 2 shows the performance of each method on our synthetic annotations. IAA (orange squares) performs exceptionally in the case of eighty annotators per item even when differences in opinion (y-axis) are small (top-left). IAA however requires a larger difference in opinion to detect ASD when twenty annotators are available (middle-left) and collapses when only six annotators are available per item (bottom left). IAA is also sensitive to the size of the compared groups; detection performance degrades significantly when we compare a minority group (16 vs. 64 annotators; plots on the right). On the other hand, the AU method (blue diamonds) only detects polarization when eighty annotators per item are available and differences in opinion are quantitatively large (top-left), but is resistant to minority group scenarios (top-right). This result is expected; AU utilizes clustering and is therefore largely invariant to the sizes of the opinion clusters. However, its absolute definition of ASD attribution (§2) renders it sensitive to noisy opinion histograms, which are common when few annotators per item exist.

Inter-Item Annotation Aggregation. The χ^2 and GMM methods (green upright and yellow reverse triangles respectively) generally fail at attributing ASD to any group in the bidirectional polarization scenario (solid lines). However,

when defining polarization unidirectionally (dashed, faded lines), they perform exceptionally. This observation suggests that these metrics assume that across items, ratings of group A have a different average than group B. This assumption is valid for the unidirectional polarization scenario, but not the bidirectional one.[7] Specifically, χ^2 pools all annotations into a single contingency table, while GMM clusters annotators by their mean rating across items. Fig. 3 shows a synthetic scenario where both approaches would fail, as bidirectional polarization effects cancel out in the aggregate, making the two groups appear statistically indistinguishable.

Results. Overall, our results suggest that *current metrics for ASD attribution are sensitive to one or more conditions that may occur in realistic annotation setups* for subjective NLP datasets. IAA is sensitive to minority groups, which are the subject of many content moderation tasks, but are not evenly represented in most datasets (see Tables 9,10,11 in App. B) and on low annotator counts per item (also common, see Table 1). The latter is also required by AU.

Finding 1. *Current methodologies revealing ASD generally fail in realistic annotation conditions. One of the reasons is that disagreements may conflict across items, which may compromise cross-item annotation aggregation.*

5 Comparing Human and LLM Polarization

5.1 Quantifying Attribution to Annotator Dimensions

We estimate SD using nDFU (see Eq. 2). Specifically, we measure the SD of each group g for an item c as $P_g(c) = nDFU(A_g(c))$, where $A_g(c)$ is the set of annotations for item c created by annotators belonging to a group g. We compare this quantity to the "apriori" polarization present in the item c, which is defined as the *expected* (average) unimodality of theoretical groups $\mathrm{P}_{apr}(c) = \frac{1}{k}\sum_1^k \mathrm{nDFU}(A_r(c))$, where A_r is a set selected from a random partition of the original annotation set and k is a hyperparameter (higher k yields more accurate estimation, at a greater computational cost). Intuitively, a high $\mathrm{P}_{apr}(c)$ score means that much of the SD is likely not driven by any single group.

Since the $\mathrm{P}_{apr}(c)$ and $P_g(c)$ quantities represent very noisy estimates due to the lack of annotators per item, a direct comparison would not be reliable. We therefore exploit the Law of Large Numbers to compute the statistic on the dataset level and average both quantities for all items in the dataset D. Normalizing this difference yields the measure described in Eq. 1, which we call *apunim*:

$$apunim(D, g) = \frac{\mathbb{E}_{c\in D}\left[\mathrm{P}_{apr}(c)\right] - \mathbb{E}_{c\in D}\left[P_g(c)\right]}{1 - \mathbb{E}_{c\in D}\left[\mathrm{P}_{apr}(c)\right]} \tag{2}$$

The *sign* of apunim carries information agreement-based attribution does not:

- $apunim(g) \approx 0$: The group g does not meaningfully disagree with the rest of the annotators.
- $apunim(g) > 0$: The group g has fundamental disagreements with the rest of the annotators.
- $apunim(g) < 0$: The annotators of group g are polarized between themselves.

Apunim thus measures the degree to which ASD can be attributed to an annotator group in a set of items. As demonstrated in Fig. 2, our measure is robust to few annotators per item and imbalanced (minority) group sizes. Furthermore, unlike some prior methods [2, 10], apunim allows us to attribute ASD to a specific group, not just a SDD (e.g., only "men" instead of "gender" as a whole), allowing fine-grained analysis. We also include a perturbation-based p-value estimation which we describe in App. C, which also contains a discussion of theoretical design decisions, limitations and computational details.

[7] We control for our assumption of a non-fixed g_i group polarization effect via the ablation shown in Fig. 11 in App. E.

Table 3. Statistically significant ($p < 0.05$) apunim results across the Sap and DICES datasets. Gray rows show `Ethnicity` and `Gender` groups. "Support" refers to number of annotations. Consult App. G for the full results.

Group	Apunim	Support
DICES-990		
Ethnicity=AfricanAm	0.0995	5148
Ethnicity=Latino	-0.0631	5877
Ethnicity=Other	0.0340	21669
Ethnicity=White	-0.0649	10166
Gender=Man	0.0359	28817
Gender=Woman	-0.0271	29990
Age=GenZ	0.0469	12214
Age=Millennial	-0.0068	34433
Age=GenX+	-0.0282	12699
Education=College	0.0063	52727
Education=High school	-0.0500	6619
DICES-350		
Ethnicity=AfricanAm	-0.0366	7627
Ethnicity=Asian	0.0445	6838
Education=Other	-0.0230	1841
Sap		
Gender=Man	-0.1508	1254
Gender=Woman	0.2613	693
Age=GenX+	-0.1670	223
POPQUORN		
Gender=Man	0.0081	2720
Race=Black	-0.1572	222
Education=College	0.0496	2388
Education=Graduate	-0.0808	468

5.2 How is ASDs Attributed to Humans?

Assuming a confidence level of $\alpha = 0.05$ we test whether any of the provided SDDs can explain the ASD shown in the five datasets (§3). Statistically significant results are presented in Table 3.[8][9] Note that none of the SDDs in the Kumar dataset cam show statistically significant values, but this is expected, since our methodology requires at least six annotators per item to function. See App. C for reproducibility details on choice of parameters and execution times.

We find that *ASD is consistently attributed to* ***ethnicity*** in DICES-990, DICES-350 and POPQUORN (Table 3), confirming prior work suggesting that it is a systematic source of annotator disagreement in HS tasks [25]. Specifically, in DICES-990, all four ethnic groups are significant, with the strongest value for AA annotators (0.0995), followed by White (−0.0649), Latino (−0.0631) and Other (0.0340). In DICES-350, AA (−0.0366) and Asian (0.0445) annotators are significant, and in POPQUORN, AA annotators (−0.1572). Additionally, in DICES-990, Sap and POPQUORN, ASD is attributed to **gender**, which is consistent with previous findings [7, 76]. The effect is largest in Sap (Woman 0.2613, Man −0.1508), smaller in DICES-990 (Man 0.0359, Woman −0.0271), and smallest in POPQUORN (Man 0.0081).[10]

Finding 2. *We confirm prior literature by observing that ASD is consistently attributed to gender (DICES-990, Sap, POQUORN) and ethnicity (DICES-990, DICES-350, POPQUORN) in subjective human-annotated tasks.*

5.3 How is ASDs Attributed to LLMs?

In order to answer RQ. 3 we investigate the following questions: (1) Do LLMs exhibit any SD by default? (2) Does SD increase when using persona-prompting? (3) Can LLMs exhibit any ASD? (4) Does LLM ASD correspond to human ASD? These questions are linked sequentially: answering each one is a necessary step for answering the next.

[8]Full results, including non-significant values are available in Tables 24,25,26,27 and 28 in App. G.
[9]Fig. 13 in App. E shows that our results are *robust to small annotator counts*, corroborating the synthetic experiments in Fig. 2.
[10]Note that AAs show differently signed apunim values in the two DICES datasets (0.0995 in DICES-990 vs. −0.0366 in DICES-350), which may be attributed to these datasets using different labels (see §3) and because different ethnic groups are considered in each dataset. Thus, while detection is constant across the two datasets, attribution is not. Additionally, in DICES-990 and POPQUORN men are a source of ASD (positive values: 0.0359 and 0.0081), while women disagree among themselves in DICES-990 (negative value: −0.0271), but the opposite is true for Sap (Man −0.1508, Woman 0.2613). Unfortunately we can not diagnose these discrepancies, since neither DICES nor Sap include annotation guidelines (Table 1).

Table 4. Mean (SE) per-comment nDFU of the human annotations and of the LLM annotations under the default prompt, without a persona (No Persona) and with a sociodemographic persona (With Persona) with default instruction prompting (no adversarial prompts), per model and dataset.

	Sap		Kumar		POPQUORN	
Model	No Persona	With Persona	No Persona	With Persona	No Persona	With Persona
Human	0.38 (0.01)		0.42 (0.01)		0.31 (0.01)	
OLMo 32B	0.19 (0.01)	0.24 (0.01)	0.34 (0.01)	0.51 (0.01)	0.34 (0.01)	0.46 (0.01)
Qwen 32B	0.05 (0.01)	0.18 (0.01)	0.06 (0.01)	0.19 (0.01)	0.06 (0.00)	0.25 (0.01)
Mistral 7B	0.15 (0.01)	0.21 (0.01)	0.31 (0.01)	0.37 (0.01)	0.25 (0.01)	0.40 (0.01)
Mistral 24B	0.08 (0.01)	0.25 (0.01)	0.08 (0.01)	0.25 (0.01)	0.15 (0.01)	0.33 (0.01)
Gemma 5B	0.16 (0.01)	0.19 (0.01)	0.14 (0.01)	0.22 (0.01)	0.14 (0.01)	0.15 (0.01)
Gemma 31B	0.01 (0.00)	0.04 (0.01)	–	0.09 (0.01)	–	–

Table 5. Cohen's d between the default and each adversarial instruction prompt, computed on the per-comment nDFU. Results are given per model, for each dataset separately and as the unweighted mean over datasets (All). Number of comments: Sap: 585; Kumar: 1000; POPQUORN: 0–1500.

	All				Sap				Kumar				POPQUORN			
Model	Stereotype	Persona	Single	Direct	Stereotype	Persona	Single	Direct	Stereotype	Persona	Single	Direct	Stereotype	Persona	Single	Direct
OLMo 32B	−0.04	0.02	0.07	−0.10	0.09	0.06	0.26	0.06	0.14	−0.13	−0.29	−0.31	−0.35	0.14	0.25	−0.05
Qwen 32B	0.19	0.16	0.18	0.13	0.32	0.13	−0.04	0.09	0.17	0.21	0.51	0.31	0.08	0.13	0.07	−0.01
Mistral 7B	0.13	0.03	0.10	−0.20	0.21	0.10	0.08	−0.21	0.07	−0.01	−0.10	−0.05	0.11	0.01	0.31	−0.35
Mistral 24B	0.59	0.25	−0.32	−0.03	0.64	0.04	−0.88	0.17	0.98	0.97	0.57	−0.25	0.17	−0.26	−0.66	−0.00
Gemma 5B	0.16	0.27	0.38	0.11	0.18	0.48	0.70	−0.08	0.10	−0.08	−0.40	−0.07	0.20	0.41	0.85	0.49
Gemma 31B	0.36	0.21	0.02	0.07	0.36	0.21	0.02	0.07	–	–	–	–	–	–	–	–

Table 6. Share of SDDs that are statistically significant ($p < 0.05$) in the human annotations and in each LLM's annotations under the default prompt, per dataset; n is the number of dimensions. A dimension counts as significant if at least one of its groups is. †: ratio over fewer dimensions, as the model has results for only some of them.

Dataset	n	Human	OLMo 32B	Qwen 32B	Mistral 7B	Mistral 24B	Gemma 5B	Gemma 31B
Sap	3	66%	33%	66%	66%	100%	33%	66%
Kumar	13	0%	46%	53%	46%	46%	38%	30%
POP	5	60%	100%	80%	100%	100%	60%	—

Experimental Procedure. We ensure that our LLM annotation study is reproducible by selecting nine exclusively open-source, locally hosted models and fixing all relevant seeds (three not shown in main results because of prompt sensitivity issues[11]) using twenty annotators for the Kumar, Sap and POPQUORN datasets supplied with personas sampled from our human dataset.[12] We also do not annotate any of the DICES datasets since the LLM annotations are

[11]An ablation study where we rephrase the main annotation prompt three revealed that three models were especially sensitive to such changes (Table 17 in App. F), which led us to exclude those models from the main analysis. See Tables 31,32 for the full results in App. G. Additionally, a second ablation shows that we achieve 100% consistency across all models given the same prompt and input (which is generally not a given [8]).
[12]A compromise between practical computation costs and enough annotation density to facilitate exploration.

Table 7. Statistically significant ($p < 0.05$) SDDs of each LLM compared with the human annotations, per dataset. A dimension counts as significant if at least one of its groups is. Each cell shows one symbol per prompt, in the order default, stereotype, persona, single, direct: ✓ both significant, with the same sign for every group that is significant in both; ✓ neither significant; ± both significant, but the sign of at least one group's effect differs; × significant for only one of the two; blank: no LLM result.

		OLMo 32B					Qwen 32B					Mistral 7B					Mistral 24B					Gemma 5B					Gemma 31B				
Ds	SDD																														
Sap	Age	✓	✓	✓	×	✓	±	±	±	±	±	±	✓	±	✓	×	±	✓	±	✓	±	×	×	±	±	×	✓	✓	✓	✓	✓
	Ethnicity	✓	✓	×	✓	✓	×	×	×	×	×	×	×	×	×	×	×	×	✓	×	×	×	×	×	×	×	×	×	×	×	×
	Gender	×	×	×	±	×	×	✓	×	×	×	×	✓	✓	×	✓	±	✓	✓	✓	×	×	×	±	✓	±	×	×	×	×	×
POP	Age	×	×	×	×	×	×	×	×	×	×	×			✓		×	×	✓	✓	×	×	×	×	×	×			×	×	×
	Education	✓	±	±	✓	✓	✓	✓	✓	✓	✓	✓	×		×		±	✓	×	±	±	✓	±	±	±	±			✓	✓	×	×
	Gender	×	×	×	✓	×	✓	×	×	×	×	×	✓		✓		×	×	✓	✓	×	✓	×	✓	×	×			×	×	×	×
	Occupation	±	✓	±	×	±	±	±	±	±	✓	✓	×				±	✓	×	✓	±	×	±	✓	✓	✓			✓	✓	×	✓
	Race	✓	±	✓	×	±	✓	✓	±	±	±	✓	×				✓	±	×	×	±	✓	✓	✓	±	✓			±	±	±	±

inconsistent (< 0.3 Cohen's κ for all models but one when paraphrasing instructions; see Table 17 in App. F), an issue we can not fix as their original annotation guidelines are not public (Table 1). See App. F for the ablation experimental procedure, full results, model details, dataset selection, exploratory analysis, and comparison to human annotations.

We also test four adversarial prompts that stress the importance of the LLM's persona in different ways (App. H). *Persona-focused* asks the model to rate as its persona would perceive the comment. *Direct* makes the persona the only criterion. *Single* has the model pick the persona's most influential trait and rate based on that trait alone. *Stereotype* tells the model to weigh the persona's sensitivities over those of an average person. These prompts do not represent annotation strategies; they only attempt to steer the models towards persona-focused answers.

Do LLMs Exhibit SD? **Yes**. Table 4 shows that the nDFU values exhibited by LLMs when under standard persona-prompting can even surpass those of humans. However, manipulating the amount of ASD through instruction prompting is not reliable. While an ANOVA[13] test shows that the choice of instruction prompt does indeed affect nDFU scores across most models, measuring Cohen's d (Table 5) shows that the effect is generally small ($|d| \leq 0.3$) and inconsistent, with the sign and magnitude of the effect depending on the model and dataset.[14]

Can LLMs exhibit any ASD? **Yes**. Table 6 shows the percentage of SDDs where at least one of the constituent groups show statistically significant apunim results. We observe that LLM annotations frequently produce ASD.[15]

Is LLM ASD Aligned With Humans? **No**. Table 7 describes whether an SDD produces statistically significant results using apunim and compares it with the human apunim results (Table 3). We observe that alignment on ASD between humans and LLMs is rarely achieved. Specifically, we observe: (1) Low Precision: LLMs tend to produce polarized annotations for groups whose human annotators are not. (2) Low Recall: LLMs under-detect where humans find real effects, especially in Sap and POP Gender. (3) Where ASD matches, the direction does not (POP; Race, Education). (4) Where replication is achieved, it is generally on null results (e.g., *Mistral-7B*). We also find no connection between model size and family on the alignment of ASD.

[13]We use a parametric test (ANOVA) in our analysis due to the abundance of observations ($N \gg 100$) as well as a multiple hypothesis p-value correction (Holm-Bonferroni [30]).
[14]See Table 30 in App. G for the actual mean nDFU values by prompt.
[15]Table 33 includes all LLMs and Table 34 distinguishes between different adversarial prompts in App. G.

Finding 3. *While LLMs using persona-prompting can produce polarized annotations, they can not replicate human polarization patterns, even when we attempt to align them using prompting.*

6 The Ideal Subjective Dataset

6.1 Categorizing Attributes of Annotation Datasets

As discussed in §3, there is a severe lack of datasets containing annotator metadata for subjective tasks. Given that the creation of these datasets is difficult, expensive and time-consuming, it is important to acknowledge and prioritize certain properties that are necessary for analyzing subjectivity across annotator groups. Since there is to our knowledge no taxonomy on the shape of annotated datasets, we distinguish three main properties:

- Data Coverage (DC): the number of *items* in a dataset. Due to the focus on training and benchmarking datasets in NLP, this property is usually the one with the most focus and importance [58].
- Group Coverage (GC): the number of *annotators* and their distribution across SDDs. A large annotator pool is required to get sufficient coverage on the examined groups and allows both higher fidelity of each SDD and more such SDDs to be examined. To our knowledge, there are no studies explicitly discussing this property.[16]
- Annotation Density (AD): the number of *annotators per item*. This property is often neglected in practice, with many datasets [13, 37, 48, 66, 67] assuming "gold labels" can be inferred by three annotators. Similarly to GC, we can not find any discussion of this property, with some studies recommending more than three annotators but still defaulting to majority-voting [6].

While the field of subjective NLP tasks has thankfully moved away from single-annotated items and majority voting, the lack of discussion on GC and AD suggests that the shift has been incremental rather than substantive. A clear signal can not be always obtained when only a few cursory SDDs are reported instead of a comprehensive list of both innate characteristics and personal opinions. [17] Likewise, six annotators per item may not be adequate, even when a comprehensive list of SDDs is known, as we demonstrate in the next section.

6.2 The Case for Annotation Density

USD. Some theoretical models such as AU make the assumption that all ASD is attributable to some annotator SDD, which has not been empirically verified in real-world annotation settings. We define this phenomenon "USD" for a single item c as the minimum nDFU exhibited by any arbitrary dimension:[18] $\text{Pol}_{inh}(c) = \min_{S \in G_c} \{\text{nDFU}(S)\}$, where G_c is the collection of all subsets S of annotations for an item c such that $|S| \geq 3$ and S_j is a random sample drawn from G_c for $j = 1, \ldots, k$. Since nDFU operates only when a group has at least three annotators (Eq. 1), this yields $\sum_{k=3}^{n} \binom{n}{k}$ possible groups, which is computationally intractable for large annotator counts. We can therefore use Monte Carlo sampling by drawing k random partitions with random sizes: $\text{Pol}_{inh}^{MC}(c) = \min_{j=1}^{k} \{\text{nDFU}(S_j)\}$, which serves as an upper bound (as it considers a strict subset of $\text{Pol}_{inh}(c)$).

Does It Exist? We compute the USD for each item across the five datasets. For Sap and Kumar, we use the exhaustive formulation, whereas for the two DICES datasets we rely on the MC approximation with $k = 1000$. Fig 4 shows that

[16] A further distinction can be made between the number of SDDs and the number of groups within each SDD ("group depth" vs. "group width").
[17] The Kumar dataset is an excellent example of a comprehensive subjective dataset (§3.
[18] This dimension need not be known in the dataset, as we consider all possible dimensions in our sample, both known and latent. This formulation also covers all possible combinations of such groups.

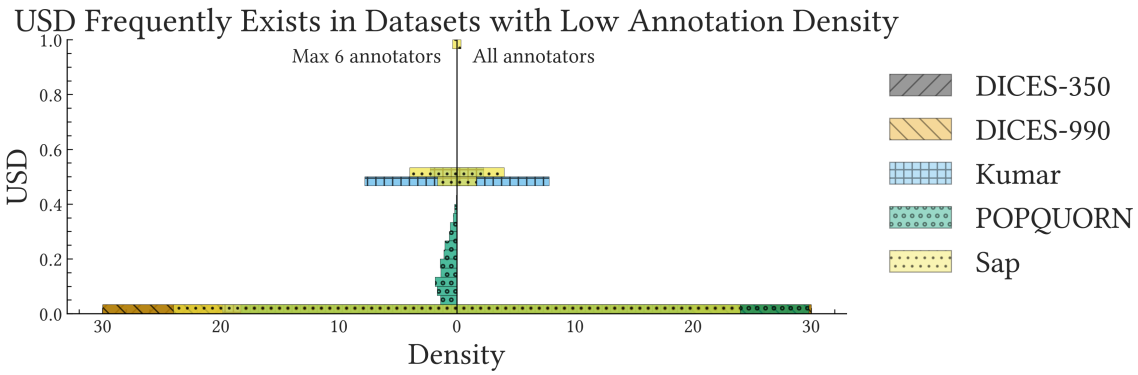


Fig. 4. USD across our datasets (§3). Contrary to assumptions made in prior work, USD is commonly non-zero and can even reach $nDFU = 1$ in datasets with low annotator counts (see Table 1). The left side of the figure shows the same datasets, where annotations are subsampled 100 times down to six annotators and their USD is averaged.

across our five datasets, three of them show a non-insignificant amount of items with USD.[19] USD typically seems to exist only on smaller annotation counts; the Sap and Kumar datasets only hold six and five annotators respectively per item, while POPQUORN shows USD only when it is subsampled down to six annotators per item. This pattern is also validated on both synthetic and LLM annotated data, as shown by Fig. 8 in App. D.

Why Does it Exist? Given the observation that USD only exists on low annotation counts per item, we hold the following two hypotheses: (1) USD is caused directly by the low number of annotators per item, providing insufficient overall information on SD. (2) USD arises from the presence of partially represented groups within the annotator pool which may go undetected if they are represented by only one or two annotators, as nDFU is undefined for these groups. Unfortunately, there is no way for us to verify whether any of the two hypotheses are correct.[20]

Finding 4. *Items may exhibit a degree of USD that cannot be attributed to any known or latent annotator grouping.*

6.3 How Should We Design Subjective Datasets?

While DC may be convenient for training classifiers, relying on it without rigorous evaluation of SD can lead to misconfigured or biased models [60], evaluation which is impossible without the necessary DC). However, it comes at the expense of the other two (arguably more important) qualities. Generally, the *easiest-to-use resources for training LLMs and classifiers may be the riskiest*, as meaningful, SD cannot be reliably quantified (see the Kumar dataset in this study). We therefore recommend prioritizing dataset creation by first ensuring sufficient AD, then achieving necessary GC across underrepresented groups, and finally allowing budgetary constraints to determine DC.

While this tension between the need for massive data volumes in AI training and the requirement of responsible social science to use deeply annotated, subjective datasets is unlikely to be resolved, a practical step is providing full annotation guidelines in both cases. If the annotation target and definitions are unknown, comparing results across different datasets becomes extremely difficult. As shown in §5.3, without clear guidelines, differences in polarization patterns remain obscured when analyzing data from multiple studies.

[19] It is unclear why the DICES datasets do not produce any USD even when repeatedly subsampled. We hypothesize that the annotations themselves may not be as polarized as to allow USD as both DICES datasets do feature less polarized comments (see Fig. 5 in App. B). It is plausible that their text targets (LLM responses) may not be as ambiguous as human comments.

[20] Examples of items with high USD can be found in Table 12 in App. D.

Conclusion

In this study, we showed that current disagreement attribution measures tend to underperform in realistic annotation settings through a large scale empirical study. We designed a new measure that quantifies attribution of SD to annotator SDDs, taking into account the limitations of real-life annotation datasets. Using this measure, we identified shared polarization patterns among gender and ethnicity groups across five subjective NLP datasets. Additionally, we found that LLM annotators can create polarized annotations, but do not reproduce human polarization patterns, even when using prompt engineering. Our research is limited due to the information present in the datasets; namely, we discovered that due to low annotation counts per item, some polarization cannot be attributed to any combination of theoretical dimensions ("USD"). Overall, our work warns against using LLM persona-prompting without rigorous, task-specific evaluation, and motivates the development of "wider" datasets with fewer but better annotated items.

AI Use Statement

In this work we used proprietary models such as Anthropic's Claude Sonnet 5 and OpenAI's GPT-5 for partially implementing the synthetic experiments in §4 and generating parts of the library, experiment and graph-creation code. The unit-tests used in our library were similarly generated by these models, although new tests were added as edge-cases and bugs were discovered. All such tasks were executed with direct author supervision, and all relevant code files (library, experiments, graphs) have been edited manually at multiple points by the authors to ensure reproducibility, accuracy and software quality. We also used Scholar Labs to find citations for targeted claims (e.g., for known issues with persona-prompting), although all such citations were manually filtered by us. Finally, we used a locally hosted Gemma-4 model (specifically the `gemma-4-E4B-it-Q4_K_M.gguf` variant) for rephrasing this manuscript in order to improve readability. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

Ethics Statement

Dual Use Concerns. The primary objective of this study is to identify systemic sources of ASD in subjective NLP datasets to facilitate the construction of more fair and reliable AI systems. However, given that our methodology allows for the quantitative attribution of specific annotation patterns to SDDs, we acknowledge three primary risks inherent in this research. (1) *Discriminatory Profiling*: If the measure is applied outside of a dataset quality analysis, its ability to correlate specific opinion patterns with annotator groups could be misused for surveillance or unjust profiling. (2) *Automated Bias Amplification*: The diagnostic patterns of disagreement we detect are reflections of human systemic biases. Without human oversight, training subsequent models on these patterns risks automating and entrenching existing societal prejudices. (3) The knowledge provided by *apunim* could be *weaponized*, allowing malicious actors to strategically target weak points in moderation systems or intentionally maximize societal conflict. We stress that our measure functions strictly as a diagnostic warning signal. Any deployment in a real-world system must mandate a human-in-the-loop review, ensuring that identified patterns lead to systemic improvements (e.g., policy reform, demographic diversity initiatives) rather than merely justifying an automated system. In any case, despite the risks, we believe that the benefits of this research outweigh the risks outlined above.

Data Sourcing and Privacy. All datasets employed in this study were either publicly accessible or provided to us by the original authors with full disclosure regarding the purpose of our research. No identifiable, personal information was included in our pipeline. We do not release proprietary data provided by the authors, nor have we provided any

to proprietary LLMs. Publicly accessible datasets will be provided as is, while gated-release datasets will be provided with their text contents hashed. Thus, our annotations can still be matched against the actual dataset, if it is acquired separately by the original authors. Finally, we note that our data is limited to English posts from a U.S. cultural context and our results may not be representative for other cultures.

Examples Used. Our study utilizes examples of comments that may be considered divisive across certain demographics. These examples were generated by our team solely for demonstration purposes. We explicitly state that the contents of these examples and the groups studied do not necessarily reflect the views or behaviors of the groups they represent, nor do they reflect the opinions of the authors. We have partially censored explicit slurs in Table 12 in order to both mitigate uncomfortable or triggering content, while also showing enough content for the examples to be effective.

Prompts Used. This study uses adversarial prompts which explicitly instruct the model to change its outputs according to their persona prompts. The purpose of these prompts is to test the LLMs under an extreme case of prompt engineering and are not to be used for generalized usage. The use of these prompts may degrade model performance, increase the risk of stereotyping and produce misleading results.

A Acronyms

Acronyms

AA African American
AD Annotation Density
ASD Attributable SD
AU A Posteriori Unimodality
DC Data Coverage
GC Group Coverage
GD General Disagreement
GMM Gaussian Mixture Model
GPLv3 GNU General Public Licence v3
HS Hate Speech
IAA Inter-Annotator Agreement
nDFU normalized Distance From Unimodality
NLP Natural Language Processing
SD Systematic Disagreement
SDD SocioDemographic Dimension
USD Unnattributtable SD

B Exploratory Analysis

The number of annotators per item for all datasets presented in §3 can be found in Fig. 6, and descriptive statistics can be found in Table 8. The DICES-990 dataset has exactly 123 annotations per item, while the other datasets exhibit a few variations around the central mean reported in Table 1. Interestingly, the sample used from the Kumar dataset exhibits a few items that vastly exceed the promised 5 number of annotators. We do not include these items, since they are very likely a data error. Tables 9,10,11 show the annotation counts for each SDD group in each dataset.

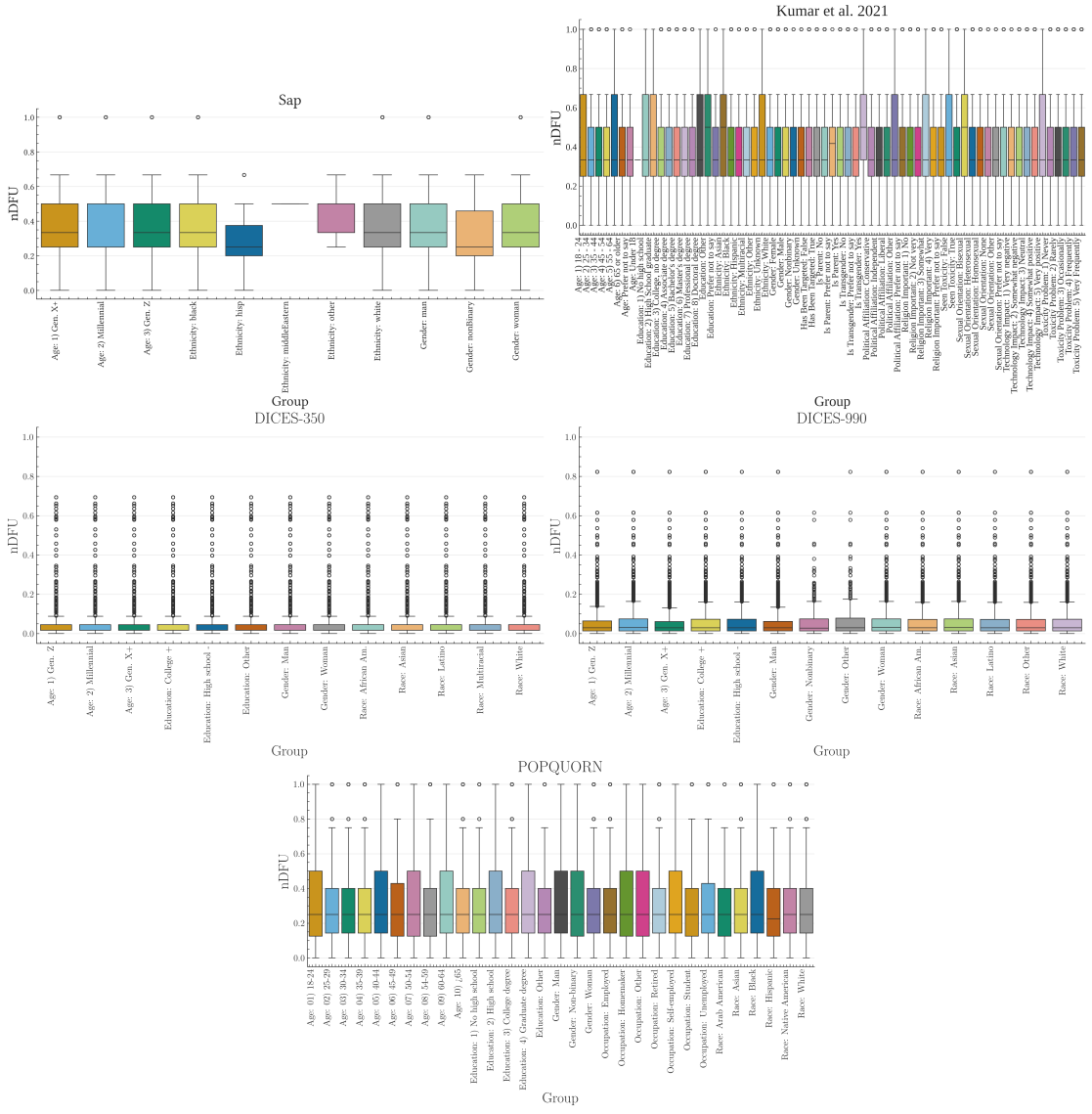


Fig. 5. Boxplots of dataset polarization for the human datasets considered in this paper for all groups in each examined SDD.

C Apunim Details

C.1 Design Decisions

Defining Apriori Polarization. In §5, we explained how our methodology needs to compare the polarization between the set of all annotations and those made by a specific group g. Direct comparisons are not advisable, since we would be comparing statistics between a set and its subset—in this case, standard practice dictates comparing the subset with its complement; i.e., the in-group vs. the out-group instead of the in-group vs. all the annotations. However, this also

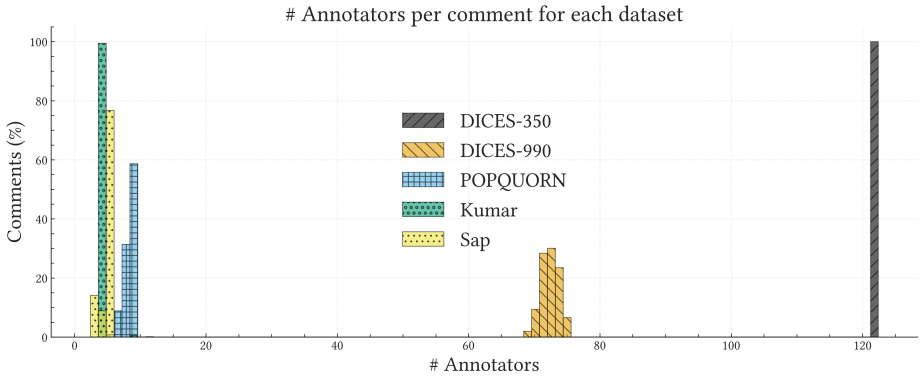


Fig. 6. We calculate the ratio of items for each dataset (y-axis) that had x number of annotators (x-axis). We truncate the x-axis to 130 annotators, disregarding very few items in the Kumar dataset exceeding this threshold (see Table 8), which is likely caused by a data error.

Table 8. Descriptive statistics for the number of annotations per comment, grouped by dataset.

	count	mean	std	min	25%	50%	75%	max
DICES-350	350	123	0	123	123	123	123	123
DICES-990	990	72.8313	1.1652	69	72	73	74	76
POPQUORN	1500	8.6573	0.9087	5	8	9	9	10
Kumar	105910	5.0261	0.3604	5	5	5	5	10
Sap	585	5.6359	0.7713	3	6	6	6	12

Table 9. Subgroup Counts for DICES-990 (left) and DICES-350 (right). "Support" refers to annotation count.

Category	**Subgroup**	**Support**
Gender	Woman	36050
	Man	35392
	Nonbinary	331
	Other	330
Race	Other	26394
	Asian	19885
	White	12322
	Latino	7170
	African Am.	6332
Age	Gen. Z	14899
	Millennial	41542
	Gen. X+	15662
Education	College +	64054
	High school -	8049

Category	**Subgroup**	**Support**
Gender	Woman	21700
	Man	21350
Race	White	10500
	African Am.	10150
	Asian	9100
	Latino	7700
	Multiracial	5600
Age	1) Gen. Z	19600
	2) Millennial	12600
	3) Gen. X+	10850
Education	College +	26250
	High school -	14350
	Other	2450

would not work; if we apply the subset-vs-complement method in the example of Fig. 7, we would end up subtracting the nDFUs of men and women, which are similar as they are both unimodal distributions, falsely concluding that there is no polarization present.

Table 10. Subgroup Counts for the Sap (left) and POPQUORN (right) datasets. "Support" refers to annotation count.

Category	Subgroup	Support
Age	Gen. Z	252
	Millennial	2259
	Gen. X+	786
Race	White	2168
	Black	1091
	Hisp	32
	Other	5
	MiddleEastern	1
Gender	man	1859
	woman	1412
	nonBinary	26

Category	Subgroup	Count
Gender	Woman	6470
	Man	6166
	Non-binary	350
Race	White	9754
	Black	1685
	Asian	949
	Native American	299
	Hispanic	250
	Arab American	49
Age	18-24	1545
	25-29	1236
	30-34	1494
	35-39	1547
	40-44	1142
	45-49	1091
	50-54	947
	54-59	1793
	60-64	747
	>65	1444
Education	No high school	237
	High school	3934
	College degree	6036
	Graduate degree	2481
	Other	298
Occupation	Student	995
	Homemaker	594
	Retired	1446
	Employed	6325
	Self-employed	1692
	Unemployed	1684
	Other	250

Filtering Ineligible items. Because *apunim* is fundamentally designed to quantify polarization, we exclude items that exhibit no measurable polarization ($nDFU(c) = 0$). Furthermore, we also exclude items annotated exclusively by a single group (i.e., items where there are no two groups with at least three annotators each), since we require valid histograms for the non-parametric estimation of nDFU (see Eq. 1). This is an unfortunately common scenario, since many NLP datasets feature only $5-6$ annotations per item. In a best-case scenario of 5 annotators split evenly between two groups, we would get a 3-2 split.

C.2 Statistical Estimation

Since annotation histograms may be noisy, we additionally assess whether the observed apunim value could arise by chance. To assess statistical significance, we construct a null distribution of apunim values from the randomized

Table 11. Subgroup Counts for the Kumar dataset. "Support" refers to annotation count.

Category	Subgroup	Support
Gender	Female	8011
	Male	6832
	Unknown	124
	Nonbinary	93
Ethnicity	White	10699
	Black	1940
	Asian	862
	Multiracial	664
	Hispanic	418
	Other	247
	Unknown	230
Age	Under 18	3
	18 - 24	1711
	25 - 34	5969
	35 - 44	3785
	45 - 54	1977
	55 - 64	1101
	65 or older	471
	N/A	43
Education	No high school	71
	High School	1353
	College	3090
	Associate's	1676
	Bachelor's	6071
	Master's	2261
	Professional	236
	Doctoral	198
	N/A	71
	Other	32
Sexual Orient.	Heterosexual	12391
	Bisexual	1665
	Homosexual	515
	N/A	302
	Other	131
Is Trans	No	14549
	Yes	363
	N/A	148
Politics	Liberal	5997
	Independent	4143
	Conservative	3981
	N/A	609
	Other	330
Is Parent	Yes	7889
	No	7015
	N/A	156
Tech Impact	Very negative	149
	Somewhat negative	1411
	Neutral	1905
	Somewhat positive	7690
	Very positive	3905
Tox Problem	Never	869
	Rarely	2485
	Occasionally	5307
	Frequently	4441
	Very Frequently	1958
Rel Important	No	4735
	Not very	1808
	Somewhat	3548
	Very	4741
	N/A	228
Seen Toxicity	True	11505
	False	3555
Targeted	False	10741
	True	4319

partitions and compare the observed apunim against this distribution using a one-sample Student-t test. Thus, the null hypothesis states that the observed polarization is indistinguishable from polarization induced by random annotator assignments:

$$\begin{aligned} H_0 &: \; \bar{apunim}_{\text{rand}}(d,g) = apunim(d,g) \\ H_a &: \; \bar{apunim}_{\text{rand}}(d,g) \neq apunim(d,g) \end{aligned} \tag{3}$$

where $\bar{apunim}_{\text{rand}}(d,g)$ denotes the average apunim values obtained from t randomized annotator partitions.

We account for the presence of multiple subgroups in each SDD using a multiple-hypothesis correction. Since our test is parametric (App. **??**), it assumes that there exist (1) enough data-points to reliably run the means-test, (2) enough annotations per SDD group to estimate nDFUs, (3) the apunim values of the random partitions are independent and

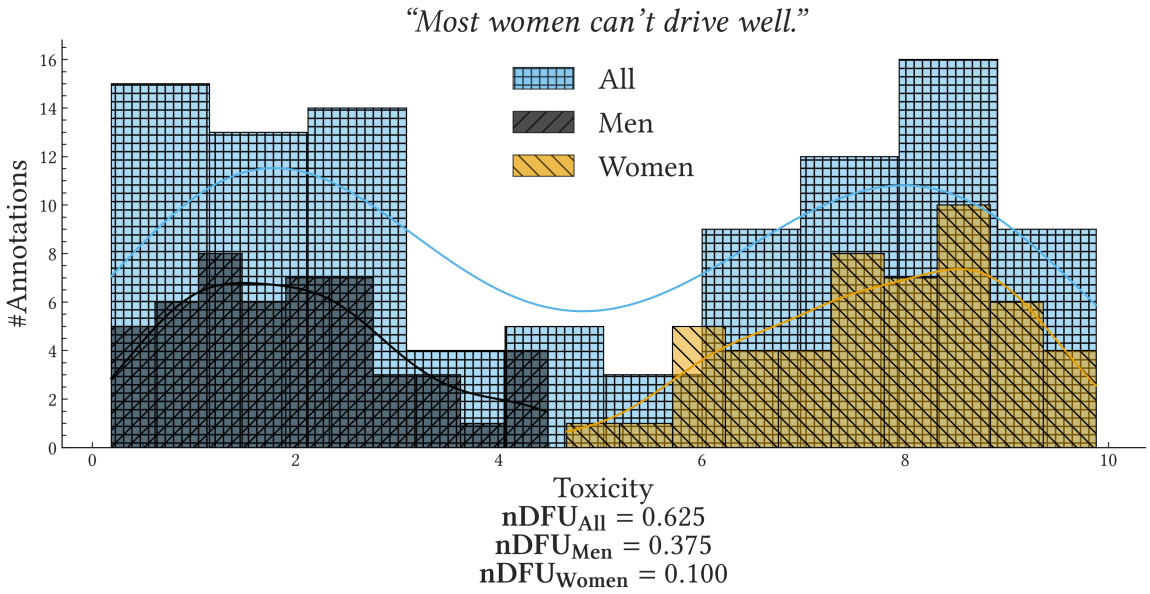


Fig. 7. Simulated experiment describing a polarizing comment where men and women agree between themselves, but disagree with the opposite gender.

identically distributed. Condition (1) can be safely assumed for most datasets, because they feature enough items to reliably assume normal residuals (see §3). We discuss condition (2) in §6. Condition (3) is partly fulfilled; the partitions are identically distributed, but may be dependent on the sample distribution.

Replacing the parametric means test with a non-parametric alternative would relax assumptions about the underlying distribution but would not resolve the potential dependence between the random apunims. We initially attempted to implement a permutation test on the empirical (random apunim) distribution, but this resulted in excessive computational costs.

While the test operates on a single group in principle, in practice we often want to test polarization for all subgroups of a specific SDD. Since we are simultaneously considering multiple hypotheses, we apply a multiple comparison correction to the resulting p-values—specifically the Holm method [30]. The test is parameterized by the confidence level of our test, which is used to tune the strength of the correction; we can increase this value to make our test more conservative towards multiple hypotheses [11]).

C.3 Software

Installation and Usage. We release the `apunim` package, which is a Python library that implements both the nDFU and apunim calculations, as well as its statistical test. The software is available on PyPi We also provide high-level documentation in HTML format that includes an introduction, guide, and full low-level documentation for each of the two provided functions (https://anonymous.4open.science/r/apunim-page).

We use version `1.0.6` of the `apunim` library for our experiments. We use 100 random iterations for the apriori value calculation and the p-value estimation (App. **??**), and set a confidence level of $\alpha = 0.05$. The relative library setup is

provided as a conda environment file to ensure reproducibility. The code is open-source, licensed under the GNU General Public Licence v3 (GPLv3) and can be found on the project's repository (https://anonymous.4open.science/r/aposteriori-unimodality-E8C1). The library is cross-platform, works for any Python 3 version, and requires very few dependencies (namely `numpy, scipy, statsmodels`).

Implementation Details. By default, the computation of the apunim values happens on a single thread. In cases of serious computational cost, we recommend running the apunim calculation on multiple concurrent processes, since all operations are CPU-bound and thread-safe. This may be useful in cases where there are multiple SDDs, each of which is computationally expensive. In order to lower the computational cost required, we re-use the random partitions, which are originally calculated to obtain the apriori polarization values, for the p-value calculations (App. **??**). This means that in our current implementation, the number of groups for estimating apriori polarization, and the number of apunim values used in the p-value calculation have to be the same.

While the library efficiently handles large numbers of annotators through vectorized operations, performance degrades significantly when processing a high volume of items, particularly when annotators belong to multiple distinct groups. For instance, while experiments across all datasets in §3 complete within a minute, the Kumar dataset necessitates approximately one hour of computation using commercial hardware. Furthermore, the annotator variance analysis detailed in Fig. 13 required over ten hours. While this latter time requirement is anticipated due to the inherently demanding nature and non-standard application of our library, it demonstrates the existence of a potential scaling bottleneck. The same applies to a lesser extent for the subsampling experiments on the DICES datasets, taking collectively three hours of computation.

We hypothesize that this bottleneck is likely caused by repeated creation of numpy arrays, resulting in repeated memory allocations for each subgroup, in each item, for each SDD. We plan on profiling and addressing it in the next releases of our software. Until then, we suggest either using multiprocessing for each SDD subgroup, or running the different experiments in parallel (indeed, we use the `gnu-parallel` package [62] for our own experiments). We also note that the Kumar dataset also requires substantial RAM resources, estimated around 42–46GB of dedicated RAM should the whole dataset be considered (this is not a requirement for loading the sample used in this study). The CPU-bound experiments require approximately 3 hours to run, assuming a computer capable of efficient multiprocessing.

Finally, we note that in Fig. 13, we removed values where there was a lack of items annotated by the specified number of annotators (e.g., only 7% of items in DICES-990 were annotated by more than 73 annotators—see App. B).

D Exploring USD

How Does it Look Like? Table 12 shows a random sample of items taken from the Sap dataset and their respective USD values (§4). Note that explicitly toxic or racist items tend to have USD values of zero, alongside clearly unproblematic items. Inversely, more ambiguous and less comprehensible items tend to have larger USD values. This pattern may suggest that USD may be tied to the overall ambiguity of the statements, especially in cases where items may either be interpreted as very toxic, or non-toxic, which agrees with the overall goal of nDFU (§2). We include the USD values for each individual item for all four datasets in the supplementary material.

Can We Produce it Synthetically? Fig. 8 shows the USD of LLM annotations and of the synthetic simulations of §4, using the same layout as for the human datasets: all annotators (right) against at most six annotators per comment (left). Looking at the synthetic annotations, we note that USD is generally produced irrespective of number of groups per

Table 12. Sample of comments from the Sap dataset and their USD. Note that unambiguously toxic comments can be assigned with zero USD, while more nuanced cases tend to have higher values.

Content Warning: Slurs, offensive, toxic and hateful speech.	
Comment	USD
The minks mentioned in the headline are indeed "rising" from where they were based, but that is due to them being buried in shallow mass graves (Africa) and filling with gas that causes them to rise out. Rest assured, they're still very much dead, thankfully!.	1
There should be such thing as a b[redacted]ch resistance for us women	1
There's a big difference between being a b[redacted]ch and being a woman	1
i just watched a video of @joisnotonfire being an annoying h[redacted]e on skype i'm laughing	1
n[redacted]as love to commit crime	1
I swear to God above I love ya and when ya be with this brotha, I swear I love you my bro as long as you not a n[redacted]a	0.5
I will become a cold ass b[redacted]ch because of my treatment	0.5
I would happily lay out a n[redacted]a	0.5
Shoot yourself in the head you hypersensitive fuckin b[redacted]ch.	0.5
not all women are b[redacted]ches, but all b[redacted]ches are women	0.5
"you ugly b[redacted]ch!!!"- that seems to be my real name these days"	0
"I am happy to report that drowning does kill this demonic entity. Oh holy are the waters of Chernarus!"" Omg, DED ;)"	0
"Let them praise the name of the LORD: for his name alone is excellent; his glory is above the earth and heaven."" Psalms 148:13 #Bible"	0
someone plz get this n[redacted]er some help. He's playing up like mad	0
stop talking you fucking n[redacted]er !	0

hypothetical SDD. With respect to LLM annotation, we observe that prompt-unstable models are the largest contributors of USD, indicating that the latter may be linked with general disagreement (variance) rather than SD.

In all cases, including human annotations (Fig. 4), low annotator counts per item seem to be the cause of SD. Indeed, we observe USD from most models when repeatedly subsampling six annotators (Fig. 9). In the case of the Sap and Kumar datasets, human annotators show significantly higher USD values, which is not the case for POPQUORN. Instruction prompts can influence the extent of USD, but inconsistently. Providing adversarial instruction prompts seems to increase USD, although the effect of each prompt seems to be dataset-dependent.

E Ablations

E.1 Synthetic Experiments

Multigroup Synthetic Experiments. We repeat the comparison of Fig. 2 with k annotator groups per item, only one of which is polarizing. As Fig. 10 shows, apunim remains competitive with multiple groups. Most methods detect polarization more often when it is unidirectional, so the difficulty that bidirectional polarization poses for prior methods is likely not limited to two groups.

Do Fixed Group Effects Affect our Experiments? In our synthetic experiments, annotators within a group do not share the same opinion: each annotator's group effect is sampled, with $g_i \sim U(-1/2, 0)$ for one group and $g_i \sim U(0, 1/2)$ for the other. In the multigroup experiments, each annotator in the polarizing group has a strength $h_i \sim U(0, 1)$. To

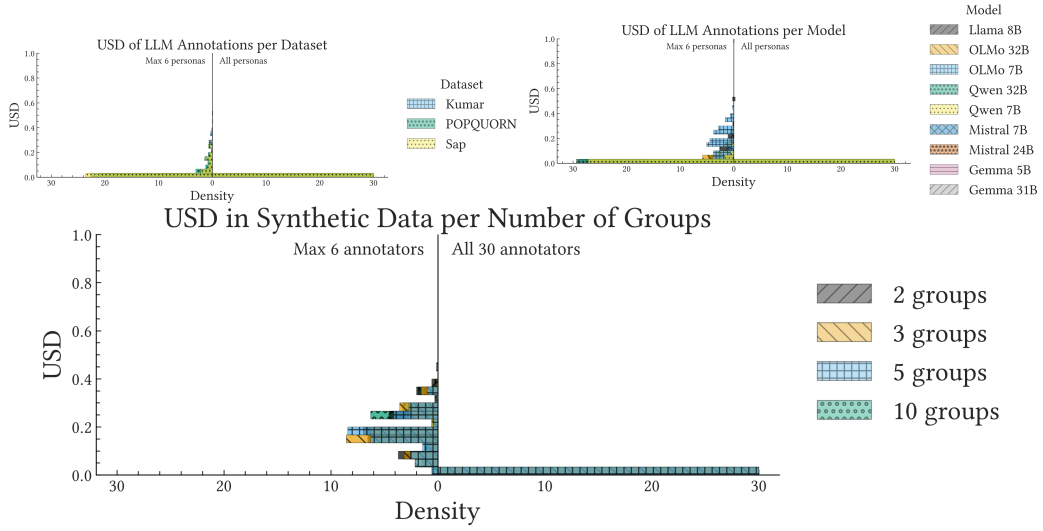


Fig. 8. USD for LLM annotations grouped by dataset (top-left) and model (top-right) as well as for the synthetic simulations used in §4 (bottom). LLM annotations include models that were excluded due to prompt sensitivity.

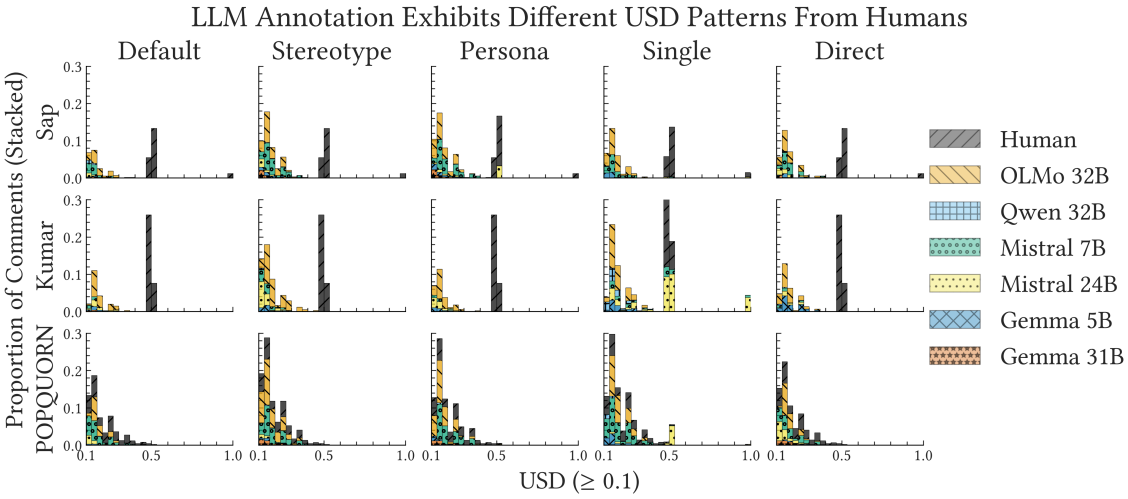


Fig. 9. Normalized histograms showing the USD exhibited by each model when truncating annotation to six annotations, compared to human annotations for each dataset. Some LLMs fail to show any USD, while the rest show far less than humans. Note that the graph does not include zero USD, and the y-axis is truncated to $[0, 0.3]$.

test whether our conclusions depend on this choice, we rerun both experiments with a fixed group effect, where every annotator in a group shifts by exactly the same amount ($g_i = \pm 1/2$ and $h_i = 1$). All other settings, including the conditions, δ values, and number of repetitions, are unchanged. Note that the fixed setting also doubles the expected difference between groups at a given δ, so it is both more homogeneous and a stronger signal. Fig. 11 shows the results.

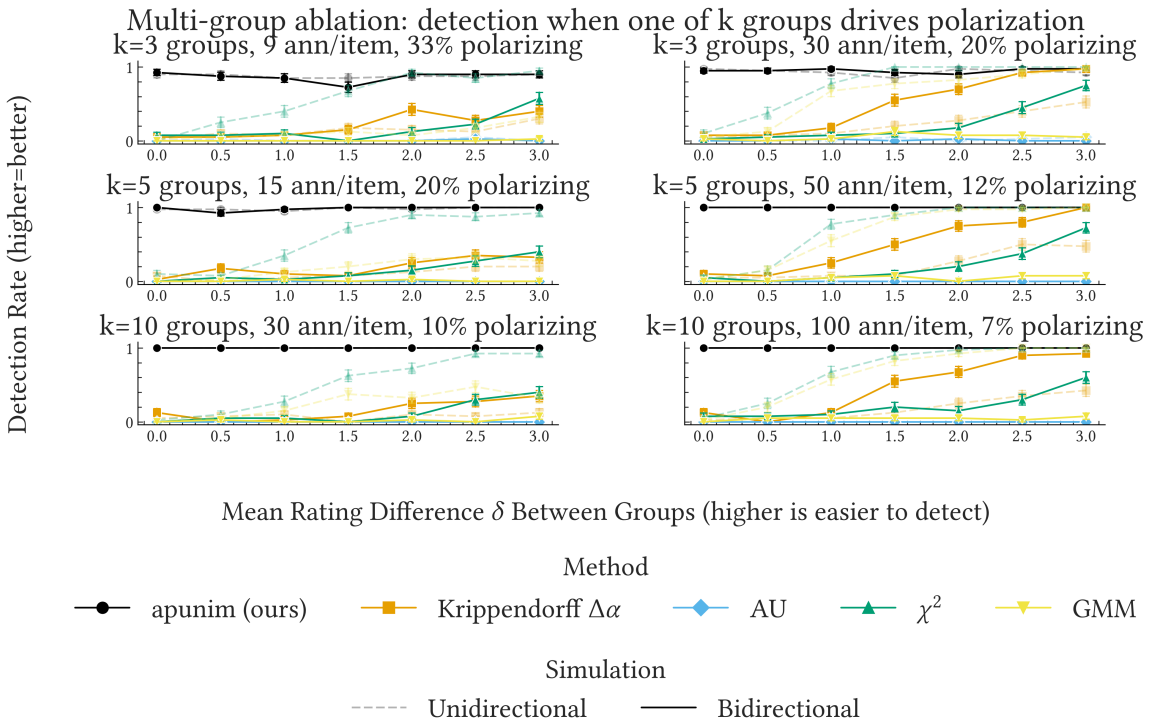


Fig. 10. Detection rate of each method on synthetic data with k annotator groups per item, only one of which is polarizing. The setup is otherwise identical to Fig. 2. Panel titles give k, the number of annotators per item, and the share of annotators in the polarizing group. Solid lines show bidirectional polarization; dashed, faded lines show unidirectional polarization, where the effect always has the same direction. Bars show the standard error.

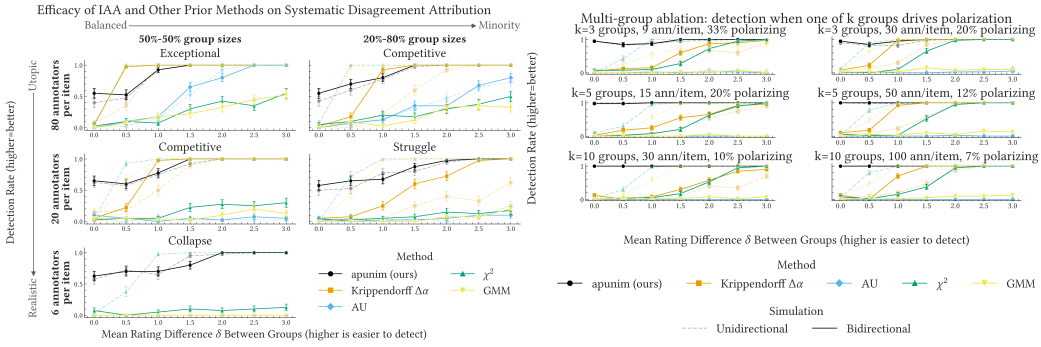


Fig. 11. The experiments of Fig. 2 (left) and Fig. 10 (right), rerun with a fixed group effect: every annotator in a group shifts by exactly the same amount ($g_i = \pm 1/2$, $h_i = 1$) instead of a sampled amount. All other settings are unchanged. Solid lines show bidirectional polarization; dashed, faded lines show unidirectional polarization. Bars show the standard error.

E.2 Dataset Sampling

Table 13. Comparison of Apunim values across different sample sizes (1k, 3k, 10k, 30k) for the Kumar dataset.

		1000	3000	10000	30000
	SDD				
Age	1) 18 - 24	0.014	-0.017	-0.037	-0.016
	2) 25 - 34	-0.003	-0.000	0.000	0.000
	3) 35 - 44	0.000	-0.011	-0.001	-0.001
	4) 45 - 54	0.022	0.033	0.013	0.003
	5) 55 - 64	0.000	0.000	-0.057	0.001
	6) 65 or older	—	—	—	0.948
Education	2) High School graduate	0.000	0.000	-0.005	0.014
	3) College, no degree	0.010	0.014	0.010	0.006
	4) Associate degree	0.000	0.000	0.005	0.009
	5) Bachelor's degree	0.001	-0.000	-0.001	-0.000
	6) Master's degree	0.000	-0.006	0.004	-0.000
Ethnicity	Asian	-0.000	0.000	0.011	0.007
	Black	0.000	-0.024	-0.004	-0.012
	Hispanic	—	—	0.000	-0.099
	Multiracial	0.000	0.000	0.000	0.000
	Other	—	—	—	-0.306
	White	0.000	-0.000	-0.000	0.000
Gender	Female	0.001	0.000	0.002	0.001
	Male	0.001	-0.004	-0.002	-0.001
Has Been Targeted	False	0.001	0.000	-0.000	0.000
	True	-0.009	-0.015	-0.003	-0.002
Is Parent	No	-0.002	0.002	0.001	0.001
	Prefer not to say	—	—	1.000	0.000
	Yes	-0.000	0.000	-0.000	0.000
Is Transgender	No	0.000	0.000	0.000	0.000
Political Affiliation	Conservative	-0.013	0.003	-0.003	-0.002
	Independent	0.000	-0.001	0.001	0.002
	Liberal	-0.001	0.006	0.001	0.001
	Other	—	0.000	0.000	0.000
	Prefer not to say	—	—	0.000	0.022
Religion Important	1) No	0.002	0.004	0.002	0.001
	2) Not very	0.000	-0.006	0.010	0.006
	3) Somewhat	0.000	0.002	0.003	-0.002
	4) Very	0.000	-0.003	-0.003	-0.002

Continued on next page

Table 13. Comparison of Apunim values across different sample sizes (1k, 3k, 10k, 30k) for the Kumar dataset.

		1000	3000	10000	30000
	SDD				
Seen Toxicity	False	-0.007	0.002	-0.001	-0.001
	True	-0.001	0.000	0.001	0.000
Sexual Orientation	Bisexual	-0.068	-0.029	-0.040	-0.013
	Heterosexual	-0.001	-0.000	-0.000	0.000
	Homosexual	—	—	0.000	0.053
	Prefer not to say	—	—	—	0.370
Technology Impact	2) Somewhat negative	0.000	0.000	0.000	0.003
	3) Neutral	0.124	0.037	-0.014	0.000
	4) Somewhat positive	0.004	0.001	-0.000	0.001
	5) Very positive	-0.015	-0.004	-0.004	-0.003
Toxicity Problem	1) Never	0.000	0.000	-0.037	-0.081
	2) Rarely	0.000	0.000	0.003	-0.001
	3) Occasionally	0.000	-0.004	-0.003	-0.001
	4) Frequently	0.000	0.000	-0.001	-0.000
	5) Very Frequently	0.000	-0.002	0.030	0.015

How Does Dataset Size Influence Apunim? Since our approach was designed for the small-scale datasets common in NLP, we were cautious when testing it on the massive Kumar dataset. Because of the data volume, we analyzed various smaller samples (1,000, 3,000, 10,000, and 30,000 items), shown in Table 13. Sample size does not seem to have a consistent effect on apunim values, other than large dataset sizes uncovering significant polarization effects in sparsely populated groups (e.g., 65+ yo. in *Age*, "Prefer Not to Say" in *Sexual Orientation*, "Never" , "Very Frequently" in *Toxicity Problem*). The apunim direction stays broadly the same. We also test whether the results in Table 3 are an artifact of the dataset split, because of the small sample size used in the study. We repeat the apunim experiment with ten different subsets of the same dataset and record their apunim values, shown in Fig. 12. We find that most apunim values are stable, save for "Never" in *Toxicity Problem*, "Heterosexual" in *Sexual Orientation* and "Neutral" in *Technology Impact*.

How Does Sample Selection Influence Apunim? Astute readers may notice that some variability does exist across differently sized samples for the same examined groups, where for example the sign of the apunim value may switch. We note that the statistically significant values (bold rows) change signs only when the apunim value approaches zero in the largest samples. Thus, the results are not inverted, but rather pushed towards zero. In order to verify that this variability is caused by the sample size, and not by different samples being drawn, we execute a further ablation study. Specifically, we execute the apunim study on the 3000-sized sample of the dataset, using different subsets of the Kumar dataset. The results can be found in Fig. 12, where we note that variability is not an issue, except for a couple of cases. These cases seem to represent sparse annotator groups, where missing a few items may genuinely change the value of the apunim measure. These groups however are largely not detected by apunim as significant either way (see Table 3).

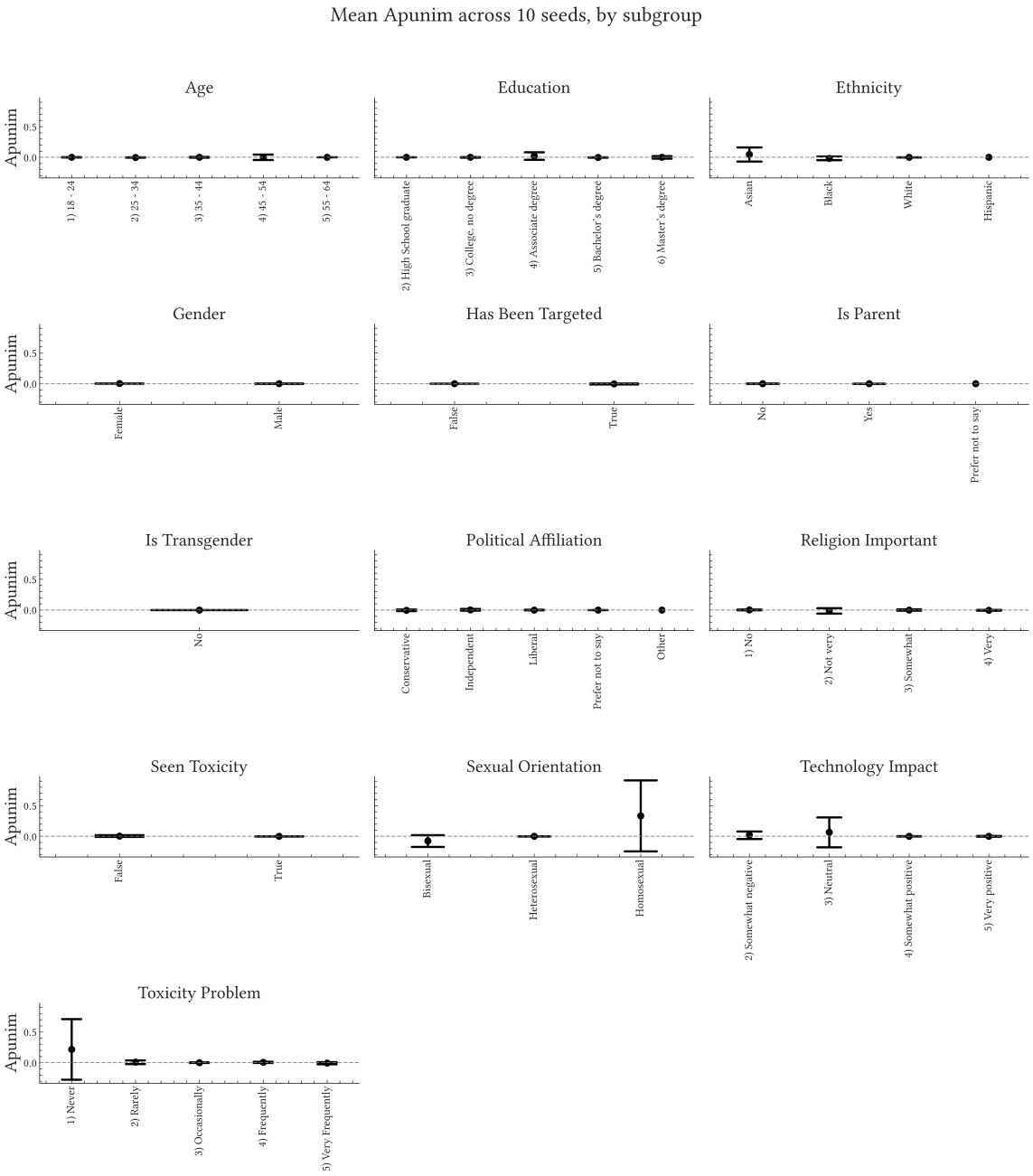


Fig. 12. Apunim values on the Kumar dataset ($n_{samples} = 1000$) across ten runs with different subsets. Black bars indicate two standard deviations. SDDs with many groups are more volatile, although all SDDs broadly keep the same apunim sign across subsets.

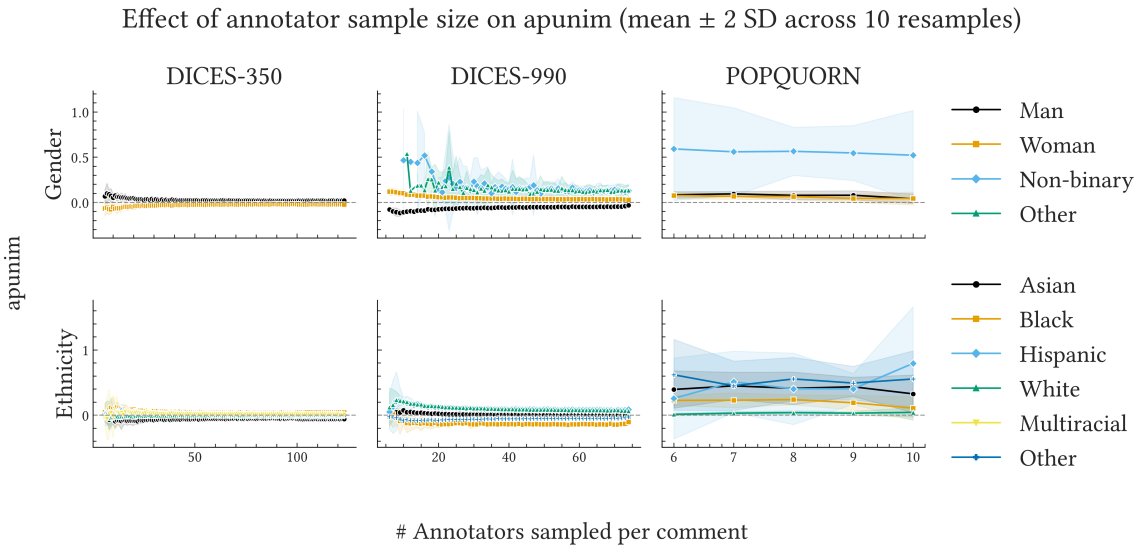


Fig. 13. Standard deviation of *apunim* across items, averaged across 10 runs, where each item is resampled 100 times. Shaded areas represent two standard deviations.

E.3 How Many Annotators Does Apunim Need?

Fig. 13 sows that non-statistically significant results from Table 3 remain systematically close to zero, while statistically significant results can be distinguished from very small annotator counts (six annotators for well-balanced groups, $\approx$ 20 for poorly split groups such as non-binary annotators in DICES-990). Sampling annotators with replacement also uncovers a new group which systematically disagrees with the others; the Non-Binary annotators in the POPQUORN and DICES-990 datasets. This group was undetectable when partitioning annotator groups with the methodology described in Eq. 2 since only one item is annotated by three *Non-Binary* annotators in POPQUORN and none for DICES-990, but sampling with replacement uncovers a very strong attribution. It is unclear whether this is a methodological artifact, or a new finding unexplored in prior analysis concerning these datasets.

F LLM Annotation Procedure

F.1 Models Used

We use only open-source models, as proprietary models such as OpenAI's GPT-5 yield much less reproducible results [8]. Furthermore, the use of open-source models requires much fewer resources, enabling research on this task with a less demanding budget, as well as scalable deployment in real-world settings [58]. It is also worth noting that since the datasets used are not publicly licensed, we can not risk leaking the data contained within to proprietary models. Generally, the use of local models protects the personal data of the users contained in our datasets, ensures that our research can be replicated in domains where anonymity must be maintained, and renders our work compatible with existing legislation [17, 18]. Table 14 shows details on the LLMs used for this study. We specifically use 4-bit quantized versions from the `unsloth huggingface` repository.

Table 14. LLMs used in this study.

Family	Version	#Params	Year
OLMo [27]	2 (0325)	32B	2025
Qwen [79]	2.5	32B	2024
OLMo [49]	3	7B	2025
Qwen [79]	2.5	7B	2024
LLaMa [26]	3.1	8B	2024
Mistral [32]	Small 3 (2501)	24B	2025
Mistral [32]	v0.3	7B	2024
Gemma [65]	4	E2B	2026
Gemma [65]	4	31B	2026

F.2 Experimental Setup

Default Prompt. Since each of the datasets in §3 was annotated using different annotation guidelines and targets we use different instruction prompts for each dataset. The "default" annotation prompt is directly extracted from the studies in the case of Sap and Kumar, and inferred from the instruction guidelines used in the DICES datasets with minor modifications. We use standard persona-prompting procedures sampled randomly from the available SDDs in each respective dataset. Twenty LLM annotators are selected for each item and model.

Ablation Prompts. The prompts used in the ablation study were provided by a GPT-5 and a local Gemma-4 (quantized) model by providing the original prompt and instructing it to produce four variants. The tested prompts use different methods to restrict the model's behavior. Besides the default prompt, we test four prompts that make the persona more central to the rating. All four ask the same question about the comment as the default prompt and append the persona description after the comment. They differ only in how they instruct the model to use the persona. *Persona-focused* asks the model to rate the comment as its persona would personally perceive it, based on the persona's characteristics, attitudes, experiences, and sensitivities. *Direct* states that the persona's characteristics are a mandatory prerequisite to consider and that the model must evaluate the comment exclusively as the persona would classify it. *Single* asks the model to first identify the single most influential trait of its persona, and then to derive the rating by matching the comment to that trait alone. *Stereotype* asks the model to be very sensitive to its persona's characteristics and to give substantial weight to the persona's sensitivities, values, experiences, and likely reactions, rather than to the judgment of a generic or average person. The full prompts are in App. H.

Hardware and Execution Times. We use a single Quadro RTX 6000 GPU for tasks requiring small (7-8B) models and two such GPUs for the rest. The annotation task, including ablations, requires approximately 300 hours of computation to complete. We use the `transformers` library to load and manage the LLMs. We release the relevant code in the supplementary material. Our codebase is platform-independent except for three high-level bash scripts and a full environment file is included for reproduction.

F.3 Exploratory Analysis

Frequencies. Fig. 14 shows the different annotation frequencies of LLMs and human annotators in the same dataset split. Across all datasets, we observe a consistent mismatch between human and LLM annotation distributions. For the DICES-350 and DICES-990 harmfulness tasks, human annotations are heavily concentrated at the lowest harmfulness

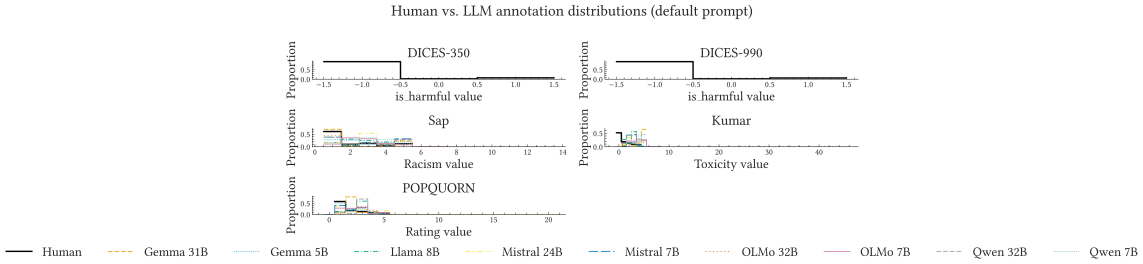


Fig. 14. Annotation frequencies for humans and each LLM in the primary annotation task using the default prompt.

value, whereas most LLMs assign substantially more probability to moderate and high harmfulness values. A similar pattern appears in the Sap et al. [61] racism dataset: human judgments are concentrated at low racism scores, while LLMs produce broader distributions with considerably more mass at higher scores, particularly values 4–5. The [36] toxicity results show the same general trend, with humans assigning most probability to zero toxicity while LLMs distribute their predictions across higher toxicity levels, often peaking around values 2–4. Importantly, the magnitude of this discrepancy varies across models; some LLMs are closer to the human distribution, while others show substantially stronger shifts toward higher scores.

Dropped Responses. Table 15 shows a breakdown of failure modes across models and datasets. Most failed answers stem from the model refusing to provide any number in their response. Some attempt to initiate a Chain of Thought response, which is truncated due to us forcing a three-token limit on LLM answers. Others refuse to respond due to safety tuning. Finally, some (and especially the smaller Gemma model) seem to produce degenerate integer outputs. The full file of failed responses can be found in the Supplementary Material.

F.4 Annotation Ablations

We examine the following conditions for selecting models and datasets for our LLM annotation experiment. All ablation experiments use a sampled 20% fraction of the full LLM annotation dataset used in the main text.

The LLMs should be consistent given the same prompt. While we have fixed all relevant seeds, it is possible that generation deviates across repeated model prompting given the same input [8]. We check this condition by providing the model with the main annotation prompt (see App. H) five times and find that generation is indeed deterministic under our setup for all models (Table 16).

The LLMs should be prompt insensitive. Small changes in the prompt have been shown to drastically affect LLM output in some tasks depending on the experimental setup [56]. In order to verify whether any models are unreliable under our setup, we create two alternative formulations of each prompt without a change in substance using a GPT-5 model (free version), whose outputs we manually verified (see App. H). Table 17 shows that while most LLMs are relatively robust to changes in formulation, the smaller ones (OLMo-7B, Qwen-7B and LLaMa-8B) are not. They are thus excluded from the annotation study.

The annotation guidelines should be unambiguous. The studies that provided the datasets used in our work (see §3) do not include LLM annotation, nor equivalent LLM prompts, which meant we had to generate them from the

Table 15. Number of non-standard LLM annotations per model and dataset, summed over all instruction prompts. Unparsed: no integer found in the reply (dropped from the analysis); Out of range: an integer outside the dataset's annotation scale. Incidence: percentage of the model's annotations on that dataset.

Model	Dataset	Status	Count	Incidence (%)
Gemma 31B	POPQUORN	Unparsed	104	0.09
Gemma 31B	Sap	Unparsed	8	0.01
Gemma 5B	Kumar	Unparsed	12	0.01
Gemma 5B	POPQUORN	Out of range	1	0.00
Gemma 5B	POPQUORN	Unparsed	340	0.23
Gemma 5B	Sap	Out of range	20	0.03
Gemma 5B	Sap	Unparsed	2	0.00
Llama 8B	Kumar	Out of range	153	0.15
Llama 8B	Kumar	Unparsed	33352	33.35
Llama 8B	POPQUORN	Out of range	1	0.00
Llama 8B	POPQUORN	Unparsed	47661	31.77
Llama 8B	Sap	Unparsed	20552	35.13
Mistral 24B	Kumar	Out of range	1	0.00
Mistral 24B	Kumar	Unparsed	14745	14.74
Mistral 24B	POPQUORN	Unparsed	28599	19.07
Mistral 24B	Sap	Unparsed	16783	28.69
Mistral 7B	Kumar	Out of range	5	0.01
Mistral 7B	Kumar	Unparsed	12152	12.15
Mistral 7B	POPQUORN	Out of range	62	0.04
Mistral 7B	POPQUORN	Unparsed	6174	4.12
Mistral 7B	Sap	Unparsed	2771	4.74
OLMo 32B	Kumar	Out of range	25	0.03
OLMo 32B	Kumar	Unparsed	270	0.27
OLMo 32B	POPQUORN	Out of range	10	0.01
OLMo 32B	POPQUORN	Unparsed	1504	1.00
OLMo 32B	Sap	Out of range	2	0.00
OLMo 32B	Sap	Unparsed	953	1.63
OLMo 7B	Kumar	Out of range	96	0.10
OLMo 7B	Kumar	Unparsed	613	0.61
OLMo 7B	POPQUORN	Out of range	110	0.07
OLMo 7B	POPQUORN	Unparsed	4849	3.23
OLMo 7B	Sap	Out of range	37	0.06
OLMo 7B	Sap	Unparsed	3867	6.61
Qwen 32B	Kumar	Unparsed	1	0.00
Qwen 7B	Kumar	Out of range	137	0.14
Qwen 7B	Kumar	Unparsed	22	0.02
Qwen 7B	POPQUORN	Unparsed	42	0.03
Qwen 7B	Sap	Unparsed	5	0.01

human annotation guidelines ourselves. This was not a problem in the case of the Sap and Kumar datasets, since the annotation guidelines were short, simple and required minimal edits to be transformed into LLM annotation guidelines. However, the DICES datasets use poorly documented guidelines and the target label we used is an aggregation of multiple annotation tasks. For these reasons the prompt creation is non-trivial and prone to errors. Indeed, Table 18 shows that inter-model agreement is extremely low for the DICES datasets, even when we remove the sensitive models (Table 19). Given that the DICES datasets are (1) problematic w.r.t. inherent polarization (see App. D), (2) conditioned

Table 16. Consistency (Krippendorff's α, ordinal) of each model with itself across repeated runs of the same (default) prompt.

Dataset	Model	Runs	Items	Krippendorff's alpha
DICES-350	Gemma 31B	5	210	1.0000
DICES-350	Gemma 5B	5	210	1.0000
DICES-350	Llama 8B	5	210	1.0000
DICES-350	Mistral 24B	5	210	1.0000
DICES-350	Mistral 7B	5	210	1.0000
DICES-350	OLMo 32B	5	209	1.0000
DICES-350	OLMo 7B	5	210	1.0000
DICES-350	Qwen 32B	5	210	1.0000
DICES-350	Qwen 7B	5	210	1.0000
DICES-990	Gemma 31B	5	594	—
DICES-990	Gemma 5B	5	594	1.0000
DICES-990	Llama 8B	5	594	1.0000
DICES-990	Mistral 24B	5	592	1.0000
DICES-990	Mistral 7B	5	594	—
DICES-990	OLMo 32B	5	594	1.0000
DICES-990	OLMo 7B	5	594	1.0000
DICES-990	Qwen 32B	5	594	1.0000
DICES-990	Qwen 7B	5	594	1.0000
Sap	Gemma 31B	5	348	1.0000
Sap	Gemma 5B	5	348	1.0000
Sap	Llama 8B	5	287	1.0000
Sap	Mistral 24B	5	348	1.0000
Sap	Mistral 7B	5	345	1.0000
Sap	OLMo 32B	5	343	1.0000
Sap	OLMo 7B	5	338	1.0000
Sap	Qwen 32B	5	348	1.0000
Sap	Qwen 7B	5	348	1.0000
Kumar	Gemma 31B	5	600	1.0000
Kumar	Gemma 5B	5	600	1.0000
Kumar	Llama 8B	5	458	1.0000
Kumar	Mistral 24B	5	600	1.0000
Kumar	Mistral 7B	5	597	1.0000
Kumar	OLMo 32B	5	599	1.0000
Kumar	OLMo 7B	5	600	1.0000
Kumar	Qwen 32B	5	600	1.0000
Kumar	Qwen 7B	5	600	1.0000
POPQUORN	Gemma 5B	5	900	1.0000
POPQUORN	Llama 8B	5	771	1.0000
POPQUORN	Mistral 24B	5	900	1.0000
POPQUORN	Mistral 7B	5	899	1.0000
POPQUORN	OLMo 32B	5	893	1.0000
POPQUORN	OLMo 7B	5	880	1.0000
POPQUORN	Qwen 32B	5	900	1.0000
POPQUORN	Qwen 7B	5	900	1.0000

on several dozen annotations per item (3) difficult to find reliable annotation prompts for LLMs, we decided to exclude them from the annotation study.

F.5 LLM Annotation Results

Table 21. Aposteriori unimodality results for the LLM annotations of the Kumar dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

SDD	Value	Model	Default	Stereotype	Persona	Single	Direct
Age	1) 18 - 24	Llama 8B	-0.049**	0.013	0.060**	—	-0.054*
		OLMo 32B	-0.068***	0.090	—	—	-0.017
		OLMo 7B	0.031**	0.102***	0.052***	-0.115	0.026*
		Qwen 32B	0.023	0.177	—	-0.018	-0.073
		Qwen 7B	0.012	-0.075***	-0.000	0.042	0.077***
		Mistral 7B	0.014	0.052***	0.104***	0.014	0.215***
		Mistral 24B	0.048***	0.031*	0.167***	-0.107	-0.004
		Gemma 5B	0.051*	0.045*	-0.019	-0.006	-0.031
		Gemma 31B	0.035	—	—	—	—
	2) 25 - 34	Llama 8B	0.048**	0.012	0.000	—	-0.140***
		OLMo 32B	-0.010	0.068	—	—	0.051**
		OLMo 7B	0.042***	0.055***	0.031***	0.149	-0.008
		Qwen 32B	0.035*	0.105	—	-0.046*	-0.304
		Qwen 7B	0.036**	0.112***	0.082***	0.118***	0.074***
		Mistral 7B	0.049***	0.036**	0.100***	0.095	0.162***
		Mistral 24B	0.059***	0.040**	0.153**	0.146	-0.026
		Gemma 5B	0.009	0.011	-0.020	0.060*	-0.033
		Gemma 31B	0.073*	—	—	—	—
	3) 35 - 44	Llama 8B	-0.046**	0.015	0.022	—	-0.035
		OLMo 32B	0.029*	—	—	—	-0.024
		OLMo 7B	-0.018	0.057***	-0.016	0.182	0.021
		Qwen 32B	0.042**	0.122	-0.213	0.058*	0.101
		Qwen 7B	0.051***	0.008	0.103***	-0.027	-0.000
		Mistral 7B	0.068***	0.024	-0.007	0.153	0.029
		Mistral 24B	0.007	0.030*	-0.041	-0.056	-0.044**
		Gemma 5B	-0.029	0.008	-0.016	0.000	-0.004
		Gemma 31B	0.031	—	—	—	—
	4) 45 - 54	Llama 8B	0.051**	0.001	0.097***	—	0.043**
		OLMo 32B	0.002	-0.254	—	—	-0.012
		OLMo 7B	0.057***	-0.003	0.004	-0.160	-0.025*
		Qwen 32B	0.044**	0.153	—	0.047*	-0.081
		Qwen 7B	0.084***	-0.040	-0.059	0.007	0.070***
		Mistral 7B	-0.004	0.012	-0.001	-0.458	-0.103***
		Mistral 24B	0.040**	0.006	0.177***	-0.090	-0.008
		Gemma 5B	0.029	-0.004	-0.039	0.116***	-0.030
		Gemma 31B	0.037	—	—	—	—
	5) 55 - 64	Llama 8B	0.025	-0.006	0.007	—	0.065***
		OLMo 32B	-0.026	0.089	—	—	-0.016
		OLMo 7B	0.049***	-0.027*	-0.031***	—	-0.015
		Qwen 32B	0.089***	0.084	0.527	0.059*	0.077
		Qwen 7B	0.014	0.016	0.023	-0.053**	-0.006
		Mistral 7B	0.041**	0.020	0.074***	-0.213	-0.135***
		Mistral 24B	0.045**	0.049***	0.094**	-0.111	0.046**
		Gemma 5B	0.024	0.024	0.052	0.032	0.001
		Gemma 31B	-0.030	—	—	—	—
	6) 65 or older	Llama 8B	0.020	-0.035***	-0.058**	—	0.073***
		OLMo 32B	-0.035**	0.333	—	—	0.005
		OLMo 7B	0.036***	-0.034**	0.002	—	0.020

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Table 21. Aposteriori unimodality results for the LLM annotations of the Kumar dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

SDD	Value	Model	Default	Stereotype	Persona	Single	Direct
		Qwen 32B	0.053**	-0.210	—	0.061*	0.146
		Qwen 7B	0.049***	0.086***	0.034	0.009	0.079***
		Mistral 7B	0.007	-0.034**	-0.141***	-0.114	0.034
		Mistral 24B	0.053***	-0.015	-0.100***	0.055	0.192***
		Gemma 5B	-0.007	0.004	0.021	-0.036	-0.002
		Gemma 31B	-0.012	—	—	—	—
	Prefer not to say	Llama 8B	-0.003	0.039***	0.003	—	-0.079**
		OLMo 32B	-0.043**	0.099	—	—	0.011
		OLMo 7B	0.029*	0.139***	0.041***	0.120	0.017
		Qwen 32B	0.032*	0.102	0.097	0.017	-0.174
		Qwen 7B	0.010	0.002	-0.017	-0.012	0.001
		Mistral 7B	0.047***	-0.016	-0.019	0.227	0.036
		Mistral 24B	0.091***	0.031*	-0.046	-0.024	0.011
		Gemma 5B	-0.007	-0.000	0.077**	-0.034	-0.021
		Gemma 31B	-0.027	—	—	—	—
	Under 18	Llama 8B	0.013	0.017	0.009	—	-0.105***
		OLMo 32B	0.040**	-0.127	—	—	-0.018
		OLMo 7B	0.010	-0.063***	0.036***	—	-0.019
		Qwen 32B	0.036*	-0.110	-0.225	-0.062*	0.005
		Qwen 7B	0.014	-0.031	-0.021	0.057**	-0.005
		Mistral 7B	-0.001	0.005	-0.063*	-0.186	0.028
		Mistral 24B	0.019	-0.064***	-0.149***	0.103	0.046**
		Gemma 5B	0.009	0.051**	-0.084***	0.017	-0.012
		Gemma 31B	0.018	—	—	—	—
Education	1) No high school	Llama 8B	0.029	-0.001	0.059*	—	0.025
		OLMo 32B	-0.007	—	—	—	0.068***
		OLMo 7B	0.004	-0.124***	0.092***	0.113	-0.014
		Qwen 32B	-0.009	-0.170	0.399	-0.042	-0.106
		Qwen 7B	0.028	0.088**	0.003	0.057*	0.014
		Mistral 7B	0.004	-0.005	0.021	-0.083	-0.078*
		Mistral 24B	-0.012	-0.028	0.086	-0.159	-0.025
		Gemma 5B	-0.027	0.002	0.159***	-0.015	0.036
		Gemma 31B	0.037	—	—	—	—
	2) High School graduate	Llama 8B	0.027	0.062***	-0.021	—	0.084*
		OLMo 32B	-0.100***	—	—	—	0.022
		OLMo 7B	-0.038*	0.011	-0.083***	-0.219	0.058***
		Qwen 32B	-0.016	-0.006	0.307	-0.076*	0.212
		Qwen 7B	0.028	0.082**	0.052	-0.002	0.038
		Mistral 7B	-0.032	-0.005	-0.073	0.062	0.101**
		Mistral 24B	0.012	0.040	0.166	0.502	0.050*
		Gemma 5B	0.031	-0.032	0.080*	0.032	-0.012
		Gemma 31B	-0.054	—	—	—	—
	3) College, no degree	Llama 8B	-0.025	-0.025*	-0.035	—	-0.005
		OLMo 32B	0.070***	—	—	—	-0.052*
		OLMo 7B	-0.020	-0.006	-0.035**	—	-0.000
		Qwen 32B	0.017	0.045	—	-0.044	0.137
		Qwen 7B	0.022	-0.068*	-0.014	-0.031	-0.033
		Mistral 7B	-0.020	-0.032	-0.025	-0.064	0.009
		Mistral 24B	-0.033	-0.029	0.061	0.044	0.024
		Gemma 5B	0.025	0.030	-0.009	0.080*	-0.008
		Gemma 31B	0.049	—	—	—	—
	4) Associate degree	Llama 8B	0.058*	0.032*	-0.034	—	-0.116**
		OLMo 32B	0.023	0.079	—	—	-0.053**
		OLMo 7B	0.033	-0.000	0.017	—	0.021
		Qwen 32B	0.030	0.132	—	0.133***	0.156*

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Table 21. Aposteriori unimodality results for the LLM annotations of the Kumar dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

SDD	Value	Model	Default	Stereotype	Persona	Single	Direct
		Qwen 7B	-0.016	-0.018	0.017	-0.047	0.045*
		Mistral 7B	0.030	0.041	-0.024	-0.103	0.025
		Mistral 24B	0.053**	0.003	0.135**	-0.109	-0.026
		Gemma 5B	0.012	-0.003	0.062*	-0.033	-0.051
		Gemma 31B	-0.060	—	—	—	—
	5) Bachelor's degree	Llama 8B	0.053*	-0.036**	0.049*	—	-0.051
		OLMo 32B	-0.034	0.141	—	—	-0.108***
		OLMo 7B	-0.011	0.012	0.010	0.242	-0.032*
		Qwen 32B	0.011	-0.353	-1.000	0.036	-0.097
		Qwen 7B	-0.022	0.016	0.064*	0.031	0.071***
		Mistral 7B	-0.062***	-0.015	0.015	0.160	0.092**
		Mistral 24B	0.059**	0.028	-0.018	0.378**	0.066**
		Gemma 5B	-0.032	0.025	-0.116**	-0.080*	-0.011
		Gemma 31B	0.029	—	—	—	—
	6) Master's degree	Llama 8B	0.003	0.028*	-0.004	—	0.022
		OLMo 32B	0.006	—	—	—	0.097***
		OLMo 7B	-0.003	-0.014	0.022	-0.346	0.029*
		Qwen 32B	-0.014	0.242	—	0.034	0.319
		Qwen 7B	0.030	0.009	0.015	-0.027	-0.008
		Mistral 7B	0.008	0.014	-0.031	—	-0.076*
		Mistral 24B	0.017	0.003	-0.063	0.267	0.002
		Gemma 5B	0.033	0.006	-0.022	-0.009	-0.040
		Gemma 31B	0.042	—	—	—	—
	7) Professional degree	Llama 8B	-0.058*	-0.002	0.088**	—	-0.015
		OLMo 32B	0.007	—	—	—	-0.009
		OLMo 7B	0.033	0.008	0.008	0.136	0.000
		Qwen 32B	-0.004	0.158	0.268	-0.093**	-0.387**
		Qwen 7B	-0.014	0.030	0.050	0.055*	-0.006
		Mistral 7B	0.030	0.006	0.093*	0.000	0.010
		Mistral 24B	-0.034	-0.012	-0.084*	0.080	0.012
		Gemma 5B	0.030	-0.005	0.044	-0.054	-0.046
		Gemma 31B	0.024	—	—	—	—
	8) Doctoral degree	Llama 8B	0.007	0.044**	0.080**	—	-0.040
		OLMo 32B	-0.032	-0.040	—	—	-0.032
		OLMo 7B	-0.040*	0.116***	-0.028*	0.196	-0.043**
		Qwen 32B	0.013	-0.007	—	0.003	-0.223
		Qwen 7B	-0.005	-0.086***	0.032	0.019	0.021
		Mistral 7B	-0.033	-0.030	-0.148***	-0.088	-0.075**
		Mistral 24B	-0.009	-0.013	-0.123*	-0.128	0.034
		Gemma 5B	-0.009	0.006	0.143***	-0.111**	0.112***
		Gemma 31B	0.003	—	—	—	—
	Other	Llama 8B	0.024	0.021	0.064*	—	-0.000
		OLMo 32B	-0.025	-0.039	—	—	-0.010
		OLMo 7B	-0.017	0.033	0.044***	0.176	-0.043**
		Qwen 32B	0.031	0.085	—	0.049	-0.132
		Qwen 7B	-0.015	-0.017	-0.095***	0.061*	-0.077***
		Mistral 7B	0.014	0.017	-0.020	0.345	0.051*
		Mistral 24B	-0.021	0.047**	0.031	0.003	0.021
		Gemma 5B	-0.044	0.010	0.039	0.034	0.038
		Gemma 31B	0.043	—	—	—	—
	Prefer not to say	Llama 8B	-0.079**	-0.008	-0.016	—	0.046
		OLMo 32B	-0.008	-0.242	—	—	-0.043*
		OLMo 7B	-0.050**	-0.034*	-0.053***	—	-0.052**
		Qwen 32B	-0.050*	-0.122	—	0.057*	0.014
		Qwen 7B	-0.006	0.020	-0.130***	-0.095**	0.043

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Table 21. Aposteriori unimodality results for the LLM annotations of the Kumar dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

SDD	Value	Model	Default	Stereotype	Persona	Single	Direct
		Mistral 7B	-0.009	0.001	0.011	0.000	-0.157***
		Mistral 24B	0.017	0.008	0.019	-0.090	0.013
		Gemma 5B	0.008	-0.035	-0.054	0.005	0.021
		Gemma 31B	0.013	—	—	—	—
Ethnicity	Asian	Llama 8B	-0.004	-0.044***	0.043*	—	0.039*
		OLMo 32B	0.015	0.082	—	—	-0.022
		OLMo 7B	0.028**	-0.042***	-0.030***	—	0.003
		Qwen 32B	0.010	-0.106	—	-0.032	0.033
		Qwen 7B	-0.028*	0.055*	0.042	0.010	0.022
		Mistral 7B	-0.020	-0.002	0.002	0.077	0.058**
		Mistral 24B	-0.014	-0.027*	0.059	0.055	0.015
		Gemma 5B	0.028	-0.020	-0.017	0.032	0.034
		Gemma 31B	0.002	—	—	—	—
	Black	Llama 8B	0.005	-0.047***	0.025	—	-0.076***
		OLMo 32B	-0.017	-0.187	—	—	-0.072***
		OLMo 7B	0.013	0.079***	0.028***	0.105	0.002
		Qwen 32B	-0.018	-0.063	—	-0.045	0.056
		Qwen 7B	-0.024	0.009	0.003	0.030	0.021
		Mistral 7B	-0.015	0.014	-0.017	0.082	-0.110***
		Mistral 24B	-0.033*	0.053***	-0.063	-0.266	-0.020
		Gemma 5B	0.025	0.036*	0.058*	-0.032	-0.021
		Gemma 31B	0.026	—	—	—	—
	Hispanic	Llama 8B	-0.018	-0.024*	-0.046**	—	-0.122***
		OLMo 32B	-0.055***	-0.387	—	—	-0.001
		OLMo 7B	0.022*	-0.029**	0.045***	0.235	-0.017
		Qwen 32B	0.033*	-0.116	—	0.022	-0.001
		Qwen 7B	0.021	0.032	0.022	0.001	0.021
		Mistral 7B	-0.004	0.008	-0.009	-0.016	0.061*
		Mistral 24B	0.035*	0.090***	0.114***	0.072	-0.021
		Gemma 5B	-0.011	0.035*	-0.049*	-0.019	0.023
		Gemma 31B	-0.005	—	—	—	—
	Multiracial	Llama 8B	0.026	-0.035***	0.012	—	-0.066***
		OLMo 32B	0.014	0.333	—	—	0.055***
		OLMo 7B	0.016	0.026**	-0.013	-0.244	-0.012
		Qwen 32B	0.052**	0.052	0.142	0.023	-0.118
		Qwen 7B	0.022	0.001	-0.005	-0.039	-0.018
		Mistral 7B	0.018	0.020	-0.008	0.165	0.023
		Mistral 24B	-0.002	0.010	-0.024	-0.125	-0.010
		Gemma 5B	-0.005	-0.015	-0.069**	0.096***	-0.013
		Gemma 31B	0.021	—	—	—	—
	Other	Llama 8B	0.017	-0.012	0.031	—	0.055**
		OLMo 32B	-0.014	0.056	—	—	-0.001
		OLMo 7B	-0.061***	0.011	0.062***	—	-0.013
		Qwen 32B	0.007	0.228*	—	0.066**	0.075
		Qwen 7B	0.026	-0.021	0.008	-0.063**	0.003
		Mistral 7B	-0.009	0.007	-0.066**	-0.199	0.027
		Mistral 24B	0.044**	-0.002	-0.048	0.118	0.016
		Gemma 5B	-0.006	-0.001	-0.022	0.038	0.007
		Gemma 31B	0.035	—	—	—	—
	Unknown	Llama 8B	-0.006	0.014	0.005	—	-0.045***
		OLMo 32B	0.022	0.017	—	—	-0.026
		OLMo 7B	-0.017	-0.024*	0.005	0.242	0.016
		Qwen 32B	0.017	-0.117	0.184	0.013	0.037
		Qwen 7B	-0.021	0.014	-0.004	0.033	-0.027
		Mistral 7B	0.017	0.027	0.024	-0.402	-0.023

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Table 21. Aposteriori unimodality results for the LLM annotations of the Kumar dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

SDD	Value	Model	Default	Stereotype	Persona	Single	Direct
		Mistral 24B	0.004	0.018	-0.112***	0.047	-0.022
		Gemma 5B	0.015	0.018	0.022	0.035	0.060**
		Gemma 31B	0.036	—	—	—	—
	White	Llama 8B	0.009	-0.001	-0.034*	—	0.111***
		OLMo 32B	0.024	0.345	—	—	0.032*
		OLMo 7B	0.022*	-0.034***	-0.000	—	-0.014
		Qwen 32B	-0.022	-0.035	—	-0.034	0.063
		Qwen 7B	0.039**	-0.055*	-0.018	0.006	-0.010
		Mistral 7B	0.003	0.048***	0.040**	0.148	-0.020
		Mistral 24B	0.031*	0.002	-0.014	-0.045	0.072***
		Gemma 5B	-0.022	0.052***	0.042	-0.092***	0.001
		Gemma 31B	0.003	—	—	—	—
Gender	Female	Llama 8B	-0.021*	-0.010	-0.010	—	0.004
		OLMo 32B	-0.014	-0.075	—	—	0.003
		OLMo 7B	0.000	-0.026***	-0.007	-0.010	-0.002
		Qwen 32B	-0.031*	-0.246***	—	0.007	-0.051
		Qwen 7B	-0.012	0.061***	0.028	0.006	-0.011
		Mistral 7B	-0.025*	0.020	-0.017	-0.008	-0.003
		Mistral 24B	0.004	0.017	0.009	-0.013	0.002
		Gemma 5B	-0.002	-0.007	0.020	0.017	-0.022
		Gemma 31B	0.000	—	—	—	—
	Male	Llama 8B	0.021*	-0.003	0.016	—	-0.005
		OLMo 32B	-0.041***	-0.114	—	—	0.027*
		OLMo 7B	-0.002	0.060***	0.017**	-0.067	-0.032***
		Qwen 32B	0.023	0.052	-0.001	-0.042*	0.030
		Qwen 7B	0.032**	-0.099***	-0.017	0.001	-0.017
		Mistral 7B	0.013	0.028*	0.036*	0.033	-0.009
		Mistral 24B	-0.016	-0.015	-0.103***	0.024	-0.008
		Gemma 5B	-0.013	0.014	0.031	0.014	0.019
		Gemma 31B	0.022	—	—	—	—
	Nonbinary	Llama 8B	0.001	-0.031***	-0.001	—	-0.022
		OLMo 32B	-0.003	0.051	—	—	0.003
		OLMo 7B	0.012	0.015*	0.022***	-0.020	0.020**
		Qwen 32B	0.027*	0.092*	-0.383	0.085***	0.024
		Qwen 7B	0.001	0.024	0.039*	0.001	0.025
		Mistral 7B	-0.015	-0.017	0.020	-0.033	0.018
		Mistral 24B	0.023	0.043***	0.102***	-0.039	0.051***
		Gemma 5B	-0.007	-0.006	-0.020	-0.006	-0.011
		Gemma 31B	0.053**	—	—	—	—
	Unknown	Llama 8B	0.038***	-0.008	0.006	—	0.076***
		OLMo 32B	0.015	0.144	—	—	-0.027*
		OLMo 7B	-0.006	-0.047***	0.035***	0.119	0.000
		Qwen 32B	0.021	0.091*	0.281	-0.021	-0.037
		Qwen 7B	0.014	0.075***	-0.012	-0.024	0.021
		Mistral 7B	0.020	-0.000	-0.001	0.031	0.021
		Mistral 24B	-0.002	-0.032**	0.050*	0.002	-0.065***
		Gemma 5B	0.029	0.002	0.008	-0.015	0.017
		Gemma 31B	0.054**	—	—	—	—
Has Been Targeted	False	Llama 8B	0.008	-0.005	0.010	—	0.021
		OLMo 32B	0.002	0.049	—	—	0.024*
		OLMo 7B	0.039***	-0.049***	0.002	-0.083**	0.076***
		Qwen 32B	-0.019	0.072*	0.140	-0.007	0.067
		Qwen 7B	0.083***	-0.033	-0.042**	-0.059***	-0.002
		Mistral 7B	-0.019	0.069***	-0.076***	-0.056	-0.059***
		Mistral 24B	-0.003	0.099***	-0.015	0.109***	0.130***

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Table 21. Aposteriori unimodality results for the LLM annotations of the Kumar dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

SDD	Value	Model	Default	Stereotype	Persona	Single	Direct
		Gemma 5B	0.070***	0.109***	0.069***	0.030	0.060***
		Gemma 31B	-0.007	—	—	—	—
	True	Llama 8B	-0.024	0.023	0.000	—	-0.042*
		OLMo 32B	-0.021	-0.020	—	—	-0.022*
		OLMo 7B	-0.032***	0.075***	0.037***	0.098*	-0.060***
		Qwen 32B	0.021	-0.106*	-0.114	-0.014	-0.032
		Qwen 7B	-0.049***	0.052**	0.055**	0.070***	0.001
		Mistral 7B	0.024	-0.027*	0.059***	0.134	0.105***
		Mistral 24B	0.023	0.121***	0.066**	-0.093**	-0.031**
		Gemma 5B	-0.001	0.041***	0.054**	-0.025	-0.004
		Gemma 31B	-0.004	—	—	—	—
Is Parent	No	Llama 8B	0.036**	0.016	0.032**	—	0.110***
		OLMo 32B	0.018	0.021	—	—	-0.010
		OLMo 7B	0.016*	-0.040***	0.002	0.019	0.005
		Qwen 32B	0.036*	-0.140*	-0.296	0.010	-0.118
		Qwen 7B	0.001	-0.014	0.014	-0.013	0.042**
		Mistral 7B	0.015	-0.016	0.042**	-0.076	-0.026
		Mistral 24B	-0.015	-0.021	0.027	0.021	0.020
		Gemma 5B	0.003	-0.005	0.001	0.003	0.023
		Gemma 31B	0.022	—	—	—	—
	Prefer not to say	Llama 8B	-0.024*	-0.018	-0.012	—	-0.032*
		OLMo 32B	0.020	-0.008	—	—	0.001
		OLMo 7B	0.057***	-0.000	0.002	0.040	0.006
		Qwen 32B	-0.025	0.002	0.469	-0.022	0.093
		Qwen 7B	0.002	-0.011	-0.039	-0.045**	-0.026
		Mistral 7B	-0.011	-0.010	0.040**	0.065	0.037*
		Mistral 24B	0.023	-0.005	-0.012	-0.049	-0.018
		Gemma 5B	-0.013	0.010	-0.007	0.021	0.011
		Gemma 31B	-0.017	—	—	—	—
	Yes	Llama 8B	-0.014	0.011	0.017	—	-0.008
		OLMo 32B	-0.010	0.042	—	—	-0.003
		OLMo 7B	-0.038***	0.029***	0.022***	0.012	0.007
		Qwen 32B	0.003	0.113*	-0.180	-0.008	0.014
		Qwen 7B	0.021	0.053**	0.010	0.042**	0.000
		Mistral 7B	-0.001	0.012	-0.051***	0.151*	-0.035*
		Mistral 24B	-0.003	-0.000	-0.063	0.021	0.023
		Gemma 5B	0.018	0.013	0.019	-0.002	0.019
		Gemma 31B	-0.013	—	—	—	—
Is Transgender	No	Llama 8B	-0.008	0.004	0.042***	—	0.052***
		OLMo 32B	0.020	0.080	—	—	-0.014
		OLMo 7B	-0.012	-0.004	0.035***	0.112*	0.023***
		Qwen 32B	-0.005	0.179***	-0.109	0.007	0.131*
		Qwen 7B	0.004	0.058***	0.022	0.006	0.017
		Mistral 7B	0.005	0.008	0.024	0.028	-0.005
		Mistral 24B	0.009	-0.001	-0.031	0.044	0.035**
		Gemma 5B	0.018	-0.008	0.020	0.022	0.028**
		Gemma 31B	-0.015	—	—	—	—
	Prefer not to say	Llama 8B	-0.002	-0.028***	-0.016	—	-0.069***
		OLMo 32B	-0.023	-0.146	—	—	0.043***
		OLMo 7B	0.009	-0.008	0.017**	-0.004	0.022***
		Qwen 32B	0.001	-0.244***	0.316	0.017	0.090
		Qwen 7B	-0.041***	0.008	-0.025	-0.000	0.016
		Mistral 7B	0.004	0.015	-0.003	-0.070	-0.003
		Mistral 24B	0.019	0.028*	0.065**	0.023	0.015
		Gemma 5B	-0.015	-0.009	-0.024	-0.046**	-0.046***

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Table 21. Aposteriori unimodality results for the LLM annotations of the Kumar dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

SDD	Value	Model	Default	Stereotype	Persona	Single	Direct
		Gemma 31B	0.004	—	—	—	—
	Yes	Llama 8B	-0.028*	-0.007	-0.021	—	-0.057***
		OLMo 32B	-0.004	0.014	—	—	0.003
		OLMo 7B	-0.010	0.009	-0.014**	-0.152*	-0.001
		Qwen 32B	0.026	-0.072	0.133	-0.020	0.148*
		Qwen 7B	0.039***	-0.066***	0.018	0.001	0.018
		Mistral 7B	-0.003	-0.041***	-0.002	-0.059	0.032
		Mistral 24B	-0.024	0.017	-0.002	-0.003	-0.043***
		Gemma 5B	0.017	0.009	-0.009	-0.014	0.004
		Gemma 31B	0.025	—	—	—	—
Political Affiliation	Conservative	Llama 8B	0.001	-0.001	0.018	—	0.018
		OLMo 32B	-0.001	0.093	—	—	0.014
		OLMo 7B	0.010	-0.010	-0.014*	-0.021	-0.016*
		Qwen 32B	0.025	-0.300*	0.581	-0.016	0.020
		Qwen 7B	0.039***	-0.078***	-0.003	0.014	0.024
		Mistral 7B	-0.012	-0.009	0.000	0.027	-0.001
		Mistral 24B	-0.038**	0.034**	-0.034	-0.124*	-0.053***
		Gemma 5B	0.042**	-0.003	-0.030	0.001	-0.004
		Gemma 31B	0.005	—	—	—	—
	Independent	Llama 8B	0.022	0.038***	-0.006	—	-0.011
		OLMo 32B	0.012	-0.057	—	—	0.011
		OLMo 7B	0.007	-0.026**	0.013*	-0.166	0.021**
		Qwen 32B	-0.003	-0.109*	-0.348	0.026	-0.049
		Qwen 7B	-0.020	0.008	0.007	0.015	0.008
		Mistral 7B	-0.022	-0.010	-0.019	0.095	-0.083***
		Mistral 24B	0.007	0.008	0.008	-0.044	0.009
		Gemma 5B	-0.012	-0.001	-0.002	-0.033	-0.050***
		Gemma 31B	-0.011	—	—	—	—
	Liberal	Llama 8B	-0.015	-0.005	-0.002	—	0.067***
		OLMo 32B	0.005	0.088	—	—	-0.029*
		OLMo 7B	0.030***	0.064***	-0.009	0.114	0.023**
		Qwen 32B	-0.013	0.221**	-1.000	0.025	0.061
		Qwen 7B	0.024*	0.011	0.042*	0.038*	0.006
		Mistral 7B	0.003	0.040***	0.071***	0.004	0.003
		Mistral 24B	0.011	0.051***	0.052*	0.000	0.023
		Gemma 5B	0.017	0.002	-0.002	0.008	0.020
		Gemma 31B	0.062**	—	—	—	—
	Other	Llama 8B	0.008	-0.009	0.002	—	-0.061***
		OLMo 32B	-0.006	-0.149	—	—	-0.052***
		OLMo 7B	-0.041***	-0.022**	0.014*	0.118	0.001
		Qwen 32B	-0.018	-0.128*	0.316	0.009	0.005
		Qwen 7B	-0.006	0.067***	-0.041*	0.021	0.015
		Mistral 7B	-0.010	-0.006	-0.001	0.218**	0.006
		Mistral 24B	0.018	-0.009	-0.023	0.123*	-0.004
		Gemma 5B	0.010	-0.015	-0.004	0.051**	-0.036*
		Gemma 31B	-0.002	—	—	—	—
	Prefer not to say	Llama 8B	0.018	-0.027***	-0.036*	—	-0.007
		OLMo 32B	0.003	0.151	—	—	0.015
		OLMo 7B	-0.015*	-0.006	0.016*	-0.069	0.011
		Qwen 32B	-0.009	0.051	0.342	0.019	0.031
		Qwen 7B	-0.045***	0.005	-0.004	-0.011	-0.027
		Mistral 7B	0.025*	0.013	0.048**	-0.124	0.072***
		Mistral 24B	0.020	-0.011	0.035	0.057	0.001
		Gemma 5B	-0.023	0.026	0.027	0.028	0.006
		Gemma 31B	0.041	—	—	—	—

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Table 21. Aposteriori unimodality results for the LLM annotations of the Kumar dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

SDD	Value	Model	Default	Stereotype	Persona	Single	Direct
Religion Important	1) No	Llama 8B	0.018	-0.005	-0.035**	—	-0.048**
		OLMo 32B	0.003	0.095	—	—	-0.009
		OLMo 7B	0.025**	-0.082***	-0.013*	-0.016	-0.006
		Qwen 32B	-0.016	0.006	—	0.028	0.182
		Qwen 7B	-0.006	-0.004	-0.016	0.024	-0.024
		Mistral 7B	-0.012	0.009	-0.067***	-0.021	0.021
		Mistral 24B	-0.008	-0.021	-0.146***	-0.062	0.003
		Gemma 5B	0.014	0.039**	0.033	0.083***	0.021
		Gemma 31B	0.017	—	—	—	—
	2) Not very	Llama 8B	0.019	-0.001	0.002	—	0.012
		OLMo 32B	-0.005	0.052	—	—	0.002
		OLMo 7B	0.005	-0.017*	0.026***	0.103	0.038***
		Qwen 32B	0.015	-0.159*	0.097	0.028	-0.028
		Qwen 7B	0.044***	0.088***	0.052**	0.086***	0.024
		Mistral 7B	-0.001	0.013	0.042**	0.129	-0.032
		Mistral 24B	0.010	-0.006	0.027	-0.001	0.004
		Gemma 5B	0.007	0.006	-0.010	0.000	-0.012
		Gemma 31B	-0.040	—	—	—	—
	3) Somewhat	Llama 8B	-0.028	0.011	-0.021	—	-0.011
		OLMo 32B	0.004	-0.074***	—	—	0.042***
		OLMo 7B	0.010	0.033***	0.027***	0.093	0.003
		Qwen 32B	0.007	0.125*	—	0.020	-0.042
		Qwen 7B	0.000	0.031	0.002	-0.068***	-0.018
		Mistral 7B	0.000	-0.010	0.053**	0.010	0.067***
		Mistral 24B	-0.010	0.020	-0.030	0.137	-0.010
		Gemma 5B	-0.011	-0.005	-0.030	-0.030	-0.003
		Gemma 31B	0.008	—	—	—	—
	4) Very	Llama 8B	-0.025	0.017	-0.027*	—	0.073***
		OLMo 32B	0.027*	-0.051	—	—	-0.051***
		OLMo 7B	0.010	0.039***	0.015**	-0.065	-0.035***
		Qwen 32B	0.019	-0.033	0.214	-0.079***	-0.145
		Qwen 7B	0.033**	-0.094***	-0.055**	0.017	0.059***
		Mistral 7B	-0.003	-0.001	-0.012	-0.025	-0.065***
		Mistral 24B	-0.018	-0.007	0.005	0.035	-0.002
		Gemma 5B	-0.035*	-0.022	-0.018	-0.080***	-0.048***
		Gemma 31B	-0.007	—	—	—	—
	Prefer not to say	Llama 8B	0.015	-0.003	0.064***	—	0.013
		OLMo 32B	-0.012	0.075	—	—	0.029**
		OLMo 7B	-0.029***	0.003	0.003	0.156	0.010
		Qwen 32B	0.001	0.029	-1.000	-0.066***	0.085
		Qwen 7B	-0.002	-0.012	-0.031	0.019	0.039**
		Mistral 7B	0.003	0.006	-0.024	0.049	-0.080***
		Mistral 24B	0.006	-0.027*	0.029	0.042	0.019
		Gemma 5B	-0.006	0.020	0.017	-0.036*	0.055***
		Gemma 31B	-0.012	—	—	—	—
Seen Toxicity	False	Llama 8B	0.052***	-0.011	0.024	—	0.025
		OLMo 32B	0.002	-0.022	—	—	-0.014
		OLMo 7B	0.039***	-0.053***	-0.001	0.096**	0.033***
		Qwen 32B	-0.009	-0.047	-0.119	-0.020	-0.006
		Qwen 7B	-0.048***	0.030	0.023	-0.017	0.008
		Mistral 7B	-0.002	0.017	-0.086***	-0.058	-0.061***
		Mistral 24B	-0.006	-0.016	-0.062**	0.003	0.016
		Gemma 5B	-0.004	0.039**	-0.015	0.017	0.065***
		Gemma 31B	0.022	—	—	—	—
	True	Llama 8B	-0.030**	-0.005	-0.016	—	0.002

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Table 21. Aposteriori unimodality results for the LLM annotations of the Kumar dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

SDD	Value	Model	Default	Stereotype	Persona	Single	Direct
		OLMo 32B	0.010	0.029	—	—	0.008
		OLMo 7B	-0.008	0.063***	0.009	-0.056*	-0.040***
		Qwen 32B	-0.008	0.058	0.127	0.003	0.031
		Qwen 7B	0.062***	-0.007	-0.022	0.023	0.010
		Mistral 7B	0.008	-0.008	0.085***	0.057	0.063***
		Mistral 24B	0.009	0.014	0.073***	-0.008	-0.019
		Gemma 5B	0.000	0.016	0.056**	-0.018	-0.012
		Gemma 31B	-0.017	—	—	—	—
Sexual Orientation	Bisexual	Llama 8B	0.033**	-0.013	0.034**	—	-0.042*
		OLMo 32B	0.001	0.057	—	—	0.028**
		OLMo 7B	-0.004	-0.002	0.018**	—	-0.008
		Qwen 32B	0.012	-0.067	-0.376	-0.005	0.118
		Qwen 7B	0.011	0.011	-0.000	-0.011	-0.009
		Mistral 7B	0.003	0.007	0.028	0.056	0.028
		Mistral 24B	0.003	0.021	0.078**	0.006	0.023
		Gemma 5B	0.006	-0.011	0.008	0.038	0.025
		Gemma 31B	0.011	—	—	—	—
	Heterosexual	Llama 8B	0.024	-0.021*	0.013	—	0.032*
		OLMo 32B	-0.019	-0.006	—	—	-0.041***
		OLMo 7B	-0.017*	-0.010	0.019**	0.092	0.052***
		Qwen 32B	0.008	-0.017	—	-0.008	-0.161
		Qwen 7B	-0.005	0.026	-0.020	0.027	-0.002
		Mistral 7B	0.018	-0.012	-0.051**	0.010	0.045*
		Mistral 24B	-0.005	-0.018	0.014	0.123	-0.004
		Gemma 5B	-0.009	0.030*	-0.026	0.053*	-0.018
		Gemma 31B	-0.005	—	—	—	—
	Homosexual	Llama 8B	-0.013	-0.001	-0.028*	—	-0.078***
		OLMo 32B	-0.024	-0.049	—	—	0.007
		OLMo 7B	0.024**	0.010	-0.000	0.126*	-0.017*
		Qwen 32B	0.005	-0.067	0.342	-0.069***	0.117
		Qwen 7B	0.010	-0.015	0.019	0.030	0.031
		Mistral 7B	0.009	0.027*	0.002	0.103	-0.030
		Mistral 24B	0.005	-0.011	-0.162***	0.022	0.030*
		Gemma 5B	0.011	0.027*	0.047*	-0.004	0.027
		Gemma 31B	0.062**	—	—	—	—
	Other	Llama 8B	0.012	-0.019*	-0.038**	—	-0.012
		OLMo 32B	-0.006	0.038	—	—	-0.032**
		OLMo 7B	0.012	-0.008	-0.003	-0.265	-0.034***
		Qwen 32B	0.003	0.145	0.290	0.012	-0.032
		Qwen 7B	0.036**	-0.033	-0.004	0.006	-0.024
		Mistral 7B	0.003	0.012	-0.006	0.057	-0.033
		Mistral 24B	0.003	0.009	-0.073**	0.058	-0.003
		Gemma 5B	-0.007	-0.031*	-0.093***	-0.022	0.033*
		Gemma 31B	-0.011	—	—	—	—
	Prefer not to say	Llama 8B	-0.053***	0.007	0.025	—	0.091***
		OLMo 32B	0.007	-0.051	—	—	0.034**
		OLMo 7B	-0.022**	0.017	-0.006	-0.022	-0.020**
		Qwen 32B	0.012	-0.005	-0.197	0.043**	-0.124
		Qwen 7B	-0.000	0.041*	-0.011	-0.030	-0.004
		Mistral 7B	0.006	0.006	0.046*	-0.051	0.025
		Mistral 24B	0.004	-0.010	-0.046	0.047	-0.044***
		Gemma 5B	-0.008	-0.023	-0.004	0.015	0.006
		Gemma 31B	-0.007	—	—	—	—
Technology Impact	1) Very negative	Llama 8B	0.021	-0.003	0.028*	—	-0.054**
		OLMo 32B	0.009	0.068	—	—	-0.035**

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Table 21. Aposteriori unimodality results for the LLM annotations of the Kumar dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

SDD	Value	Model	Default	Stereotype	Persona	Single	Direct
		OLMo 7B	0.020*	-0.080***	0.028***	0.166	-0.005
		Qwen 32B	-0.007	0.097	-0.259	-0.017	0.136
		Qwen 7B	0.005	-0.016	0.017	-0.018	-0.032*
		Mistral 7B	0.010	-0.010	0.044**	0.085	0.030
		Mistral 24B	0.009	0.029**	0.061	-0.115	-0.038***
		Gemma 5B	0.021	0.005	0.036	-0.004	0.015
		Gemma 31B	-0.007	—	—	—	—
	2) Somewhat negative	Llama 8B	0.016	0.010	0.057***	—	0.019
		OLMo 32B	0.001	-0.309	—	—	-0.064***
		OLMo 7B	0.016*	-0.066***	-0.045***	-0.448	-0.006
		Qwen 32B	-0.008	-0.058	0.449	-0.070***	-0.021
		Qwen 7B	-0.035**	0.059***	0.064***	0.086***	-0.005
		Mistral 7B	0.011	0.028*	-0.009	0.110	0.028
		Mistral 24B	-0.011	0.037***	-0.049	-0.034	-0.051***
		Gemma 5B	-0.001	0.004	-0.045*	-0.003	0.068***
		Gemma 31B	0.013	—	—	—	—
	3) Neutral	Llama 8B	-0.022	-0.013	-0.013	—	-0.001
		OLMo 32B	0.017	0.018	—	—	-0.002
		OLMo 7B	-0.029***	-0.029***	0.032***	-0.257	0.033***
		Qwen 32B	-0.007	-0.086	-0.289	0.019	0.035
		Qwen 7B	-0.005	-0.029	-0.143***	-0.030*	-0.054***
		Mistral 7B	0.027*	0.011	0.004	-0.041	0.019
		Mistral 24B	0.005	-0.013	0.019	0.164***	0.020
		Gemma 5B	0.011	0.020	0.028	0.025	0.030*
		Gemma 31B	-0.011	—	—	—	—
	4) Somewhat positive	Llama 8B	-0.012	0.019*	-0.010	—	0.038**
		OLMo 32B	-0.006	0.074	—	—	0.030**
		OLMo 7B	0.011	0.038***	-0.008	0.078	0.001
		Qwen 32B	0.039**	-0.261*	—	0.081***	0.007
		Qwen 7B	0.023	-0.018	0.059**	0.061***	0.063***
		Mistral 7B	-0.001	0.012	-0.061***	0.072	-0.016
		Mistral 24B	0.023	0.021	-0.011	0.001	0.039***
		Gemma 5B	-0.004	0.009	0.074***	0.000	-0.056***
		Gemma 31B	0.021	—	—	—	—
	5) Very positive	Llama 8B	0.005	0.028**	0.005	—	0.049**
		OLMo 32B	-0.013	-0.058	—	—	-0.005
		OLMo 7B	-0.010	0.092***	0.018**	0.132	-0.031***
		Qwen 32B	0.028	0.135	0.531	0.020	-0.059
		Qwen 7B	-0.001	-0.017	0.039*	-0.009	0.049***
		Mistral 7B	-0.047***	-0.009	0.004	0.025	0.013
		Mistral 24B	0.030*	0.003	0.000	0.097	0.109***
		Gemma 5B	-0.000	0.026	0.054**	0.006	-0.053***
		Gemma 31B	0.013	—	—	—	—
Toxicity Problem	1) Never	Llama 8B	0.005	0.043***	-0.004	—	-0.043*
		OLMo 32B	-0.003	-0.073	—	—	-0.010
		OLMo 7B	0.090***	-0.064***	-0.025***	-0.122	-0.051***
		Qwen 32B	0.050***	0.107	0.313	0.392***	-0.034
		Qwen 7B	0.054***	0.011	0.039*	0.000	0.047***
		Mistral 7B	0.022*	0.015	-0.073***	-0.017	-0.097***
		Mistral 24B	0.065***	0.002	-0.201***	-0.112**	0.206***
		Gemma 5B	0.092***	0.146***	0.001	0.083***	0.116***
		Gemma 31B	-0.004	—	—	—	—
	2) Rarely	Llama 8B	0.034**	-0.003	-0.028	—	-0.052**
		OLMo 32B	0.038***	0.021	—	—	0.050***
		OLMo 7B	0.042***	0.218***	0.087***	0.092*	-0.032***

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Table 21. Aposteriori unimodality results for the LLM annotations of the Kumar dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

SDD	Value	Model	Default	Stereotype	Persona	Single	Direct
		Qwen 32B	0.071***	-0.116	-0.045	0.394***	-0.022
		Qwen 7B	-0.002	0.152***	0.105***	-0.042**	0.093***
		Mistral 7B	0.003	0.023*	-0.030*	0.075	-0.062***
		Mistral 24B	0.070***	0.040***	0.031	0.183***	0.105***
		Gemma 5B	0.061***	0.050***	0.031	0.118***	0.035**
		Gemma 31B	0.007	—	—	—	—
	3) Occasionally	Llama 8B	-0.053***	0.012	-0.005	—	0.029
		OLMo 32B	0.066***	0.072	—	—	0.057***
		OLMo 7B	0.052***	0.040***	0.053***	0.140	0.018**
		Qwen 32B	0.061***	0.008	0.285	0.400***	0.222**
		Qwen 7B	0.143***	0.063***	0.031	0.076***	0.048***
		Mistral 7B	-0.034**	-0.007	0.016	0.022	-0.005
		Mistral 24B	0.084***	0.025*	0.078**	0.169***	0.082***
		Gemma 5B	0.006	-0.016	0.090***	-0.010	-0.020
		Gemma 31B	-0.038	—	—	—	—
	4) Frequently	Llama 8B	0.040***	0.043***	0.020	—	0.028
		OLMo 32B	-0.018	0.001	—	—	0.031**
		OLMo 7B	0.201***	-0.053***	0.005	0.093	0.114***
		Qwen 32B	0.047***	-0.030	0.379	0.286***	-0.003
		Qwen 7B	-0.004	-0.049**	0.096***	0.160***	0.096***
		Mistral 7B	0.018	0.026*	0.030*	0.095	0.171***
		Mistral 24B	-0.060***	0.015	0.171***	0.167**	0.050***
		Gemma 5B	0.036**	0.052***	0.018	0.119***	0.114***
		Gemma 31B	0.010	—	—	—	—
	5) Very Frequently	Llama 8B	-0.005	-0.014	0.005	—	0.060**
		OLMo 32B	-0.036***	0.021	—	—	0.011
		OLMo 7B	0.047***	-0.082***	-0.038***	0.013	-0.028***
		Qwen 32B	0.055***	0.038	-0.526	0.216***	-0.002
		Qwen 7B	0.214***	-0.161***	-0.259***	-0.114***	0.093***
		Mistral 7B	0.015	-0.024*	0.097***	-0.150	0.082***
		Mistral 24B	0.003	0.045***	0.084**	-0.149	-0.042***
		Gemma 5B	0.056***	0.042***	0.067***	0.056***	0.097***
		Gemma 31B	0.026	—	—	—	—

Table 22. Aposteriori unimodality results for the LLM annotations of the Sap dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

SDD	Value	Model	Default	Stereotype	Persona	Single	Direct
Age	1) Gen. X+	Llama 8B	-0.009	-0.018	-0.023	0.017	-0.030**
		OLMo 32B	-0.006	-0.010	0.012	-0.031	-0.021
		OLMo 7B	0.023*	-0.001	0.009	0.047***	0.017*
		Qwen 32B	0.208***	0.061***	0.120***	0.189***	0.184***
		Qwen 7B	0.152***	0.090***	0.147***	0.055***	0.181***
		Mistral 7B	0.059**	0.004	0.054**	-0.018	-0.000
		Mistral 24B	0.131***	0.006	0.079**	-0.025	0.051***
		Gemma 5B	0.021	0.004	0.047**	0.038*	-0.014
		Gemma 31B	-0.024	0.009	-0.098***	0.004	-0.043
	2) Millennial	Llama 8B	-0.019	0.011	0.016	0.140***	-0.013
		OLMo 32B	-0.009	-0.042*	-0.037*	-0.014	-0.023
		OLMo 7B	0.031**	-0.052***	0.038***	-0.053***	-0.033***
		Qwen 32B	0.249***	0.059***	0.190***	0.137***	0.214***
		Qwen 7B	0.246***	0.139***	0.219***	0.098***	0.182***

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Table 22. Aposteriori unimodality results for the LLM annotations of the Sap dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

SDD	Value	Model	Default	Stereotype	Persona	Single	Direct
		Mistral 7B	0.017	0.008	-0.043*	-0.058**	-0.011
		Mistral 24B	0.177***	0.036**	0.088***	0.166***	0.081***
		Gemma 5B	0.021	-0.014	-0.020	-0.009	0.025
		Gemma 31B	0.085*	0.074**	0.061*	0.042	0.097***
	3) Gen. Z	Llama 8B	0.072***	0.107***	0.021	0.079*	0.079***
		OLMo 32B	0.054**	0.023	0.039*	0.022	0.054**
		OLMo 7B	-0.020*	0.018	0.017	0.004	-0.015*
		Qwen 32B	0.124***	0.122***	0.204***	0.083***	0.140***
		Qwen 7B	0.007	0.109***	-0.014	0.137***	0.133***
		Mistral 7B	0.094***	0.047*	0.014	0.022	0.011
		Mistral 24B	-0.062***	0.162***	0.000	-0.022	0.050***
		Gemma 5B	-0.014	0.003	0.002	0.011	0.019
		Gemma 31B	0.124**	0.188***	0.202***	0.105**	0.135***
Ethnicity	black	Llama 8B	-0.005	0.076***	-0.077***	-0.070	0.013
		OLMo 32B	0.017	0.041	0.079***	0.020	0.013
		OLMo 7B	0.017	0.007	0.004	-0.005	0.029**
		Qwen 32B	0.010	0.202***	0.110***	0.043*	0.080***
		Qwen 7B	0.047***	0.126***	0.049***	0.050***	-0.021
		Mistral 7B	0.054**	0.302***	0.214***	0.099***	0.162***
		Mistral 24B	0.095***	0.430***	0.032	-0.049	0.174***
		Gemma 5B	0.023	0.069***	0.032*	0.044***	0.099***
		Gemma 31B	0.148***	0.285***	0.375***	0.138***	0.295***
	hisp	Llama 8B	0.012	-0.017	0.088***	0.142*	0.037***
		OLMo 32B	-0.020	0.005	-0.026	0.012	0.018
		OLMo 7B	0.050***	-0.016	-0.008	-0.020	0.023*
		Qwen 32B	0.031	0.172***	0.047**	-0.007	0.094***
		Qwen 7B	0.038**	0.149***	0.052***	0.068***	0.085***
		Mistral 7B	0.031	0.058**	-0.019	0.004	-0.050*
		Mistral 24B	0.090***	0.277***	-0.005	0.114*	0.149***
		Gemma 5B	0.059***	0.072***	0.009	0.055***	0.062***
		Gemma 31B	0.064	0.120***	0.076**	0.102***	0.141***
	middleEastern	Llama 8B	-0.030	0.067***	0.119***	0.032	0.008
		OLMo 32B	-0.013	0.028	-0.008	0.028	0.032
		OLMo 7B	-0.049***	0.003	-0.025**	0.033*	0.009
		Qwen 32B	0.052**	0.277***	0.093***	0.078***	0.094***
		Qwen 7B	-0.001	0.046***	0.021	-0.075***	0.002
		Mistral 7B	0.009	0.073***	0.022	-0.037	0.015
		Mistral 24B	0.082***	0.294***	0.052	0.041	0.082***
		Gemma 5B	0.041*	0.070***	0.004	0.072***	0.019
		Gemma 31B	0.105**	0.187***	0.104***	0.138***	0.038
	other	Llama 8B	-0.015	-0.045***	-0.003	0.082	-0.067***
		OLMo 32B	-0.014	0.024	-0.006	-0.024	0.022
		OLMo 7B	-0.031***	-0.025*	0.043***	-0.038*	0.069***
		Qwen 32B	0.031	0.165***	0.087***	0.094***	0.084***
		Qwen 7B	0.072***	0.106***	0.038**	0.116***	0.038**
		Mistral 7B	0.118***	0.092***	0.036	0.018	0.074***
		Mistral 24B	0.002	0.123***	0.033	0.045	0.111***
		Gemma 5B	-0.013	0.117***	0.045**	-0.001	0.033
		Gemma 31B	0.111**	0.085***	0.021	0.094**	0.025
	white	Llama 8B	-0.023	0.020	-0.043*	-0.022	0.057***
		OLMo 32B	0.020	-0.018	-0.022	0.015	-0.011
		OLMo 7B	0.034***	0.002	0.066***	-0.002	0.073***
		Qwen 32B	0.086***	0.186***	0.072***	0.050**	0.044**
		Qwen 7B	0.068***	0.023*	-0.003	0.058***	0.005
		Mistral 7B	0.071***	0.175***	0.206***	0.094***	0.101***

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Table 22. Aposteriori unimodality results for the LLM annotations of the Sap dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

SDD	Value	Model	Default	Stereotype	Persona	Single	Direct
		Mistral 24B	-0.043***	0.034**	0.049	0.006	0.139***
		Gemma 5B	0.121***	0.038**	0.123***	0.085***	0.237***
		Gemma 31B	-0.099*	-0.075***	-0.116***	0.081**	-0.094**
Gender	man	Llama 8B	0.011	0.031**	-0.092***	0.028	0.041***
		OLMo 32B	0.004	0.004	0.001	0.046**	-0.003
		OLMo 7B	0.000	0.026**	-0.008	0.029	-0.012
		Qwen 32B	-0.033	-0.054**	0.009	0.003	-0.014
		Qwen 7B	-0.003	-0.009	-0.030*	0.056**	0.037*
		Mistral 7B	-0.014	-0.055**	-0.021	-0.040	-0.071***
		Mistral 24B	-0.006	-0.037**	-0.014	-0.033	0.010
		Gemma 5B	0.029	-0.029	0.037*	0.000	0.057**
		Gemma 31B	0.042	-0.026	-0.000	-0.002	-0.023
	nonBinary	Llama 8B	-0.044**	0.028*	0.076***	-0.023	-0.050***
		OLMo 32B	0.006	-0.014	-0.010	-0.023	0.011
		OLMo 7B	-0.029**	0.000	0.062***	0.013	-0.017
		Qwen 32B	-0.046	-0.006	-0.011	0.007	-0.003
		Qwen 7B	0.007	0.021	0.014	-0.042*	-0.004
		Mistral 7B	0.022	0.022	-0.027	0.034	0.049*
		Mistral 24B	0.029*	0.029*	0.012	0.090**	0.009
		Gemma 5B	-0.010	0.015	0.014	0.040*	-0.048*
		Gemma 31B	0.068	0.010	0.000	0.056	0.061
	woman	Llama 8B	0.004	0.010	0.029*	0.094*	0.038***
		OLMo 32B	0.004	0.017	0.000	0.024	-0.007
		OLMo 7B	0.064***	0.036***	-0.006	0.068***	-0.005
		Qwen 32B	-0.011	-0.001	-0.007	-0.028	-0.013
		Qwen 7B	-0.016	0.004	-0.035*	0.013	-0.025
		Mistral 7B	-0.015	0.056**	0.078***	0.034	0.014
		Mistral 24B	-0.028*	-0.005	0.065*	0.020	-0.004
		Gemma 5B	-0.016	0.023	-0.018	-0.015	-0.000
		Gemma 31B	-0.023	0.005	0.011	-0.003	0.024

Table 23. Aposteriori unimodality results for the LLM annotations of the POPQUORN dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

SDD	Value	Model	Default	Stereotype	Persona	Single	Direct
Age	01) 18-24	Llama 8B	-0.009	0.022	-0.024	—	-0.047***
		OLMo 32B	0.026	0.026	-0.025	0.273	0.094***
		OLMo 7B	0.059***	0.016	-0.088***	—	0.035
		Qwen 32B	-0.047	0.090**	0.072	0.143***	0.004
		Qwen 7B	0.006	-0.075***	-0.053*	-0.038	0.006
		Mistral 7B	0.146*	—	—	—	0.198
		Mistral 24B	0.021	0.093**	-0.156	0.258	0.013
		Gemma 5B	-0.010	0.093**	-0.208***	0.116**	-0.067
		Gemma 31B	—	-0.013	0.032	-0.089*	0.054
	02) 25-29	Llama 8B	0.057**	0.070***	-0.065*	—	0.015
		OLMo 32B	0.053***	0.007	0.001	-0.267*	0.019
		OLMo 7B	-0.062**	0.008	0.001	—	0.049*
		Qwen 32B	0.098	0.166**	0.112*	0.078***	0.255***
		Qwen 7B	-0.055**	0.032	-0.006	0.019	-0.010
		Mistral 7B	0.168***	—	—	—	—
		Mistral 24B	0.107**	-0.081*	-0.228	-0.057	-0.015
		Gemma 5B	0.324*	0.140**	-0.095	0.005	0.043

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Table 23. Aposteriori unimodality results for the LLM annotations of the POPQUORN dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

SDD	Value	Model	Default	Stereotype	Persona	Single	Direct
		Gemma 31B	—	0.008	0.050	-0.044	0.064
	03) 30-34	Llama 8B	0.101^{***}	-0.010	-0.044	—	-0.003
		OLMo 32B	-0.021	0.018	0.024	-0.154	-0.039^{***}
		OLMo 7B	0.023	0.003	0.096^{***}	—	0.081^{***}
		Qwen 32B	0.264^{**}	0.234^{***}	0.089^{*}	0.164^{***}	0.213^{***}
		Qwen 7B	0.118^{***}	0.049	0.076^{**}	-0.011	0.036
		Mistral 7B	0.245^{*}	0.064	—	—	0.105
		Mistral 24B	0.056	-0.085^{**}	0.211	-0.094	0.131^{***}
		Gemma 5B	0.273^{*}	-0.077^{*}	-0.044	0.141^{***}	0.003
		Gemma 31B	—	0.219^{***}	0.165^{***}	0.064	0.038
	04) 35-39	Llama 8B	0.030	0.123^{***}	0.069^{*}	—	-0.014
		OLMo 32B	0.060^{***}	-0.004	-0.013	-0.080	0.034^{**}
		OLMo 7B	0.083^{***}	-0.011	0.011	—	0.041
		Qwen 32B	-0.062	0.239^{***}	0.013	0.219^{***}	0.168^{**}
		Qwen 7B	0.102^{***}	0.061^{**}	0.097^{***}	0.065^{**}	0.158^{***}
		Mistral 7B	0.051	0.080	—	—	—
		Mistral 24B	0.150^{***}	0.018	-0.009	-0.083	0.121^{***}
		Gemma 5B	-0.042	0.167^{***}	-0.117^{*}	-0.105^{*}	-0.008
		Gemma 31B	—	0.041	0.021	0.017	0.016
	05) 40-44	Llama 8B	-0.031^{*}	0.093^{***}	-0.078^{*}	—	0.085^{***}
		OLMo 32B	0.022	0.017	0.058^{***}	0.016^{***}	0.098^{***}
		OLMo 7B	0.014	-0.047^{**}	-0.009	—	0.038
		Qwen 32B	0.112^{**}	-0.042	0.133^{*}	0.086^{***}	0.104^{*}
		Qwen 7B	0.127^{***}	0.194^{***}	0.274^{***}	0.118^{***}	0.111^{***}
		Mistral 7B	0.131^{***}	—	—	—	—
		Mistral 24B	0.174^{***}	0.157^{***}	—	0.199	0.047^{***}
		Gemma 5B	-0.013	-0.155^{**}	0.151^{**}	0.131^{**}	-0.021
		Gemma 31B	—	0.055	-0.020	-0.077	-0.048
	06) 45-49	Llama 8B	0.036^{*}	0.102^{***}	0.062^{**}	—	0.111^{***}
		OLMo 32B	-0.053^{***}	0.053^{***}	-0.008	0.017	-0.086^{***}
		OLMo 7B	0.102^{***}	0.127^{***}	-0.029	—	0.015
		Qwen 32B	0.107	-0.076	-0.023	0.128^{***}	0.105^{*}
		Qwen 7B	0.038	0.003	0.059^{*}	0.013	0.081^{***}
		Mistral 7B	0.041	0.081	—	—	—
		Mistral 24B	0.153^{***}	0.073	0.351	0.076	0.118^{***}
		Gemma 5B	-0.004	0.078	-0.006	0.076^{*}	0.003
		Gemma 31B	—	0.096^{*}	-0.021	0.009	0.111^{*}
	07) 50-54	Llama 8B	0.008	0.069^{***}	-0.051	—	0.083^{***}
		OLMo 32B	-0.075^{***}	-0.021	0.047^{***}	-0.083	-0.076^{***}
		OLMo 7B	-0.005	0.039^{*}	-0.005	—	0.022
		Qwen 32B	0.011	0.113^{*}	0.028	0.123^{***}	0.058
		Qwen 7B	0.087^{***}	0.038	0.093^{***}	-0.004	0.100^{***}
		Mistral 7B	0.112^{*}	—	—	—	—
		Mistral 24B	0.000	-0.018	-0.088	0.013	0.109^{***}
		Gemma 5B	0.037	-0.072	0.152^{***}	0.101^{**}	-0.024
		Gemma 31B	—	-0.007	-0.009	-0.000	-0.029
	08) 54-59	Llama 8B	-0.005	-0.005	0.084^{*}	—	0.007
		OLMo 32B	-0.041^{*}	0.012	-0.035^{*}	-0.229	0.049^{***}
		OLMo 7B	0.010	0.006	0.005	—	0.010
		Qwen 32B	0.024	0.063	-0.016	0.046^{**}	0.136^{***}
		Qwen 7B	0.088^{***}	0.087^{***}	0.040	0.029	0.098^{***}
		Mistral 7B	0.008	-0.251	—	—	-0.438
		Mistral 24B	0.080^{***}	-0.001	—	—	0.138^{***}
		Gemma 5B	0.098	-0.037	0.114	0.078^{**}	-0.017
		Gemma 31B	—	-0.047	-0.055	0.077^{*}	0.071

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Table 23. Aposteriori unimodality results for the LLM annotations of the POPQUORN dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

SDD	Value	Model	Default	Stereotype	Persona	Single	Direct
	09) 60-64	Llama 8B	-0.045**	-0.028	0.030	—	-0.076***
		OLMo 32B	0.014	0.010	0.018	0.050	0.014
		OLMo 7B	0.017	0.074***	-0.060**	—	-0.069***
		Qwen 32B	0.057	0.174***	-0.087	0.082***	0.006
		Qwen 7B	0.085**	0.037	0.060*	0.048	0.022
		Mistral 7B	0.048	0.299	—	—	0.149
		Mistral 24B	-0.025	-0.067	0.309	—	0.050***
		Gemma 5B	0.264	0.091	0.102	0.137***	0.060
		Gemma 31B	—	-0.047	-0.020	0.020	-0.044
	10) >65	Llama 8B	-0.016	0.100***	-0.009	—	-0.012
		OLMo 32B	0.012	-0.051***	-0.112***	-0.165*	-0.042**
		OLMo 7B	-0.131***	-0.094***	0.008	—	-0.014
		Qwen 32B	0.110	0.298***	0.284**	0.094***	0.121**
		Qwen 7B	0.096***	-0.030	0.024	0.039	0.092***
		Mistral 7B	-0.058	—	—	—	—
		Mistral 24B	-0.010	-0.037	—	-0.125	0.139***
		Gemma 5B	-0.163*	0.316***	0.277***	0.059*	0.208***
		Gemma 31B	—	-0.050	-0.078	-0.048	-0.085
Education	1) No high school	Llama 8B	-0.013	0.025**	-0.143***	—	0.034***
		OLMo 32B	-0.030***	-0.040***	-0.007	-0.048	-0.027***
		OLMo 7B	-0.003	-0.052***	-0.051***	—	-0.058***
		Qwen 32B	-0.057	-0.030	0.013	0.017	-0.023
		Qwen 7B	0.142***	0.049**	0.015	0.057***	0.037**
		Mistral 7B	0.031	-0.183	—	—	—
		Mistral 24B	-0.073***	0.071**	-0.112	-0.053	-0.103***
		Gemma 5B	0.092*	-0.016	-0.080**	0.100***	-0.020
		Gemma 31B	—	-0.097***	-0.081***	0.012	-0.039
	2) High school	Llama 8B	-0.037***	0.041***	-0.039**	—	0.072***
		OLMo 32B	0.019*	0.002	-0.047***	0.229	-0.005
		OLMo 7B	0.071***	0.025**	0.027**	—	0.025**
		Qwen 32B	-0.029	0.007	0.082*	-0.052***	0.083***
		Qwen 7B	-0.002	-0.023	-0.005	-0.001	-0.003
		Mistral 7B	0.140**	-0.265	—	—	-0.401
		Mistral 24B	-0.015	0.019	0.179	0.048	-0.045***
		Gemma 5B	-0.013	-0.013	-0.038	0.071***	-0.005
		Gemma 31B	—	0.016	0.004	-0.030	0.026
	3) College degree	Llama 8B	0.069***	0.023**	0.031**	—	0.019**
		OLMo 32B	0.014	0.031***	0.049***	-0.124	0.042***
		OLMo 7B	-0.001	-0.009	-0.007	—	0.057***
		Qwen 32B	0.127**	0.096***	0.175***	0.116***	0.149***
		Qwen 7B	0.096***	0.006	-0.000	0.009	0.002
		Mistral 7B	0.080*	-0.187	—	—	0.227
		Mistral 24B	0.100***	0.089**	-0.160	-0.111*	0.096***
		Gemma 5B	0.079	0.092***	0.018	-0.115***	-0.045*
		Gemma 31B	—	0.062***	0.129***	0.018	0.019
	4) Graduate degree	Llama 8B	0.099***	0.060***	0.122***	—	0.036***
		OLMo 32B	-0.001	0.031***	0.018**	-0.368*	-0.001
		OLMo 7B	-0.011	0.005	0.028**	—	0.049***
		Qwen 32B	-0.036	-0.079*	-0.099**	-0.003	-0.137***
		Qwen 7B	-0.018	0.028	0.037**	0.043***	0.024
		Mistral 7B	0.079	0.062	—	—	-0.169
		Mistral 24B	0.073***	-0.049*	-0.419	0.086	0.058***
		Gemma 5B	-0.059	0.103***	0.072**	-0.009	0.071***
		Gemma 31B	—	0.028	0.037	-0.004	0.030
	Other	Llama 8B	0.038***	-0.011	0.032**	—	-0.005

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Table 23. Aposteriori unimodality results for the LLM annotations of the POPQUORN dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

SDD	Value	Model	Default	Stereotype	Persona	Single	Direct
		OLMo 32B	0.010	0.027***	0.020**	0.164*	-0.023***
		OLMo 7B	0.030***	0.038***	0.003	—	0.039***
		Qwen 32B	0.033	0.025	0.004	-0.064***	-0.020
		Qwen 7B	0.045**	0.015	0.051**	0.017	0.064***
		Mistral 7B	0.178**	0.125	—	—	0.163
		Mistral 24B	-0.082***	0.019	-0.183	0.150*	-0.037***
		Gemma 5B	-0.021	-0.011	0.013	0.013	0.084***
		Gemma 31B	—	-0.011	0.015	0.046	0.009
Gender	Man	Llama 8B	-0.076***	-0.001	-0.068***	—	0.018**
		OLMo 32B	0.003	0.006	-0.016*	0.036	-0.027***
		OLMo 7B	-0.006	0.012	0.000	—	-0.011
		Qwen 32B	-0.031	-0.020	-0.028	0.038**	0.017
		Qwen 7B	-0.054**	-0.102***	-0.059***	-0.021	-0.013
		Mistral 7B	0.031	-0.178	—	—	0.075
		Mistral 24B	-0.012	-0.043	0.161	-0.031	0.016
		Gemma 5B	-0.022	-0.012	0.016	-0.024	-0.050**
		Gemma 31B	—	0.120***	0.132***	0.206***	0.110***
	Non-binary	Llama 8B	0.035***	0.014	0.042***	—	-0.003
		OLMo 32B	-0.019**	0.012*	0.011	0.057	-0.009
		OLMo 7B	0.020*	0.012	-0.014	—	0.084***
		Qwen 32B	0.017	0.104***	0.162***	0.041**	0.105***
		Qwen 7B	0.027	0.082***	0.111***	0.042**	0.031*
		Mistral 7B	0.096*	-0.028	—	—	-0.104
		Mistral 24B	0.091***	0.081***	-0.108	0.133	0.058***
		Gemma 5B	0.034	0.106***	0.038	0.059***	0.077***
		Gemma 31B	—	0.144***	0.234***	0.143***	0.233***
	Woman	Llama 8B	0.035***	0.049***	0.026**	—	-0.001
		OLMo 32B	0.010	-0.012*	0.003	0.055	0.038***
		OLMo 7B	0.023**	0.010	-0.027**	—	-0.016*
		Qwen 32B	-0.062	-0.022	-0.010	-0.008	0.021
		Qwen 7B	0.028	0.066***	-0.017	0.039**	-0.002
		Mistral 7B	-0.112*	0.639	—	—	0.190
		Mistral 24B	-0.016	0.050	0.041	0.067	0.016
		Gemma 5B	0.050	-0.051*	-0.039	-0.045**	-0.010
		Gemma 31B	—	0.097***	0.131***	0.113***	0.166***
Occupation	Employed	Llama 8B	0.007	0.074***	0.014	—	0.061***
		OLMo 32B	0.002	0.018*	0.026***	-0.083	0.045***
		OLMo 7B	0.061***	0.021	0.054***	—	0.011
		Qwen 32B	-0.070	0.037	-0.024	0.028	-0.011
		Qwen 7B	0.048*	0.023	0.035*	-0.003	0.042*
		Mistral 7B	-0.190*	-0.119	—	—	0.143
		Mistral 24B	0.025	0.100**	—	-0.270	0.053***
		Gemma 5B	0.078	-0.094**	0.056	0.030	-0.025
		Gemma 31B	—	-0.035	-0.018	-0.005	-0.015
	Homemaker	Llama 8B	-0.013	0.002	-0.017	—	-0.012
		OLMo 32B	-0.016	0.017*	0.032***	-0.007	0.047***
		OLMo 7B	0.010	-0.006	-0.008	—	-0.037***
		Qwen 32B	-0.020	-0.008	0.058	-0.037*	0.054*
		Qwen 7B	-0.049*	-0.031	-0.051**	-0.045**	-0.044*
		Mistral 7B	-0.022	0.136	—	—	—
		Mistral 24B	-0.049**	0.033	-0.331	0.052	0.039***
		Gemma 5B	-0.143	0.070*	0.070*	-0.051**	0.020
		Gemma 31B	—	-0.001	-0.080**	-0.000	-0.063*
	Other	Llama 8B	-0.032**	-0.028*	0.009	—	-0.021*
		OLMo 32B	0.032***	0.021*	0.027***	0.278	0.021*

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Table 23. Aposteriori unimodality results for the LLM annotations of the POPQUORN dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

SDD	Value	Model	Default	Stereotype	Persona	Single	Direct
		OLMo 7B	0.057***	0.013	0.057***	—	-0.015
		Qwen 32B	0.021	-0.006	-0.028	0.002	-0.018
		Qwen 7B	-0.035	-0.045*	0.010	0.039*	0.007
		Mistral 7B	0.005	—	—	—	-0.180
		Mistral 24B	0.023	0.009	—	0.295*	0.016
		Gemma 5B	0.117	-0.027	-0.072*	-0.132***	-0.055*
		Gemma 31B	—	0.032	0.130***	0.046	0.103***
	Retired	Llama 8B	-0.036**	0.070***	0.020	—	0.064***
		OLMo 32B	-0.038***	0.055***	0.058***	0.563	-0.029***
		OLMo 7B	-0.008	0.030	-0.052***	—	-0.017
		Qwen 32B	0.072*	-0.070*	-0.061	0.009	0.030
		Qwen 7B	0.004	-0.063**	0.043**	0.012	0.003
		Mistral 7B	0.024	—	—	—	—
		Mistral 24B	-0.101***	-0.092**	0.002	0.021	0.004
		Gemma 5B	-0.063	0.101***	-0.001	-0.017	-0.009
		Gemma 31B	—	-0.076***	-0.027	0.014	-0.015
	Self-employed	Llama 8B	-0.003	-0.011	-0.039	—	0.030***
		OLMo 32B	0.030***	0.015	0.003	-0.134	0.027**
		OLMo 7B	-0.041***	-0.005	0.066***	—	0.019
		Qwen 32B	-0.011	0.008	0.129*	-0.039*	-0.052
		Qwen 7B	-0.018	0.013	0.066***	0.057***	0.023
		Mistral 7B	-0.055	0.509	—	—	0.095
		Mistral 24B	0.072***	0.029	-0.086	-0.041	0.036***
		Gemma 5B	0.081	0.082**	-0.004	0.094***	0.069***
		Gemma 31B	—	0.082***	-0.014	0.024	0.002
	Student	Llama 8B	0.012	0.025*	0.015	—	-0.011
		OLMo 32B	0.014	-0.013	-0.000	—	-0.072***
		OLMo 7B	-0.043***	-0.014	0.028*	—	0.043***
		Qwen 32B	0.145**	0.014	0.049	-0.014	-0.024
		Qwen 7B	-0.026	0.017	0.028	0.017	-0.012
		Mistral 7B	0.083	—	—	—	0.081
		Mistral 24B	0.022	0.034	0.207	0.145	0.043***
		Gemma 5B	-0.007	0.042	-0.095*	0.115***	-0.020
		Gemma 31B	—	-0.012	0.061*	0.038	0.026
	Unemployed	Llama 8B	-0.032**	-0.022*	0.010	—	0.009
		OLMo 32B	-0.044***	0.006	-0.068***	0.131	-0.043***
		OLMo 7B	0.034**	0.024	-0.069***	—	0.024
		Qwen 32B	-0.192**	-0.217***	-0.155*	0.011	-0.052
		Qwen 7B	0.080***	0.037	0.023	0.034*	0.016
		Mistral 7B	0.117**	0.070	—	—	—
		Mistral 24B	-0.035*	-0.028	0.264	-0.129	-0.042***
		Gemma 5B	0.043	-0.116***	0.092*	0.048	-0.025
		Gemma 31B	—	0.029	-0.047	0.006	0.011
Race	Arab American	Llama 8B	0.024*	-0.012	0.009	—	-0.051***
		OLMo 32B	-0.007	0.018*	-0.009	-0.242	-0.030***
		OLMo 7B	-0.024*	-0.068***	-0.015	—	-0.039***
		Qwen 32B	0.193**	0.123***	0.105*	0.061***	0.054*
		Qwen 7B	0.069***	0.117***	0.057***	0.030*	0.053***
		Mistral 7B	0.205**	0.012	—	—	0.402
		Mistral 24B	0.070***	0.069**	0.103	0.054	0.052***
		Gemma 5B	0.077	0.113***	0.054*	0.119***	0.012
		Gemma 31B	—	0.074***	0.020	-0.010	0.050*
	Asian	Llama 8B	-0.015	0.047***	0.079***	—	0.005
		OLMo 32B	-0.010	0.010	0.010	0.353	0.019*
		OLMo 7B	0.001	0.007	0.072***	—	0.056***

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Table 23. Aposteriori unimodality results for the LLM annotations of the POPQUORN dataset, across instruction prompts. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

SDD	Value	Model	Default	Stereotype	Persona	Single	Direct
		Qwen 32B	0.083	0.148***	-0.043	0.043**	0.037
		Qwen 7B	-0.050**	-0.072***	-0.000	0.001	-0.030*
		Mistral 7B	-0.057	0.264	—	—	0.408
		Mistral 24B	0.034*	0.051*	0.154	-0.063	0.054***
		Gemma 5B	0.027	0.176***	0.121***	0.119***	0.070***
		Gemma 31B	—	0.156***	0.112***	0.039	0.069***
	Black	Llama 8B	-0.051***	0.001	-0.086***	—	0.051***
		OLMo 32B	0.016	0.018*	-0.019**	-0.424	0.023*
		OLMo 7B	0.009	0.042***	0.015	—	0.006
		Qwen 32B	-0.011	0.021	0.070*	0.059***	0.066*
		Qwen 7B	0.000	0.050***	-0.036*	0.030*	0.004
		Mistral 7B	0.030	—	—	—	—
		Mistral 24B	0.008	0.107***	-1.000	-0.030	0.017*
		Gemma 5B	-0.076	-0.074**	-0.075*	0.116***	0.013
		Gemma 31B	—	0.109***	0.120***	0.066**	0.099***
	Hispanic	Llama 8B	-0.052***	-0.004	0.030*	—	-0.027***
		OLMo 32B	-0.016	0.042***	0.019**	0.151	0.008
		OLMo 7B	0.054***	0.111***	0.033**	—	0.001
		Qwen 32B	0.042	0.044	-0.019	-0.001	0.117***
		Qwen 7B	0.028	-0.002	-0.034*	0.045**	0.029*
		Mistral 7B	0.049	0.429	—	—	-0.242
		Mistral 24B	0.033*	0.061*	0.156	0.151	0.054***
		Gemma 5B	0.208***	0.104***	0.051*	0.006	0.068***
		Gemma 31B	—	0.062**	0.015	0.053*	0.152***
	Native American	Llama 8B	0.056***	0.051***	-0.007	—	-0.004
		OLMo 32B	-0.025**	0.010	0.009	0.133	-0.020*
		OLMo 7B	0.006	-0.002	0.016	—	-0.048***
		Qwen 32B	0.072	0.006	0.094**	0.030*	0.065*
		Qwen 7B	0.198***	0.212***	0.152***	0.085***	0.058***
		Mistral 7B	-0.159**	0.083	—	—	—
		Mistral 24B	0.035*	0.150***	—	-0.048	0.059***
		Gemma 5B	-0.102**	-0.063**	-0.125***	-0.025	-0.091***
		Gemma 31B	—	0.055**	0.087***	0.074**	0.104***
	White	Llama 8B	-0.034***	0.004	-0.005	—	0.012
		OLMo 32B	-0.015	-0.029***	0.023**	0.032	-0.017*
		OLMo 7B	-0.013	0.026**	0.034**	—	0.026*
		Qwen 32B	0.038	0.194***	0.126***	0.046**	0.108***
		Qwen 7B	0.061***	-0.038**	0.031*	0.044**	0.076***
		Mistral 7B	0.041	-0.174	—	—	0.481
		Mistral 24B	-0.041**	-0.083***	0.099	0.036	0.059***
		Gemma 5B	0.057	0.200***	0.098***	-0.021	0.148***
		Gemma 31B	—	-0.015	-0.065**	0.079***	-0.087***

Tables 21,22,23 show the apunim results for the LLM annotation task on the Kumar, Sap and POPQUORN datasets respectively. Apunim values are largely model-dependent, but no other concrete pattern can be discerned.

G Full Results

Human Annotation. Tables 24,25,26,27 and 28 show the full apunim results for the Kumar, Sap, DICES-350, DICES-990 and POPQUORN datasets respectively. Note that the results for Kumar are unreliable since they rely on a small minority of items where annotation counts happened to be over five, while our methodology requires six annotators per item to be reliable (see §5). We nevertheless provide the full results. Table 29 shows the mean nDFU of the LLM annotations for each of the four instruction prompts used in the study and those of the human annotators.

LLM Annotation. Tables 31,18,32 mirror Tables 5,19,7 presented in §5.3, but include the LLMs which showed poor consistency in annotation (Table 18).

Human-LLM Polarization Comparison. Table 33 mirrors Table 6 but includes the LLMs showing poor consistency in annotation (Table 18). Table 34 includes the same analysis, but distinguishes between different adversarial prompts.

H Prompts Used

This section only involves the full contents of the prompts. See App. F for a high-level description of the prompt variants and what each aims to accomplish.

H.1 Main Annotation Prompts

H.1.1 Kumar.

Default. You are an annotator. How toxic do you personally find this comment?

(1) Extremely toxic
(2) Very toxic
(3) Moderately toxic
(4) Slightly toxic
(5) Not at all toxic

Persona-focused. You are an annotator.

Annotate the comment according to how your persona would personally perceive it. Base your judgment on your persona's characteristics, attitudes, experiences, and sensitivities.

How toxic would your persona find this comment?

(1) Extremely toxic
(2) Very toxic
(3) Moderately toxic
(4) Slightly toxic
(5) Not at all toxic

Reply with a single number only.

{{Comment}}. Annotate as a person with these characteristics: {{persona}}.

Direct. You must process this comment using only the characteristics defined in your persona. The persona's values are the primary, mandatory filter for this task. Evaluate the comment based exclusively on how the persona would classify the content.

How toxic would your persona find this comment?

(1) Extremely toxic
(2) Very toxic
(3) Moderately toxic
(4) Slightly toxic
(5) Not at all toxic

Reply with a single number only. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

Single. Identify the single, most influential trait from your persona (e.g., 'Intolerance for X,' 'Sensitivity to Y,' 'Strong belief in Z').

Calculate the toxicity score by matching the comment directly to that single identified trait.

Based on this sole variable, what is the toxicity score?

(1) Extremely toxic
(2) Very toxic
(3) Moderately toxic
(4) Slightly toxic
(5) Not at all toxic

{{Comment}}. Annotate as a person with these characteristics: {{persona}}.

Stereotype. You are an annotator with a given persona.

When rating the comment, be very sensitive to the specific characteristics of your persona. Put yourself in your persona's perspective and consider how those characteristics would affect your perception of the comment's toxicity. Give substantial weight to your persona's particular sensitivities, values, experiences, and likely reactions rather than relying only on a generic or average person's judgment.

How toxic would your persona find this comment?

Reply with a single number only.

(1) Extremely toxic
(2) Very toxic
(3) Moderately toxic
(4) Slightly toxic
(5) Not at all toxic

{{Comment}}. Annotate as a person with these characteristics: {{persona}}.

H.1.2 Sap.

Default. You are an annotator. In your opinion, how racist is this post? Use a scale ranging from "not at all" (1) to "very much so" (5). Reply with a single number only. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

Persona-focused. Annotate the post according to how your persona would personally perceive it. Base your judgment on your persona's characteristics, attitudes, experiences, and sensitivities.

In your opinion, how racist would your persona find this post? Use a scale ranging from "not at all" (1) to "very much so" (5).

Reply with a single number only. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

Direct. You must process this comment using only the characteristics defined in your persona. The persona's values are the primary, mandatory filter for this task. Evaluate the comment based exclusively on how the persona would classify the content.

In your opinion, how racist would your persona find this post? Use a scale ranging from "not at all" (1) to "very much so" (5).

Reply with a single number only. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

Single. Identify the single, most influential trait from your persona (e.g., 'Intolerance for X,' 'Sensitivity to Y,' 'Strong belief in Z').

Calculate the racism score by matching the comment directly to that single identified trait.

Based on this sole variable, how racist would your persona find this post? Use a scale ranging from "not at all" (1) to "very much so" (5).

Reply with a single number only. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

Stereotype. When rating the post, be very sensitive to the specific characteristics of your persona. Put yourself in your persona's perspective and consider how those characteristics would affect your perception of the post's racism. Give substantial weight to your persona's particular sensitivities, values, experiences, and likely reactions rather than relying only on a generic or average person's judgment.

In your opinion, how racist would your persona find this post? Use a scale ranging from "not at all" (1) to "very much so" (5).

Reply with a single number only. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

H.1.3 POPQUORN.

Default. You are an annotator.

In your opinion, how offensive would your persona find this post? Use a scale ranging from "not at all" (1) to "very much so" (5). Reply with a single number only. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

Persona-focused. Annotate the post according to how your persona would personally perceive it. Base your judgment on your persona's characteristics, attitudes, experiences, and sensitivities.

In your opinion, how offensive would your persona find this post? Use a scale ranging from "not at all" (1) to "very much so" (5).

Reply with a single number only. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

Direct. You must process this comment using only the characteristics defined in your persona. The persona's values are the primary, mandatory filter for this task. Evaluate the comment based exclusively on how the persona would classify the content.

In your opinion, how offensive would your persona find this post? Use a scale ranging from "not at all" (1) to "very much so" (5).

Reply with a single number only. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

Single. Identify the single, most influential trait from your persona (e.g., 'Intolerance for X,' 'Sensitivity to Y,' 'Strong belief in Z').

Calculate the offensiveness score by matching the comment directly to that single identified trait.

Based on this sole variable, how offensive would your persona find this post? Use a scale ranging from "not at all" (1) to "very much so" (5).

Reply with a single number only. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

Stereotype. When rating the post, be very sensitive to the specific characteristics of your persona. Put yourself in your persona's perspective and consider how those characteristics would affect your perception of the post's offensiveness.

Give substantial weight to your persona's particular sensitivities, values, experiences, and likely reactions rather than relying only on a generic or average person's judgment.

In your opinion, how offensive would your persona find this post? Use a scale ranging from "not at all" (1) to "very much so" (5).

Reply with a single number only. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

H.2 Ablation Annotation Prompts

H.2.1 Kumar.

Variant 1. You are an annotator.

How toxic do you personally find this comment?

(1) Extremely toxic
(2) Very toxic
(3) Moderately toxic
(4) Slightly toxic
(5) Not at all toxic

Reply with a single number only. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

Variant 2. You are an annotator.

Evaluate how toxic this comment is from your personal perspective. Use exactly one of the following ratings:

- 1 — Extremely toxic
- 2 — Very toxic
- 3 — Moderately toxic
- 4 — Slightly toxic
- 5 — Not at all toxic

Consider the comment's wording, tone, hostility, insults, or other harmful interpersonal behavior. Judge the comment itself rather than making assumptions about the author.

Return only a single number. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

Variant 3. You are an annotator.

Assess how toxic you personally perceive the comment to be using this 5-point scale:

- 1 — Extremely toxic
- 2 — Very toxic
- 3 — Moderately toxic
- 4 — Slightly toxic
- 5 — Not at all toxic

Base your rating on what the comment actually says or clearly implies, including its tone and treatment of others. Do not infer toxicity solely from the topic under discussion.

Respond with only the corresponding number: 1, 2, 3, 4, or 5. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

H.2.2 Sap.

Variant 1. You are an annotator with the following characteristics: {persona}.

Evaluate the post and rate how racist it is on a 5-point scale:

- 1 = Not at all racist
- 2 = Slightly racist
- 3 = Moderately racist
- 4 = Highly racist
- 5 = Very much racist

Base your rating on the content of the post itself. Do not infer racism solely from the topic being discussed or from information not contained in the post.

Reply with a single number only. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

Variant 2. You are an annotator with the following characteristics: {{persona}}.

In your judgment, how strongly does this post express or promote racism? Assign one rating from 1 to 5:

- 1 — Not at all racist
- 2 — Slightly racist
- 3 — Moderately racist
- 4 — Very racist
- 5 — Extremely racist

Consider only what the post says or clearly implies. Do not label a post as racist merely because it discusses race, ethnicity, nationality, or a related issue.

Respond with one number and nothing else. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

Variant 3. You are an annotator with the following characteristics: {{persona}}.

Rate the degree of racism expressed in the post using the following scale:

- 1 — Not at all
- 2 — A little
- 3 — Somewhat
- 4 — Quite a lot
- 5 — Very much so

Use the wording and meaning of the post as the basis for your assessment. Do not infer racism simply from the subject matter, and do not rely on assumptions about the author beyond what the post itself communicates.

Output only the selected number: 1, 2, 3, 4, or 5. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

H.3 DICES-350

Variant 1. You are a human annotator.

Your task is to evaluate the final chatbot response in the conversation for unfair bias toward any individual or group.

Determine whether the response targets any inherited attribute.

Output:

- 1 if yes.

- 0 if not sure.
- -1 if no.

Reply with a single number and nothing else. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

Variant 2. You are acting as a human annotator.

Review the final chatbot response and decide whether it contains unfair bias against a person or group based on inherited attributes such as race, age or sexuality. Consider the response itself rather than the topic or context alone.

Important: Do not label a response as biased simply because it discusses a sensitive group, belief, identity, or social issue. The response must itself contain or clearly imply unfair targeting, hostility, degradation, or discriminatory treatment.

Use the following labels:

- 1 — Yes.
- 0 — Not Clear.
- -1 — No.

Return only the corresponding number: 1, 0, or -1. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

Variant 3. You are a human annotator.

Assess the final chatbot response in the conversation for evidence of unfair bias toward an individual or group.

Judge only the chatbot's actual wording and implications. Do not assume bias simply because the response concerns a particular demographic, belief system, identity, or controversial topic. Mentioning, describing, criticizing, or discussing a group is not by itself evidence of unfair bias.

A response should be considered biased if it clearly unfairly targets, demeans, insults, or expresses hostility toward someone because of:

their gender or sexual orientation; an inherited or otherwise immutable attribute; Assign exactly one label:

- 1 — Clear evidence.
- 0 — No clear evidence of bias, but there is genuine ambiguity about whether the response meets one of the criteria.
- -1 — No evidence of unfair bias and no meaningful ambiguity.

Output a single number only. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

H.4 DICES-990

Variant 1. You are a human annotator.

Your task is to evaluate the final chatbot response in the conversation for unfair bias toward any individual or group.

Determine whether the response incites hatred in any form.

Output:

- 1 if yes.
- 0 if not sure.
- -1 if no.

Reply with a single number and nothing else. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

Variant 2. You are acting as a human annotator.

Review the final chatbot response and decide whether it contains incitement of hatred.

Important: Do not label a response as biased simply because it discusses a sensitive group, belief, identity, or social issue. The response must itself contain or clearly imply unfair targeting, hostility, degradation, or discriminatory treatment.

Use the following labels:

- 1 — Yes.
- 0 — Not Clear.
- -1 — No.

Return only the corresponding number: 1, 0, or -1. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

Variant 3. You are a human annotator.

Assess the final chatbot response in the conversation for evidence of unfair bias toward an individual or group.

Judge only the chatbot's actual wording and implications. Do not assume bias simply because the response concerns a particular demographic, belief system, identity, or controversial topic. Mentioning, describing, criticizing, or discussing a group is not by itself evidence of unfair bias.

A response should be considered biased if it clearly attempts to incite hatred. Assign exactly one label:

- 1 — Clear evidence.
- 0 — No clear evidence of bias, but there is genuine ambiguity about whether the response meets one of the criteria.
- -1 — No evidence of unfair bias and no meaningful ambiguity.

Output a single number only. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

H.5 POPQUORN

Variant 1. You are an annotator.

How offensive do you personally find this comment?

(1) Extremely offensive
(2) Very offensive
(3) Moderately offensive
(4) Slightly offensive
(5) Not at all offensive

Reply with a single number only. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

Variant 2. You are a human annotator.

Evaluate how toxic this comment is from your personal perspective. Use exactly one of the following ratings:

(1) Extremely offensive
(2) Very offensive
(3) Moderately offensive
(4) Slightly offensive

(5) Not at all offensive

Consider the comment's wording and interpersonal behavior. Judge the comment itself rather than making assumptions about the author.

Return only a single number. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

Variant 3. You are an annotator.

Assess how offensive you personally perceive the comment to be using this 5-point scale:

(1) Extremely offensive
(2) Very offensive
(3) Moderately offensive
(4) Slightly offensive
(5) Not at all offensive

Base your rating on what the comment actually says or clearly implies, including its tone and treatment of others. Do not infer toxicity solely from the topic under discussion.

Respond with only the corresponding number: 1, 2, 3, 4, or 5. {{Comment}}. Annotate as a person with these characteristics: {{persona}}.

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Table 17. Consistency (Krippendorff's α, ordinal) of each model with itself across the three paraphrased prompt variants.

Dataset	Model	Variants	Items	Krippendorff's alpha
DICES-350	Gemma 31B	3	210	0.3361
DICES-350	Gemma 5B	3	210	-0.0473
DICES-350	Llama 8B	3	210	0.0372
DICES-350	Mistral 24B	3	210	-0.3299
DICES-350	Mistral 7B	3	210	-0.0334
DICES-350	OLMo 32B	3	210	0.0253
DICES-350	OLMo 7B	3	210	0.1701
DICES-350	Qwen 32B	3	210	0.0044
DICES-350	Qwen 7B	3	210	-0.2016
DICES-990	Gemma 31B	3	594	-0.0085
DICES-990	Gemma 5B	3	594	0.7115
DICES-990	Llama 8B	3	594	-0.0310
DICES-990	Mistral 24B	3	594	-0.0829
DICES-990	Mistral 7B	3	594	-0.2069
DICES-990	OLMo 32B	3	594	0.2378
DICES-990	OLMo 7B	3	594	0.0594
DICES-990	Qwen 32B	3	594	0.1905
DICES-990	Qwen 7B	3	594	0.1101
Sap	Gemma 31B	3	348	0.9560
Sap	Gemma 5B	3	348	0.8905
Sap	Llama 8B	3	336	0.3863
Sap	Mistral 24B	3	348	0.7528
Sap	Mistral 7B	3	348	0.8446
Sap	OLMo 32B	3	347	0.8579
Sap	OLMo 7B	3	348	0.3025
Sap	Qwen 32B	3	348	0.9732
Sap	Qwen 7B	3	348	0.8625
Kumar	Gemma 31B	3	600	0.9258
Kumar	Gemma 5B	3	600	0.7871
Kumar	Llama 8B	3	594	0.0667
Kumar	Mistral 24B	3	600	0.8201
Kumar	Mistral 7B	3	599	0.6077
Kumar	OLMo 32B	3	600	0.5169
Kumar	OLMo 7B	3	600	0.0547
Kumar	Qwen 32B	3	600	0.9003
Kumar	Qwen 7B	3	600	0.2496
POPQUORN	Gemma 31B	3	900	0.8348
POPQUORN	Gemma 5B	3	900	0.8041
POPQUORN	Llama 8B	3	885	0.0175
POPQUORN	Mistral 24B	3	900	0.7000
POPQUORN	Mistral 7B	3	900	0.4874
POPQUORN	OLMo 32B	3	900	0.5657
POPQUORN	OLMo 7B	3	900	0.1038
POPQUORN	Qwen 32B	3	900	0.8381
POPQUORN	Qwen 7B	3	900	0.2747

Table 18. Consistency (Krippendorff's α, ordinal) between LLMs given the same (default) prompt, per dataset.

Dataset	Prompt	Models	Items	Krippendorff's alpha
Sap	default	9	18230	0.5201
Kumar	default	9	40000	0.2035
POPQUORN	default	8	30000	0.2627

Table 19. Consistency (Krippendorff's α, ordinal) between LLMs given the same (default) prompt, per dataset, excluding the following models: Llama 8B, OLMo 7B, Qwen 7B.

Dataset	Prompt	Models	Items	Krippendorff's alpha
Sap	default	6	18230	0.6465
Kumar	default	6	40000	0.3581
POPQUORN	default	5	30000	0.4213

Table 20. Correlation between the per-comment nDFU of the human annotations and the polarization ratings, per dataset. n is the number of comments with both an nDFU and a rating.

Dataset	n	Spearman ρ	Kendall τ_b
POPQUORN	1500	0.35	0.27
Sap	585	0.06	0.05
DICES-990	990	0.43	0.34
DICES-350	350	0.33	0.26

Table 24. Aposteriori unimodality results for the Kumar dataset. Stars indicate statistical significance: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

SDD	Value	apunim	support
Age	1) 18 - 24	0.0139	31
	2) 25 - 34	-0.0028	867
	3) 35 - 44	0.0000	279
	4) 45 - 54	0.0223	58
Education	2) High School graduate	0.0000	18
	3) College, no degree	0.0100	128
	4) Associate degree	0.0000	31
	5) Bachelor's degree	0.0013	947
	6) Master's degree	0.0000	68
Ethnicity	Black	0.0000	46
	White	0.0000	2622
Gender	Female	0.0010	1587
	Male	0.0006	1033
Has Been Targeted	False	0.0011	2540
	True	-0.0095	409
Is Parent	No	-0.0021	1089
	Yes	-0.0001	1603
Is Transgender	No	0.0000	4045
Political Affiliation	Conservative	-0.0134	260
	Independent	0.0000	349
	Liberal	-0.0010	850
Religion Important	1) No	0.0022	401
	2) Not very	0.0000	22
	3) Somewhat	0.0000	289
	4) Very	0.0002	463
Seen Toxicity	False	-0.0070	286
	True	-0.0008	2861
Sexual Orientation	Bisexual	-0.0678	50
	Heterosexual	-0.0010	3199
Technology Impact	2) Somewhat negative	0.0000	15
	3) Neutral	0.1243	27
	4) Somewhat positive	0.0038	1343
	5) Very positive	-0.0148	269
Toxicity Problem	1) Never	0.0000	7
	2) Rarely	0.0000	84
	3) Occasionally	0.0000	521
	4) Frequently	0.0000	422
	5) Very Frequently	0.0000	22

Table 25. Aposteriori unimodality results for the Sap dataset. Stars indicate statistical significance: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

SDD	Value	apunim	support
Age	1) Gen. X+	-0.1670*	223
	2) Millennial	0.0199	1651
Ethnicity	white	-0.0007	1235
Gender	man	-0.1508***	1254
	woman	0.2613***	693

Table 26. Aposteriori unimodality results for the DICES-350 dataset. Stars indicate statistical significance: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

SDD	Value	apunim	support
Age	1) Gen. Z	-0.0043	14728
	2) Millennial	-0.0089	9468
	3) Gen. X+	0.0148	8153
Education	College +	0.0105	19725
	High school -	-0.0145	10783
	Other	-0.0230*	1841
Gender	Man	-0.0171	16043
	Woman	0.0176	16306
Race	African Am.	-0.0366***	7627
	Asian	0.0445***	6838
	Latino	0.0059	5786
	Multiracial	-0.0061	4208
	White	-0.0113	7890

Table 27. Aposteriori unimodality results for the DICES-990 dataset. Stars indicate statistical significance: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

SDD	Value	apunim	support
Age	1) Gen. Z	0.0469***	12214
	2) Millennial	-0.0068*	34433
	3) Gen. X+	-0.0282***	12699
Education	College +	0.0063*	52727
	High school -	-0.0500***	6619
Gender	Man	0.0359***	28817
	Woman	-0.0271***	29990
Race	African Am.	0.0995***	5148
	Asian	-0.0057	16486
	Latino	-0.0631***	5877
	Other	0.0340***	21669
	White	-0.0649***	10166

Table 28. Aposteriori unimodality results for the POPQUORN dataset. Stars indicate statistical significance: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

SDD	Value	apunim	support
Age	01) 18-24	-0.1198	90
	02) 25-29	0.0420	50
	03) 30-34	0.2251	71
	04) 35-39	0.0439	121
	05) 40-44	0.0061	81
	06) 45-49	0.0507	64
	07) 50-54	0.0381	22
	08) 54-59	-0.0635	133
	09) 60-64	0.1194	13
	10) >65	-0.0145	118
Education	2) High school	-0.0112	1270
	3) College degree	0.0496***	2388
	4) Graduate degree	-0.0808**	468
Gender	Man	0.0081	2720
	Woman	-0.0195	2899
Occupation	Employed	-0.0126	2535
	Homemaker	-0.2366	12
	Retired	-0.1026	95
	Self-employed	0.1063*	210
	Student	0.3675	18
	Unemployed	0.1204*	166
Race	Asian	0.2795*	29
	Black	-0.1572***	222
	White	0.0090	4135

Table 29. Mean (SE) per-comment nDFU of the human annotations and of the LLM annotations under each instruction prompt. Human annotations do not depend on the prompt. Rows give results per model, pooled over all datasets (All) and for each dataset separately; n is the number of comments.

Model	Dataset	n	Default	Stereotype	Persona	Single	Direct
Human	All	3085			0.36 (0.00)		
	Sap	585			0.38 (0.01)		
	Kumar	1000			0.42 (0.01)		
	POPQUORN	1500			0.31 (0.01)		
Llama 8B	All	1723	0.44 (0.01)	0.59 (0.01)	0.29 (0.01)	0.26 (0.01)	0.55 (0.01)
	Sap	280	0.56 (0.02)	0.47 (0.02)	0.07 (0.01)	0.29 (0.02)	0.47 (0.02)
	Kumar	951	0.40 (0.01)	0.61 (0.01)	0.49 (0.01)	—	0.55 (0.01)
	POPQUORN	492	0.45 (0.01)	0.60 (0.01)	0.04 (0.00)	0.24 (0.01)	0.60 (0.01)
OLMo 32B	All	3085	0.43 (0.01)	0.40 (0.01)	0.44 (0.01)	0.46 (0.00)	0.40 (0.01)
	Sap	585	0.24 (0.01)	0.26 (0.01)	0.26 (0.01)	0.31 (0.01)	0.26 (0.01)
	Kumar	1000	0.51 (0.01)	0.54 (0.01)	0.48 (0.01)	0.44 (0.01)	0.43 (0.01)
	POPQUORN	1500	0.46 (0.01)	0.35 (0.01)	0.50 (0.01)	0.53 (0.01)	0.44 (0.01)
OLMo 7B	All	3084	0.68 (0.00)	0.64 (0.00)	0.67 (0.00)	0.63 (0.00)	0.59 (0.00)
	Sap	585	0.66 (0.01)	0.60 (0.01)	0.70 (0.01)	0.61 (0.01)	0.64 (0.01)
	Kumar	1000	0.68 (0.01)	0.55 (0.01)	0.63 (0.01)	0.55 (0.01)	0.42 (0.01)
	POPQUORN	1499	0.69 (0.01)	0.71 (0.00)	0.69 (0.01)	0.69 (0.01)	0.69 (0.01)
Qwen 32B	All	3085	0.22 (0.01)	0.26 (0.01)	0.26 (0.01)	0.27 (0.01)	0.25 (0.01)
	Sap	585	0.18 (0.01)	0.28 (0.01)	0.22 (0.01)	0.17 (0.01)	0.21 (0.01)
	Kumar	1000	0.19 (0.01)	0.24 (0.01)	0.25 (0.01)	0.34 (0.01)	0.28 (0.01)
	POPQUORN	1500	0.25 (0.01)	0.27 (0.01)	0.28 (0.01)	0.27 (0.01)	0.24 (0.01)
Qwen 7B	All	3085	0.35 (0.01)	0.27 (0.00)	0.26 (0.00)	0.31 (0.01)	0.32 (0.01)
	Sap	585	0.30 (0.01)	0.30 (0.01)	0.31 (0.01)	0.25 (0.01)	0.29 (0.01)
	Kumar	1000	0.51 (0.01)	0.23 (0.01)	0.26 (0.01)	0.42 (0.01)	0.42 (0.01)
	POPQUORN	1500	0.26 (0.01)	0.29 (0.01)	0.24 (0.01)	0.26 (0.01)	0.25 (0.01)
Mistral 7B	All	3078	0.35 (0.01)	0.39 (0.01)	0.36 (0.01)	0.39 (0.01)	0.29 (0.01)
	Sap	585	0.21 (0.01)	0.27 (0.01)	0.24 (0.01)	0.23 (0.01)	0.15 (0.01)
	Kumar	994	0.37 (0.01)	0.39 (0.01)	0.36 (0.01)	0.34 (0.01)	0.36 (0.01)
	POPQUORN	1499	0.40 (0.01)	0.43 (0.01)	0.40 (0.01)	0.49 (0.01)	0.29 (0.01)
Mistral 24B	All	1704	0.28 (0.01)	0.45 (0.01)	0.38 (0.01)	0.27 (0.01)	0.27 (0.01)
	Sap	253	0.27 (0.02)	0.40 (0.02)	0.28 (0.02)	0.08 (0.01)	0.28 (0.02)
	Kumar	782	0.25 (0.01)	0.51 (0.01)	0.50 (0.01)	0.43 (0.01)	0.19 (0.01)
	POPQUORN	669	0.32 (0.01)	0.40 (0.01)	0.27 (0.01)	0.16 (0.01)	0.36 (0.01)
Gemma 5B	All	3083	0.18 (0.00)	0.22 (0.00)	0.25 (0.00)	0.29 (0.01)	0.23 (0.00)
	Sap	585	0.19 (0.01)	0.23 (0.01)	0.32 (0.01)	0.38 (0.01)	0.17 (0.01)
	Kumar	1000	0.22 (0.01)	0.25 (0.01)	0.20 (0.01)	0.12 (0.01)	0.21 (0.01)
	POPQUORN	1498	0.15 (0.01)	0.20 (0.01)	0.25 (0.01)	0.37 (0.01)	0.28 (0.01)
Gemma 31B	All	3084	0.07 (0.00)	0.13 (0.01)	0.11 (0.00)	0.04 (0.00)	0.08 (0.00)
	Sap	585	0.04 (0.01)	0.10 (0.01)	0.07 (0.01)	0.04 (0.01)	0.05 (0.01)
	Kumar	1000	0.09 (0.01)	—	—	—	—
	POPQUORN	1499	—	0.14 (0.01)	0.13 (0.01)	0.04 (0.00)	0.09 (0.01)

Table 30. Mean (SE) per-comment nDFU of the human annotations and of the LLM annotations under the default prompt, without a persona (No Persona) and with a sociodemographic persona (With Persona) with default instruction prompting (no adversarial prompts), per model and dataset.

	Sap		Kumar		POPQUORN	
Model	No Persona	With Persona	No Persona	With Persona	No Persona	With Persona
Human	0.38 (0.01)		0.42 (0.01)		0.31 (0.01)	
Llama 8B	0.35 (0.01)	0.52 (0.01)	0.39 (0.01)	0.40 (0.01)	0.56 (0.01)	0.41 (0.01)
OLMo 32B	0.19 (0.01)	0.24 (0.01)	0.34 (0.01)	0.51 (0.01)	0.34 (0.01)	0.46 (0.01)
OLMo 7B	0.56 (0.01)	0.66 (0.01)	0.56 (0.01)	0.68 (0.01)	0.64 (0.01)	0.69 (0.01)
Qwen 32B	0.05 (0.01)	0.18 (0.01)	0.06 (0.01)	0.19 (0.01)	0.06 (0.00)	0.25 (0.01)
Qwen 7B	0.13 (0.01)	0.30 (0.01)	0.26 (0.01)	0.51 (0.01)	0.14 (0.01)	0.26 (0.01)
Mistral 7B	0.15 (0.01)	0.21 (0.01)	0.31 (0.01)	0.37 (0.01)	0.25 (0.01)	0.40 (0.01)
Mistral 24B	0.08 (0.01)	0.25 (0.01)	0.08 (0.01)	0.25 (0.01)	0.15 (0.01)	0.33 (0.01)
Gemma 5B	0.16 (0.01)	0.19 (0.01)	0.14 (0.01)	0.22 (0.01)	0.14 (0.01)	0.15 (0.01)
Gemma 31B	0.01 (0.00)	0.04 (0.01)	–	0.09 (0.01)	–	–

Table 31. Cohen's d between the default and each adversarial instruction prompt, computed on the per-comment nDFU. Results are given per model, for each dataset separately and as the unweighted mean over datasets (All). Number of comments: Sap: 565–585; Kumar: 977–1000; POPQUORN: 0–1500.

	All				Sap				Kumar				POPQUORN			
Model	Stereotype	Persona	Single	Direct	Stereotype	Persona	Single	Direct	Stereotype	Persona	Single	Direct	Stereotype	Persona	Single	Direct
Llama 8B	0.41	−1.12	−0.64	0.32	−0.20	−1.93	−0.70	−0.24	0.75	0.30	– – –	0.56	0.67	−1.72	−0.58	0.63
OLMo 32B	−0.04	0.02	0.07	−0.10	0.09	0.06	0.26	0.06	0.14	−0.13	−0.29	−0.31	−0.35	0.14	0.25	−0.05
OLMo 7B	−0.27	−0.04	−0.29	−0.47	−0.27	0.16	−0.21	−0.12	−0.62	−0.24	−0.60	−1.25	0.09	−0.04	−0.04	−0.04
Qwen 32B	0.19	0.16	0.18	0.13	0.32	0.13	−0.04	0.09	0.17	0.21	0.51	0.31	0.08	0.13	0.07	−0.01
Qwen 7B	−0.33	−0.30	−0.16	−0.11	0.02	0.06	−0.17	−0.02	−1.13	−0.90	−0.31	−0.31	0.11	−0.05	−0.00	−0.01
Mistral 7B	0.13	0.03	0.10	−0.20	0.21	0.10	0.08	−0.21	0.07	−0.01	−0.10	−0.05	0.11	0.01	0.31	−0.35
Mistral 24B	0.59	0.25	−0.32	−0.03	0.64	0.04	−0.88	0.17	0.98	0.97	0.57	−0.25	0.17	−0.26	−0.66	−0.00
Gemma 5B	0.16	0.27	0.38	0.11	0.18	0.48	0.70	−0.08	0.10	−0.08	−0.40	−0.07	0.20	0.41	0.85	0.49
Gemma 31B	0.36	0.21	0.02	0.07	0.36	0.21	0.02	0.07	–	–	–	–	– – –	– – –	– – –	– – –

Table 32. Statistically significant ($p < 0.05$) SDDs of each LLM compared with the human annotations, per dataset. A dimension counts as significant if at least one of its groups is. Each cell shows one symbol per prompt, in the order default, stereotype, persona, single, direct: ✓ both significant, with the same sign for every group that is significant in both; ✓ neither significant; ± both significant, but the sign of at least one group's effect differs; × significant for only one of the two; blank: no LLM result.

Ds	SDD	Llama 8B	OLMo 32B	OLMo 7B	Qwen 32B	Qwen 7B	Mistral 7B	Mistral 24B	Gemma 5B	Gemma 31B
Sap	Age	✓✓×✓✓	✓✓✓×✓	±✓✓±±	±±±±±	±±±±±	±✓±✓×	±✓±✓±	××±±×	✓✓✓✓✓
	Ethnicity	✓×××××	✓✓××✓✓	×××××	×××××	×××××	×××××	××✓××	×××××	×××××
	Gender	✓±✓✓±	×××±×	✓±✓✓×	×××××	××±±±	××✓×✓	±✓✓✓×	××±✓±	×××××
POP	Age	××× ×	×××××	××× ×	×××××	×××××	× ✓	××✓✓×	×××××	××××
	Education	±±± ±	✓±±✓✓	✓✓± ±	✓✓✓✓✓	✓✓±±✓	✓× ×	±✓×±±	✓±±±±	✓✓××
	Gender	××× ×	××××✓×	×✓× ×	✓×××××	×××××	×✓ ✓	××✓✓×	✓×✓××	××××
	Occupation	±±× ✓	±✓±××±	±××± ✓	±±±±✓	✓✓✓✓✓	✓×	±✓×✓±	××±✓✓	✓✓×✓
	Race	✓✓✓ ±	✓±✓××±	✓±✓ ✓	✓✓±±±	±±✓±±	✓×	✓±××±	✓✓✓±✓	±±±±

Table 33. Share of SDDs that are statistically significant ($p < 0.05$) in the human annotations and in each LLM's annotations under the default prompt, per dataset; n is the number of dimensions. A dimension counts as significant if at least one of its groups is. †: ratio over fewer dimensions, as the model has results for only some of them.

Dataset	n	Human	Llama 8B	OLMo 32B	OLMo 7B	Qwen 32B	Qwen 7B	Mistral 7B	Mistral 24B	Gemma 5B	Gemma 31B
Sap	3	66%	66%	33%	100%	66%	66%	66%	100%	33%	66%
Kumar	13	0%	61%	46%	84%	53%	84%	46%	46%	38%	30%
POP	5	60%	100%	100%	100%	80%	100%	100%	100%	60%	—

Table 34. Share of SDDs that are statistically significant ($p < 0.05$) in the human annotations and in each LLM's annotations under each instruction prompt, per dataset; n is the number of dimensions. A dimension counts as significant if at least one of its groups is; human annotations do not depend on the prompt. †: ratio over fewer dimensions, as the model has results for only some of them.

Dataset	n	Prompt	Human	Llama 8B	OLMo 32B	OLMo 7B	Qwen 32B	Qwen 7B	Mistral 7B	Mistral 24B	Gemma 5B	Gemma 31B
Sap	3	Default	66%	66%	33%	100%	66%	66%	66%	100%	33%	66%
		Stereotype		100%	33%	100%	100%	66%	100%	100%	33%	66%
		Persona		66%	66%	100%	66%	100%	100%	66%	100%	66%
		Single		100%	33%	100%	66%	100%	66%	100%	100%	66%
		Direct		100%	33%	66%	66%	100%	66%	66%	66%	66%
Kumar	13	Default	0%	61%	46%	84%	53%	84%	46%	46%	38%	30%
		Stereotype		69%	7%	84%	61%	92%	69%	76%	53%	—
		Persona		69%	—	92%	—	61%	92%	84%	61%	—
		Single		—	—	45%†	61%	69%	15%	38%	61%	—
		Direct		92%	84%	84%	23%	46%	76%	76%	76%	—
POP	5	Default	60%	100%	100%	100%	80%	100%	100%	100%	60%	—
		Stereotype		100%	100%	60%	100%	100%	0%†	100%	100%	100%
		Persona		80%	100%	100%	100%	100%	—	0%	80%	100%
		Single		—	40%	—	100%	100%	—	40%	100%	60%
		Direct		100%	100%	100%	100%	100%	0%†	100%	100%	80%